Foundations of Pelvis and Perineum

Elizabeth A Spudich, PhD

JeffMD 2026

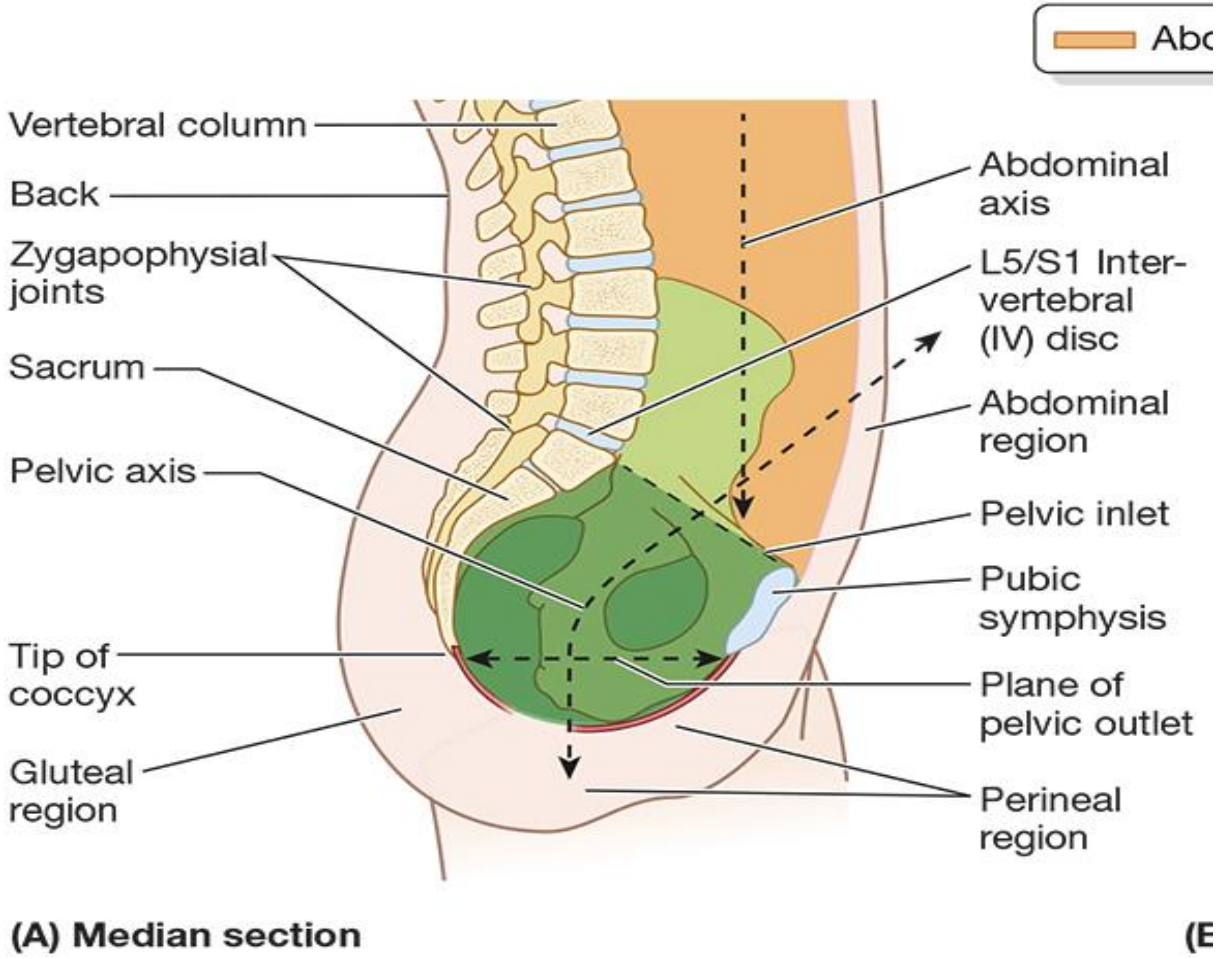
EUR Block

Learning Objectives

- Discuss the skeletal elements defining the pelvic cavity including osseous features, the differences driven by hormonally based development, and their clinical relevance
- Discuss the joints and ligaments of the pelvic girdle and their clinical significance
- Discuss structural and functional anatomy of muscles of the pelvic floor and the issues stemming from dysfunction
- Discuss the fascia defining and organs contained in the perineum

Pelvic Cavity

- Continuous with the abdominal cavity
- Boundary of the Pelvic cavity is Pelvic Inlet and the Pelvic Outlet
  - Defines Greater v. Lesser pelvis
- Lesser is considered the True Pelvis
  - Anterior border is the Symphysis Pubis
  - Posterior Border is the coccyx



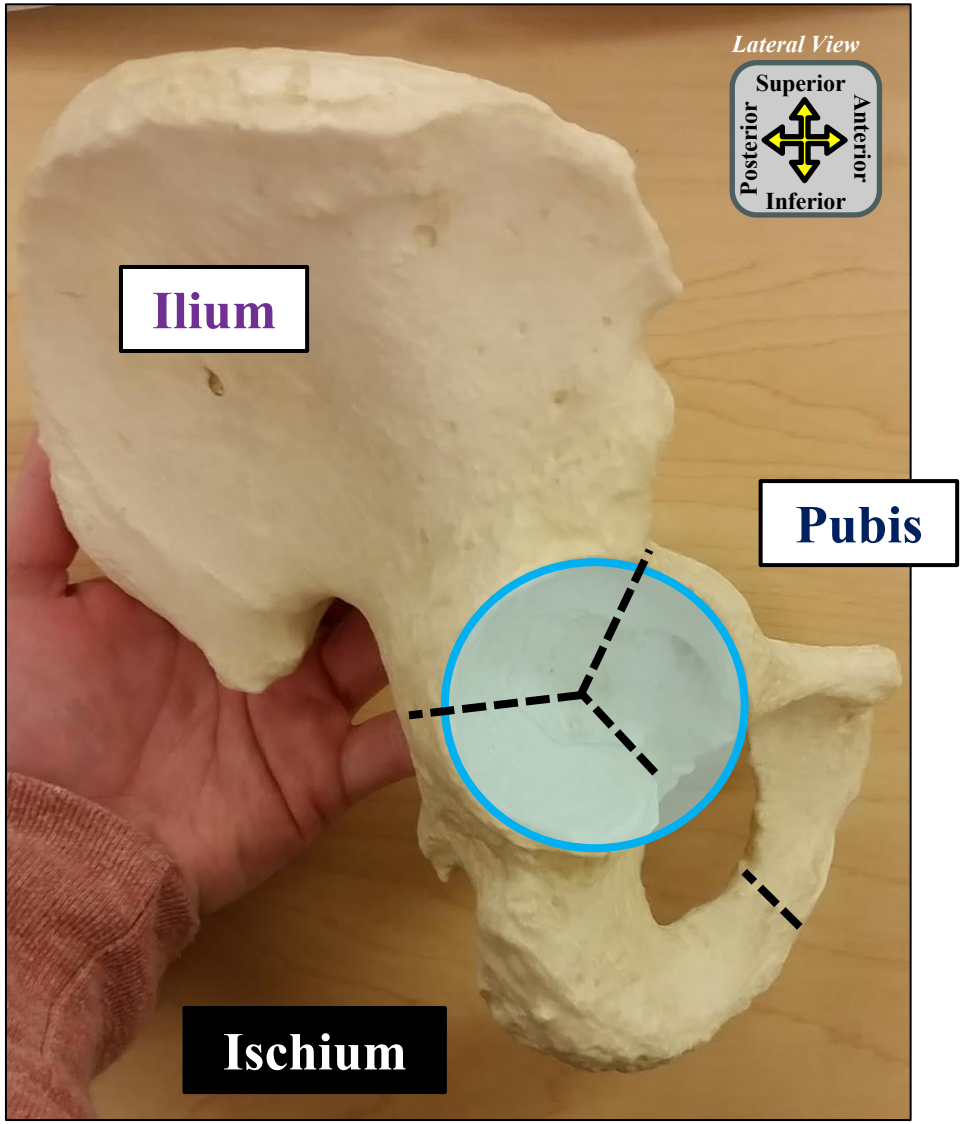
Pelvic Girdle

- Weight bearing/distribution to legs
  - Must withstand compression during walking
- Sacrum
  - 5 fused vertebrae
- Coccyx
  - Variable fused vertebrae
- Hip Bone (2 os coxae)



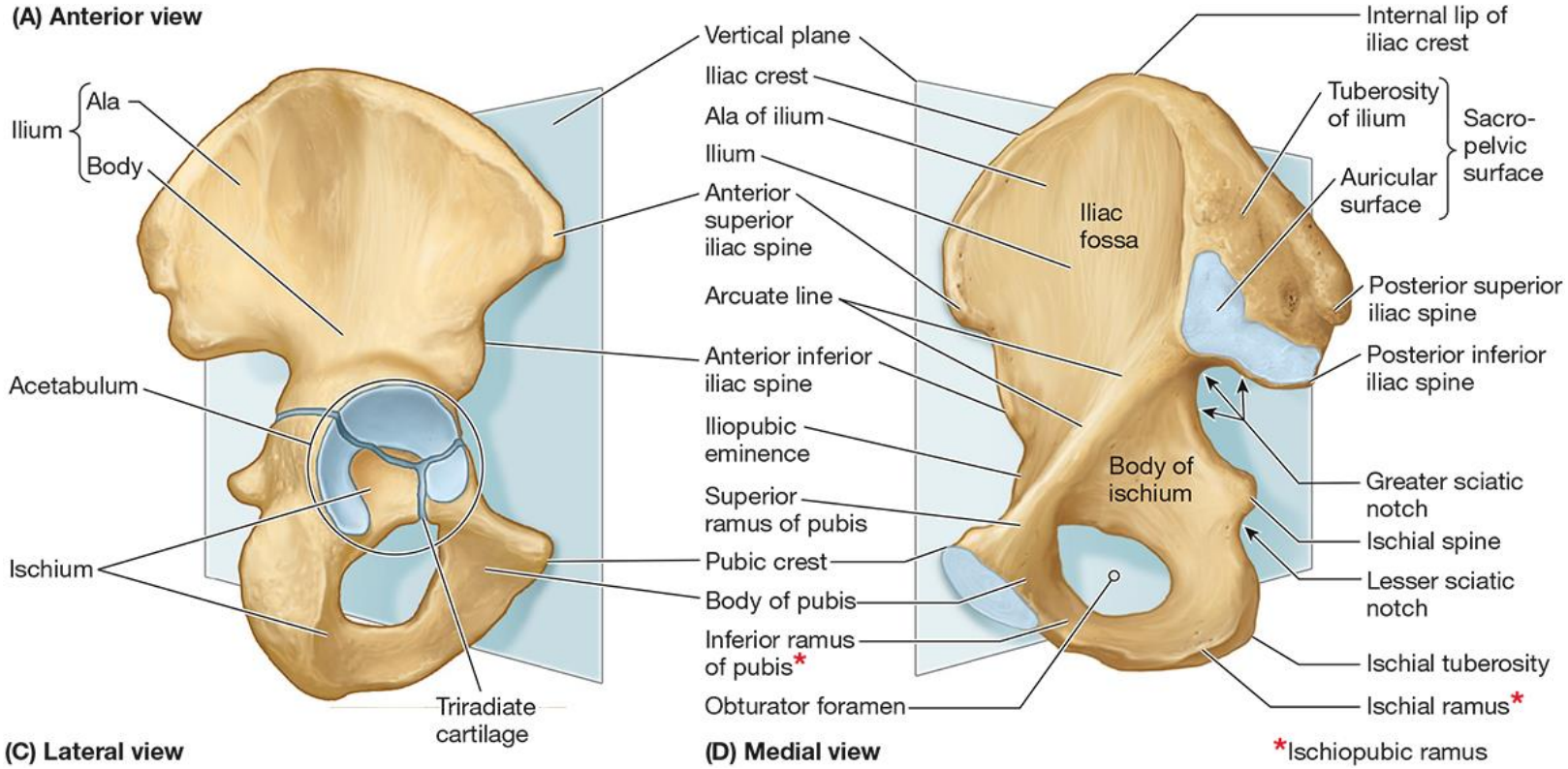
Hip Bone (Os Coxae)

- Three separate bones come together to make the Hip bone
  - Ilium
  - Ischium
  - Pubis
- Fuse to form the acetabulum
  - Depression for the hip joint



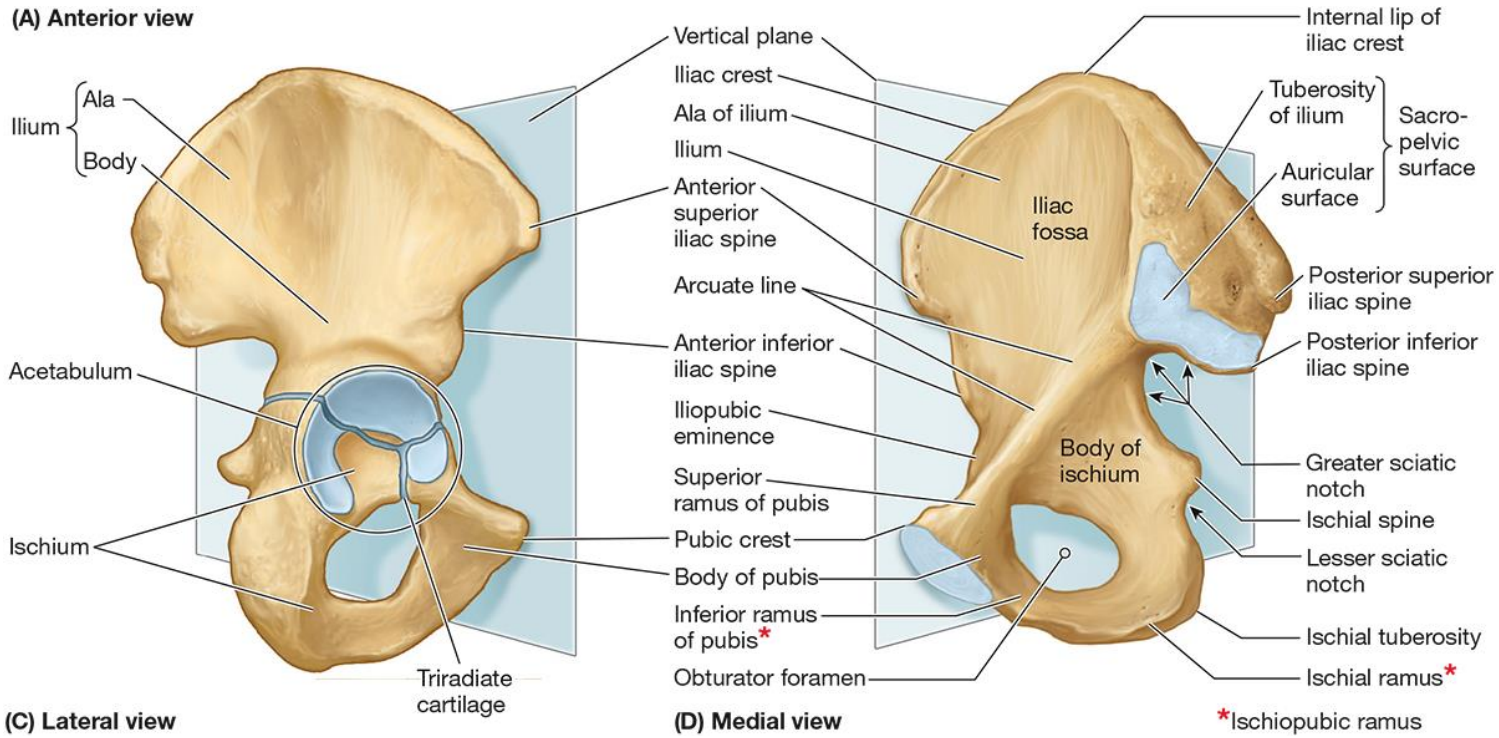
Hip Bone - Ilium

- Broad fan shaped bone
  - Ala is the fan
- Forms the superior region of the Acetabulum
- Iliac crest runs along the Ala
  - from ASIS to PSIS
- Medial surface of Ala is the Iliac Fossa
  - Iliacus muscle attaches
- Auricular surface of sacroiliac joint
- Holds part of the arcuate line

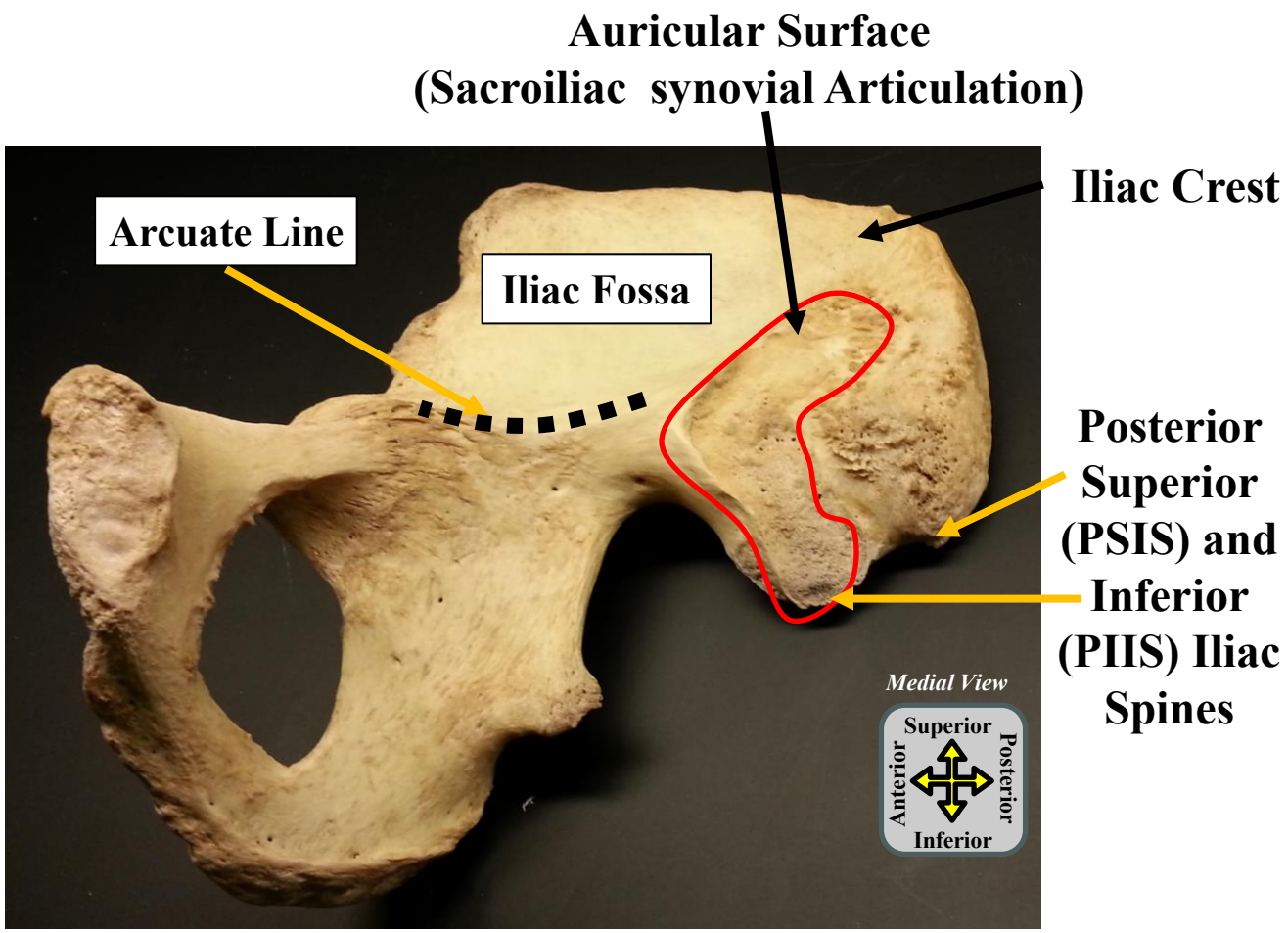


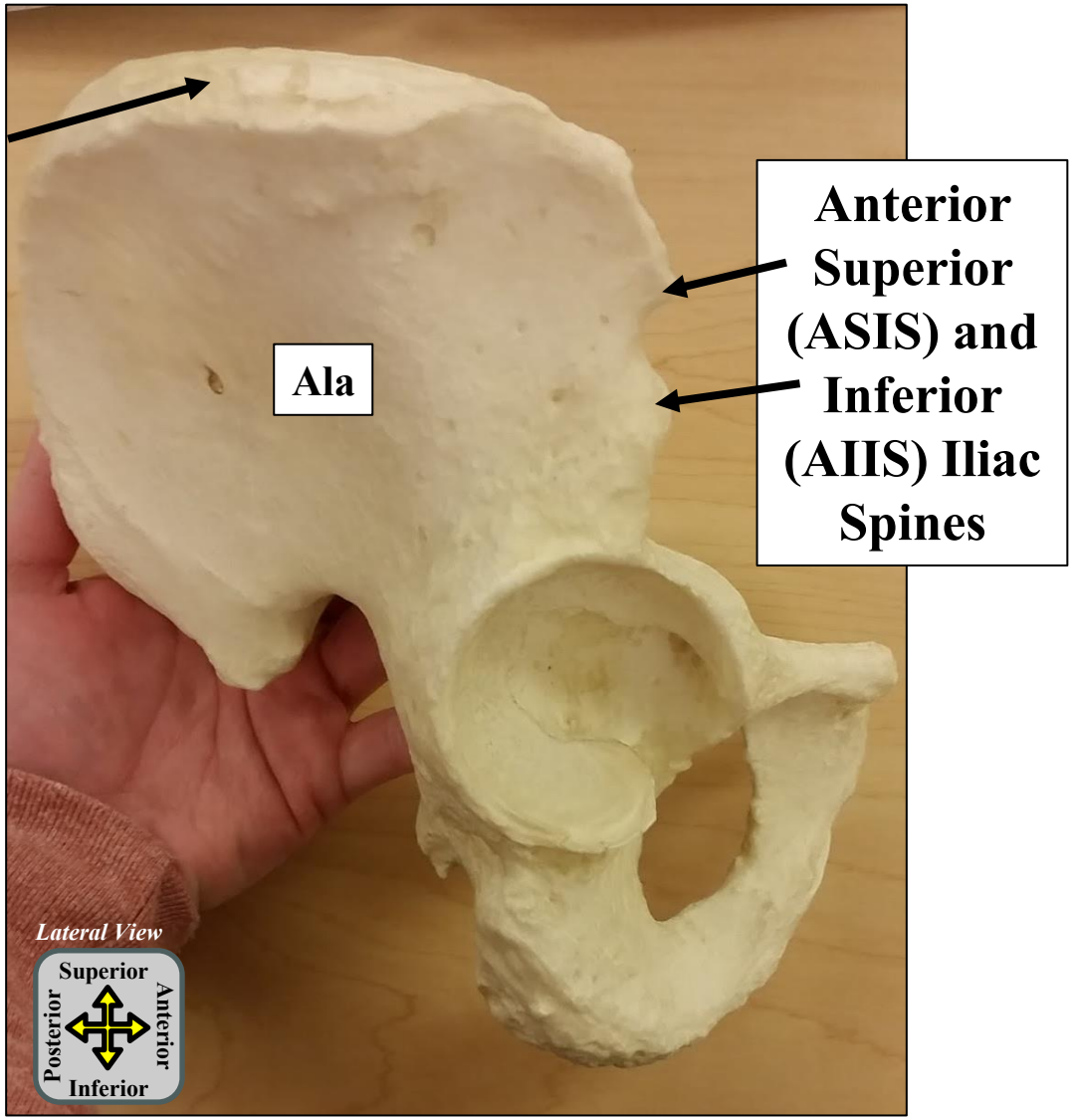
Hip Bone - Ilium

- Auricular surface of sacroiliac joint
  - Mixed joint
- Holds part of the arcuate line
  - Where the ilium meets the body of ischium
- Posterior regions marked by the Greater Sciatic Notch



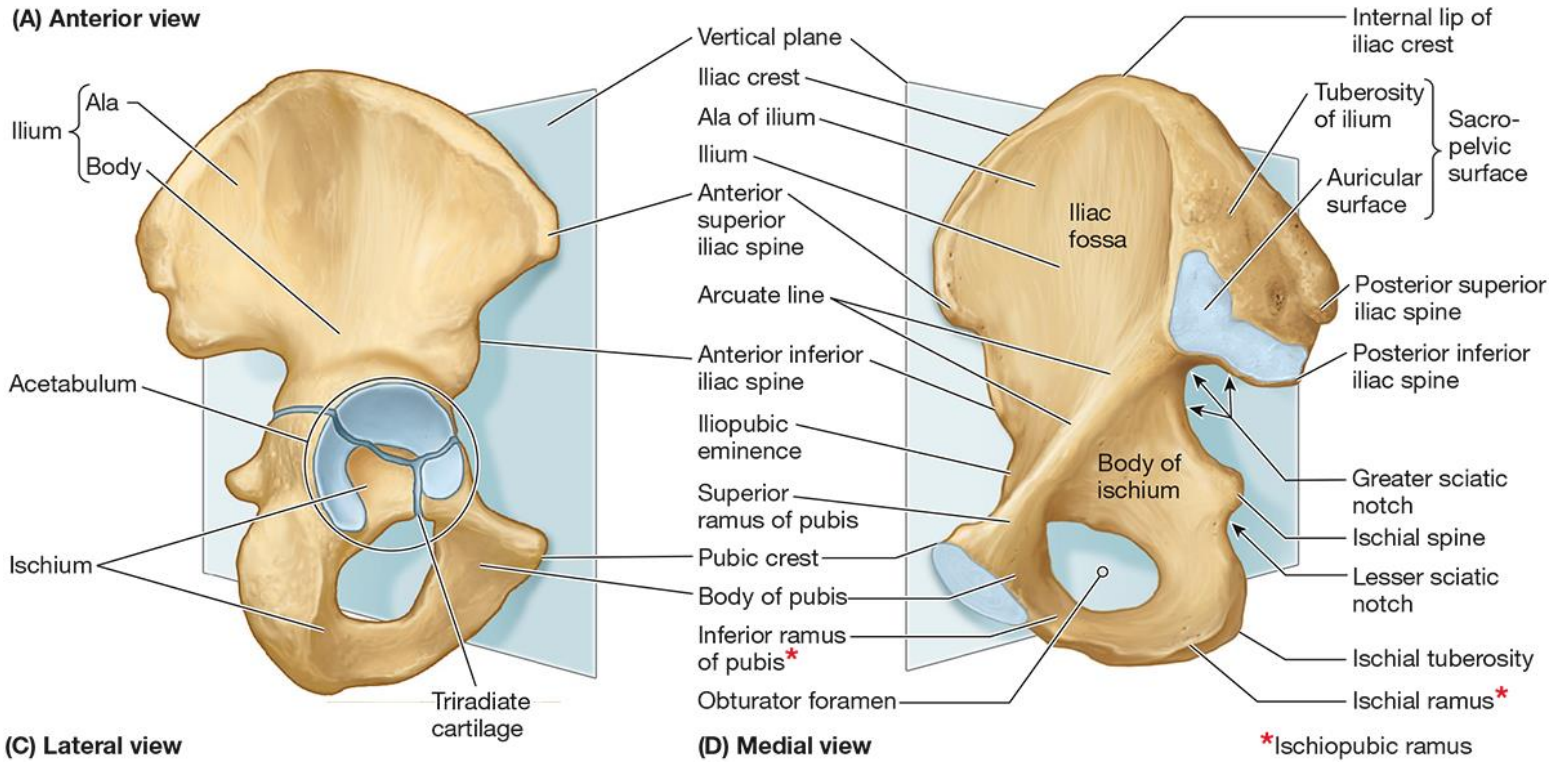
Boney Landmarks of the Ilium



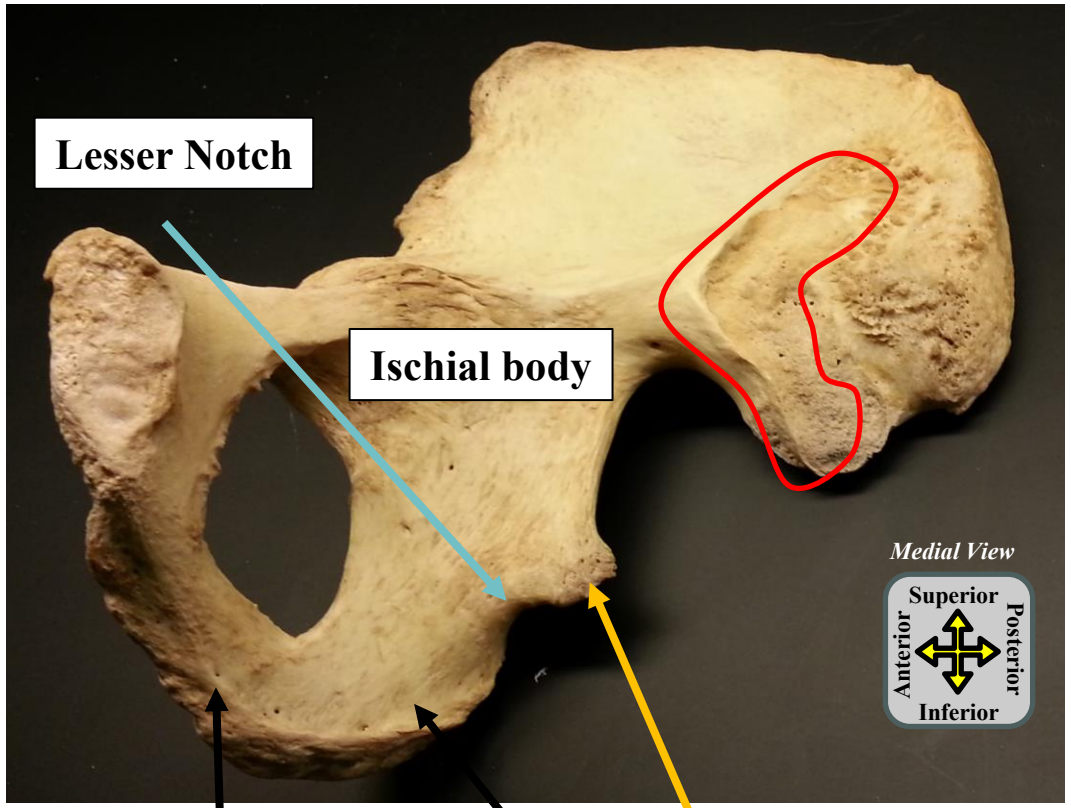


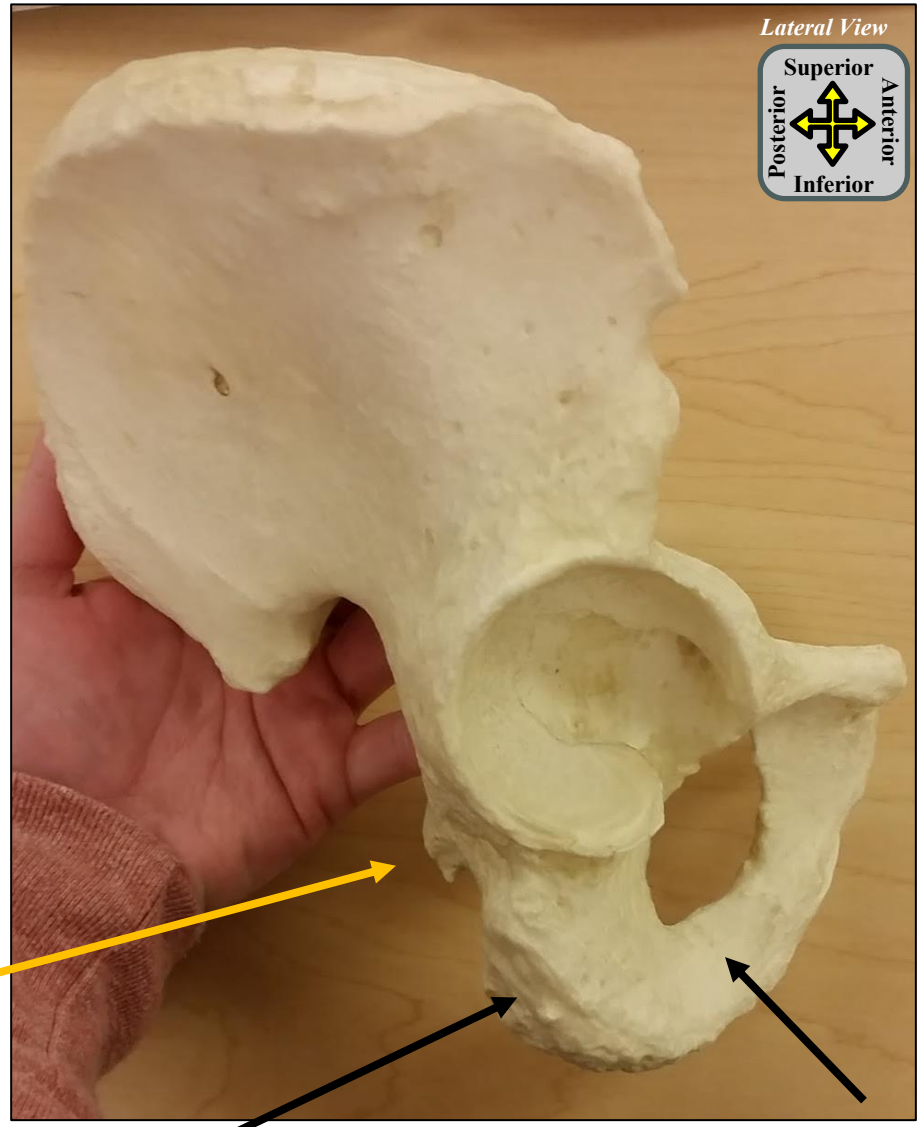
Hip Bone - Ischium

- Body and a ramus
  - Body forms the posterior region of the acetabulum
  - Ramus forms posterior Border of the Obturator Foramen
- Ischial Tuberosity runs along the ramus
- Ischial Spine projects posteriorly off the body
  - Defines the lesser sciatic notch



Boney Landmarks of the Ischium





Ischial Ramus

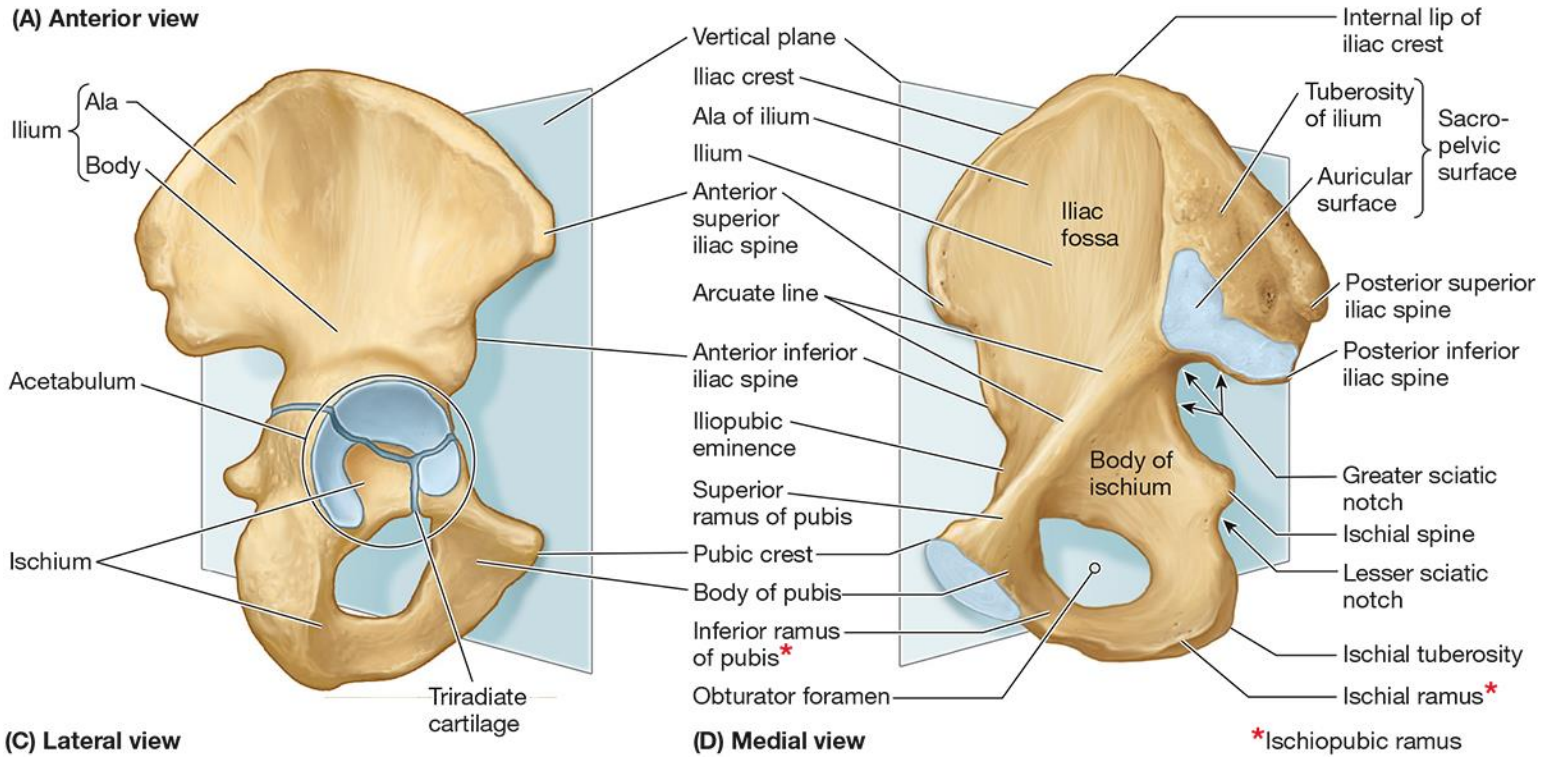
Ischial Tuberosity

Ischial spine

Ischial Ramus

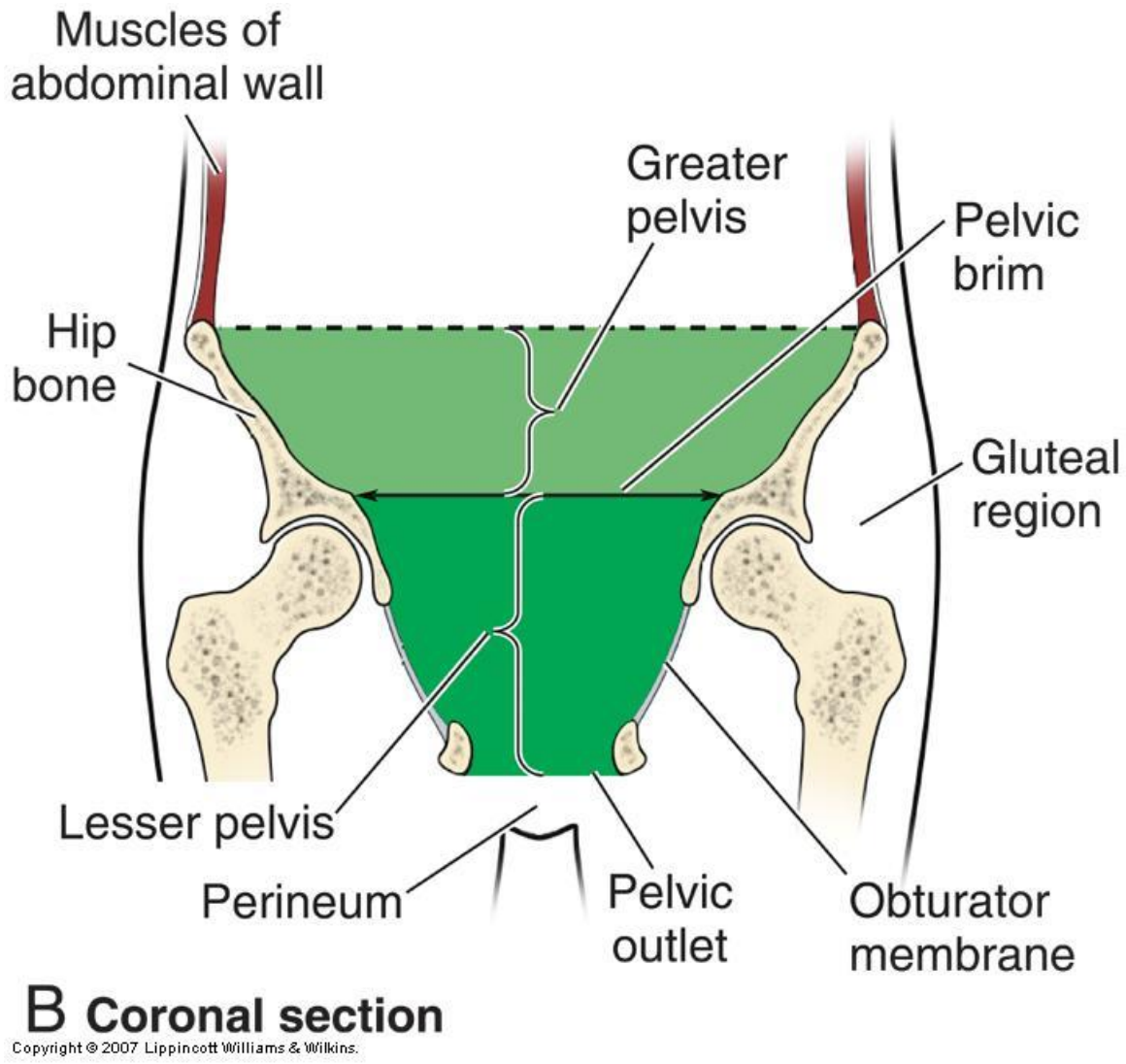
Hip Bone - Pubis

- C-Shaped
- Superior Pubic ramus forms the anterior of the acetabulum
  - Pectineal line part of the arcuate line
- Inferior Pubic Ramus forms the anterior border of the Obturator Foramen
- Body has a thickening called the Pubic Crest
  - Ends with the Pubic tubercle
    - Recall the inguinal ligament
  - Articulates with the other pubis at the pubic symphysis



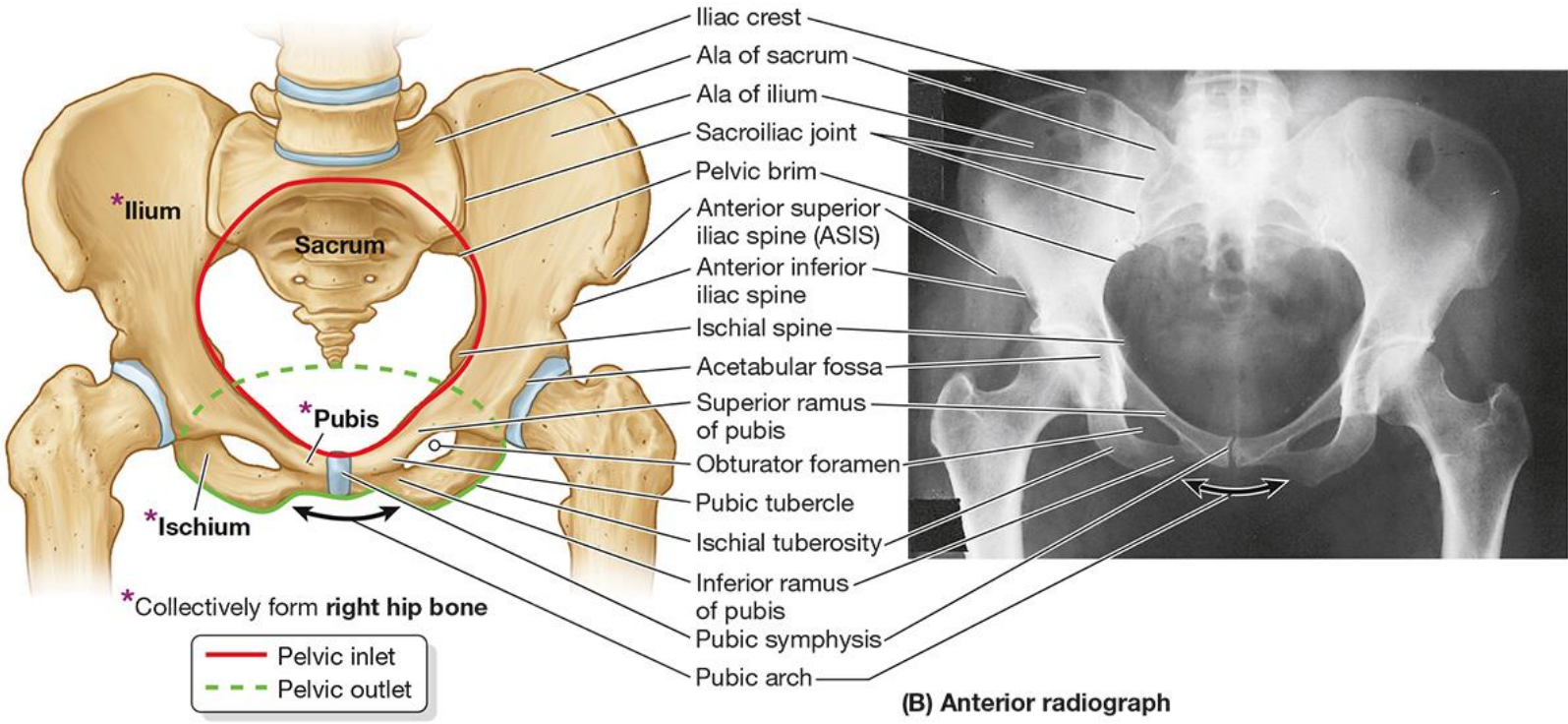
Pelvis

- Greater Pelvis
  - False Pelvis
  - Top of the iliac crests to the Pelvic inlet
- Lesser Pelvis
  - True Pelvis
  - Between the pelvic inlet and outlet



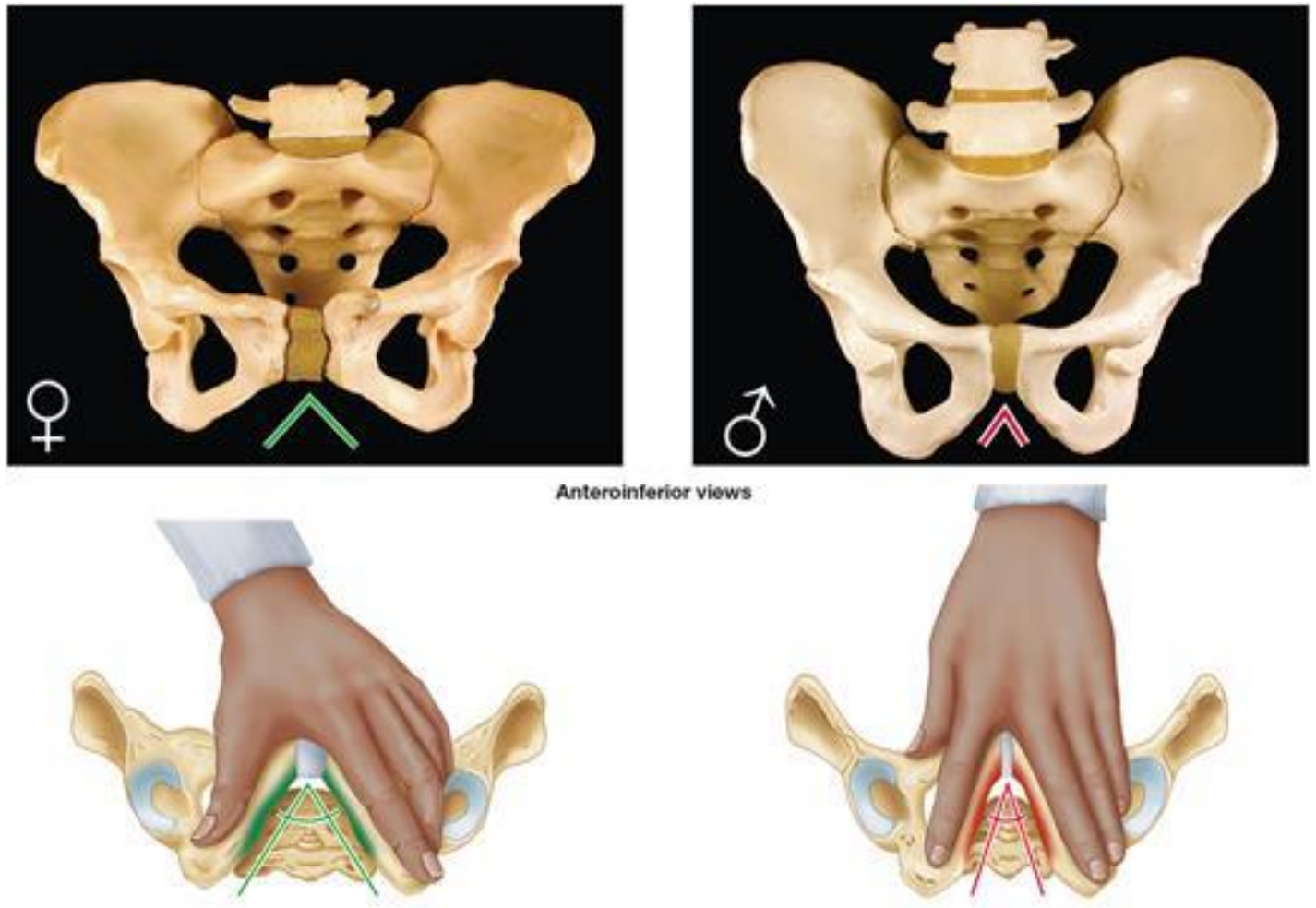
Pelvic Inlet and Outlet

- Inlet Boundary is the Linea terminalis of the Bony Pelvis
  - Superior margin of Pubic symphysis
  - Pubic crest and Pectin Pubis
  - Arcuate line of the Ilium
  - Anterior margin of the Ala of sacrum and Sacral promontory
- Outlet Boundary
  - Inferior Margin of the Symphysis Pubis
  - Inferior rami of the pubis and ischium tuberosities
  - Series of Ligaments posteriorly/laterally
  - Tip of Coccyx



Pubic Arch/Angle

- Arch is Formed from the Ischiopubic Rami
  - Anterior border of the pelvic cavity
- Rami articulate at the Symphysis Pubis
- Inferior borders define the Pubic Angle
  - >80º is estrogen driven
  - < 70º is testosterone driven



Additional Estrogen-driven Differences

Shallower greater pelvis

Thinner ilium

Rounder pelvic inlet

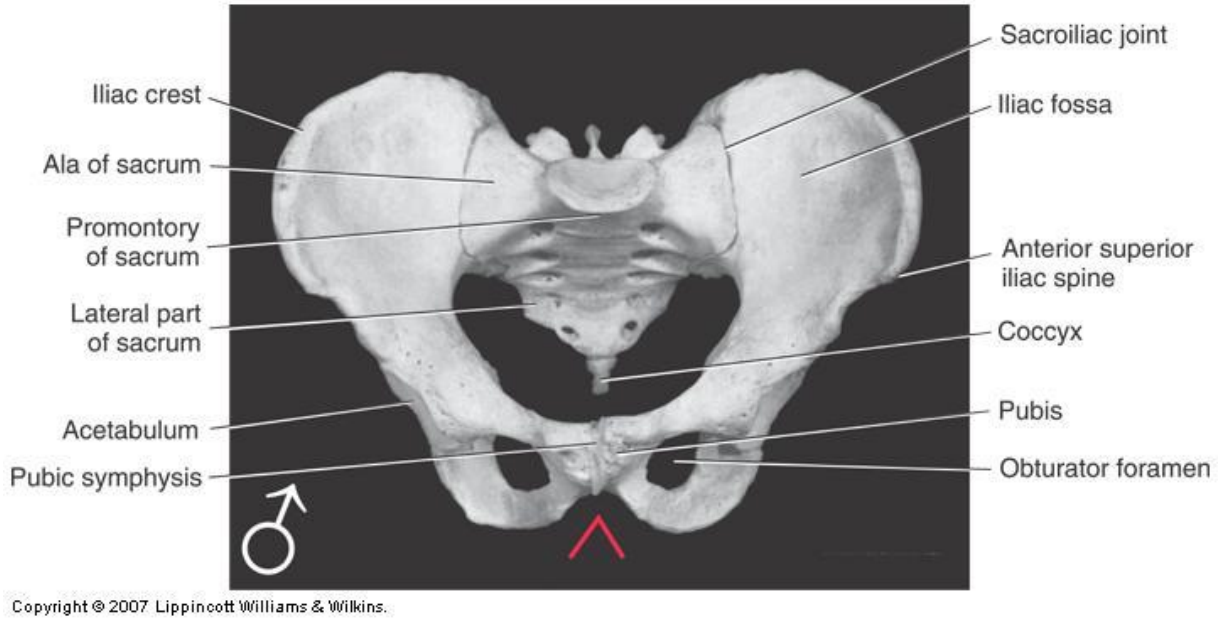
Less curvature to the sacrum

Wider pelvic outlet

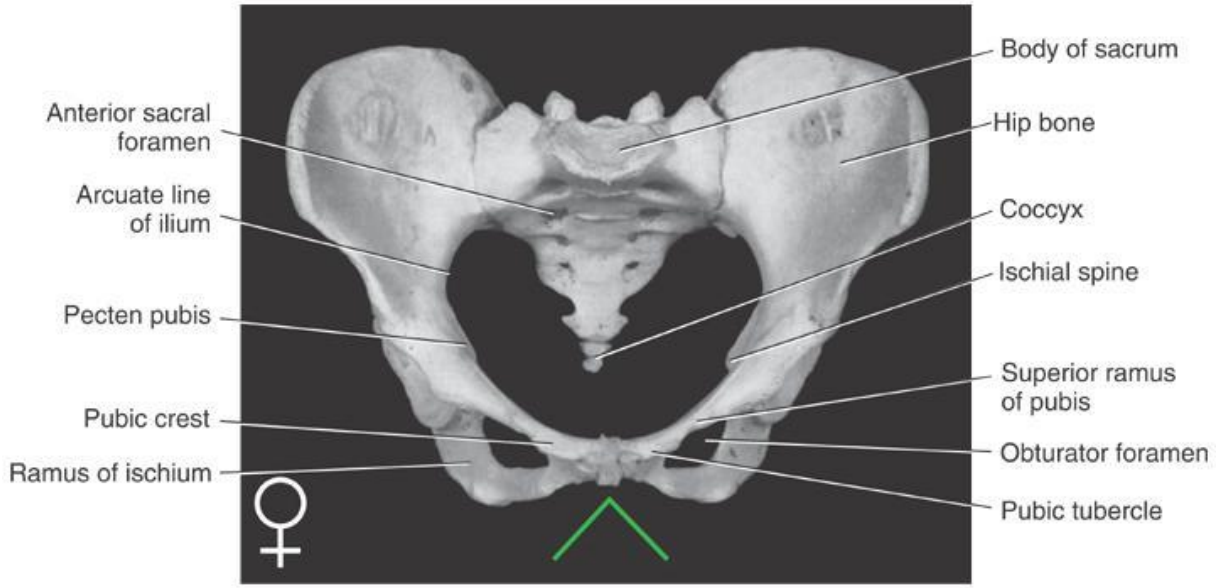
Oval obturator foramen

Small acetabulum

Greater sciatic notch is larger



Copyright © 2007 Lippincott Williams & Wilkins.



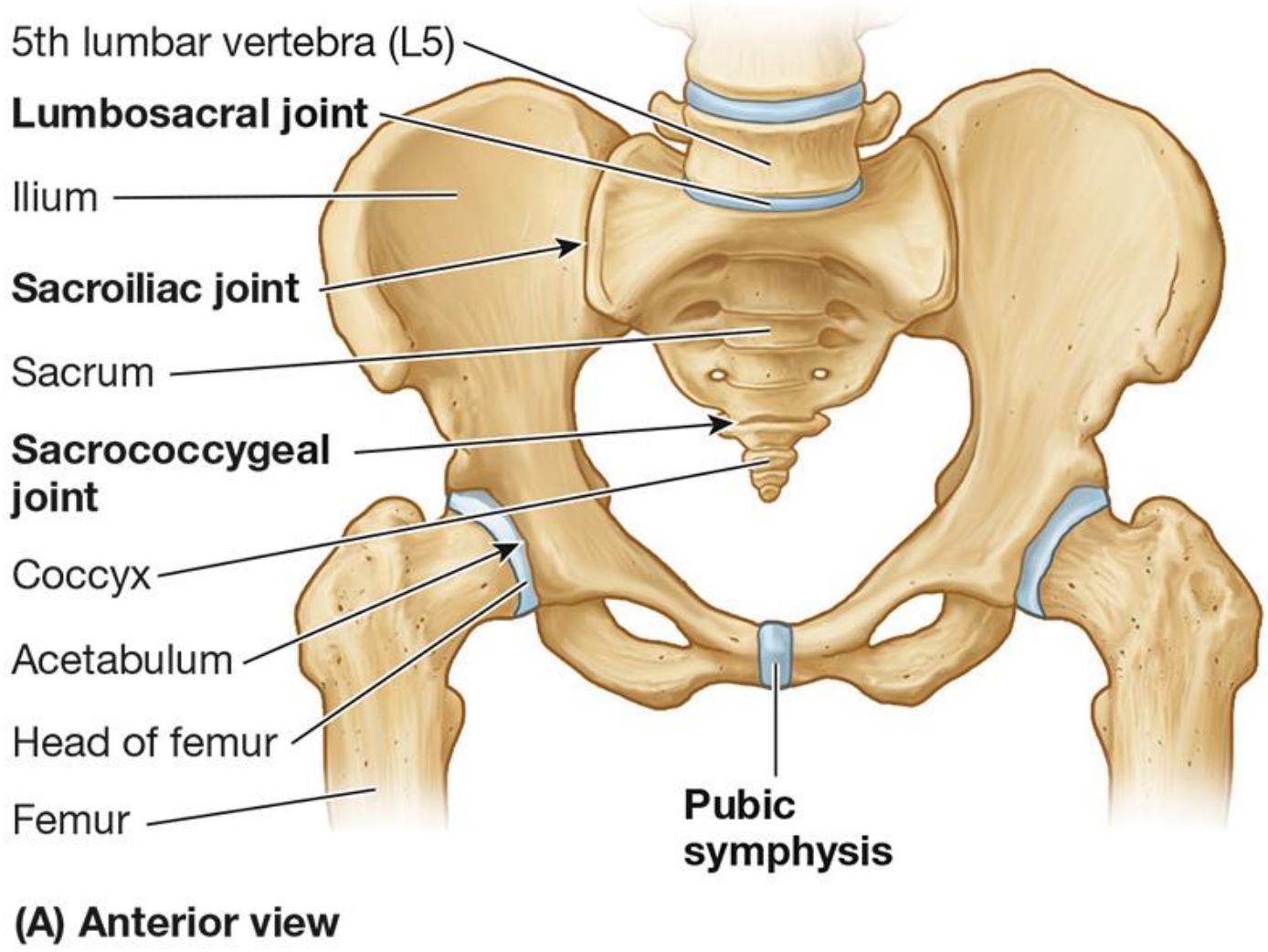
Copyright © 2007 Lippincott Williams & Wilkins.

Learning Objectives

- Discuss the skeletal elements defining the pelvic cavity including osseous features, the differences driven by hormonally based development, and their clinical relevance
- Discuss the joints and ligaments of the pelvic girdle and their clinical significance
- Discuss structural and functional anatomy of muscles of the pelvic floor and the issues stemming from dysfunction
- Discuss the fascia defining and organs contained in the perineum

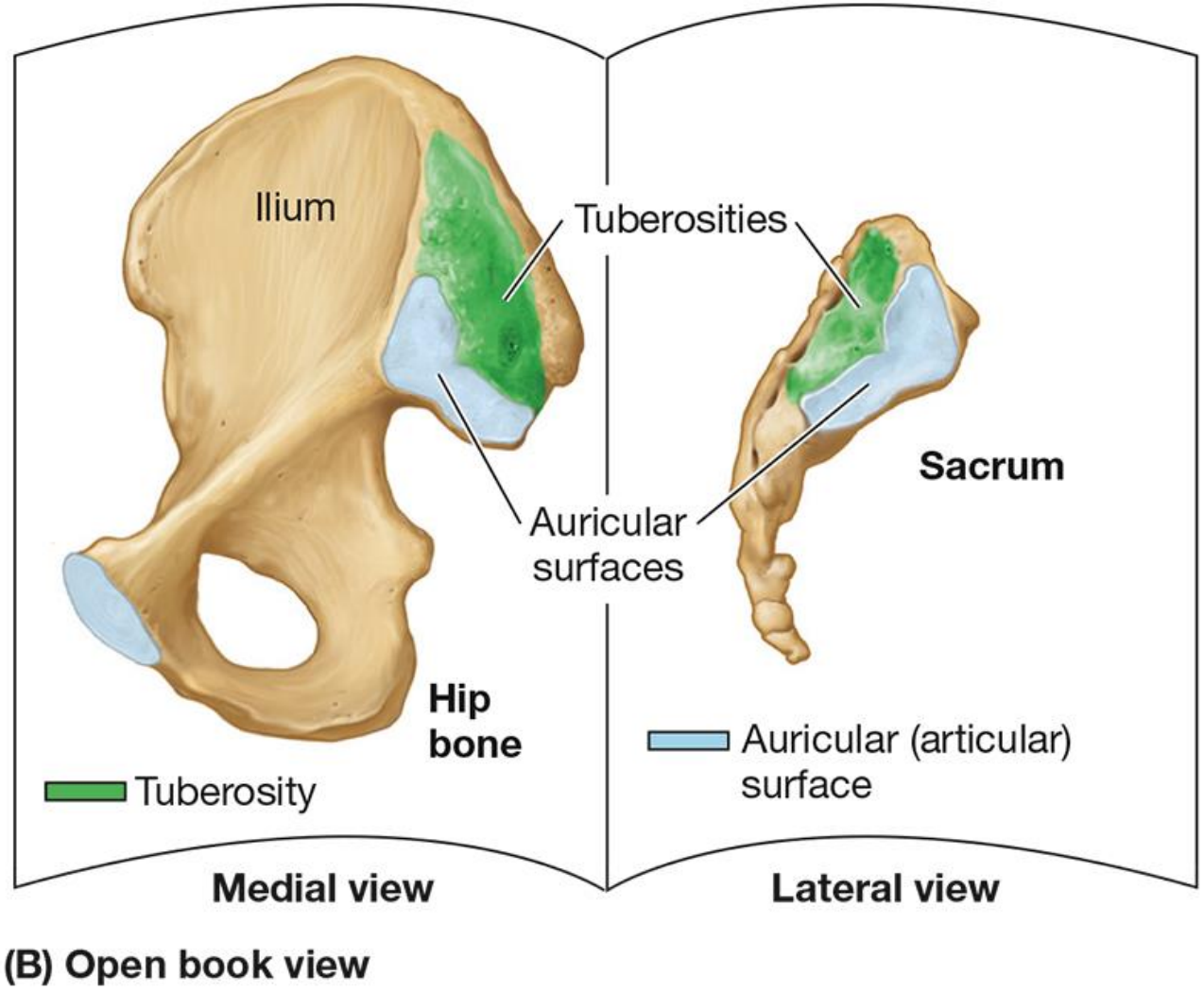
Joints and Ligaments of the Pelvic Girdle

- Four joints holding the pelvic girdle together
  - Lumbosacral joint
  - Sacroliliac joint
  - Sacrococcygeal joint
  - Pubic Symphysis



Sacroiliac Joint

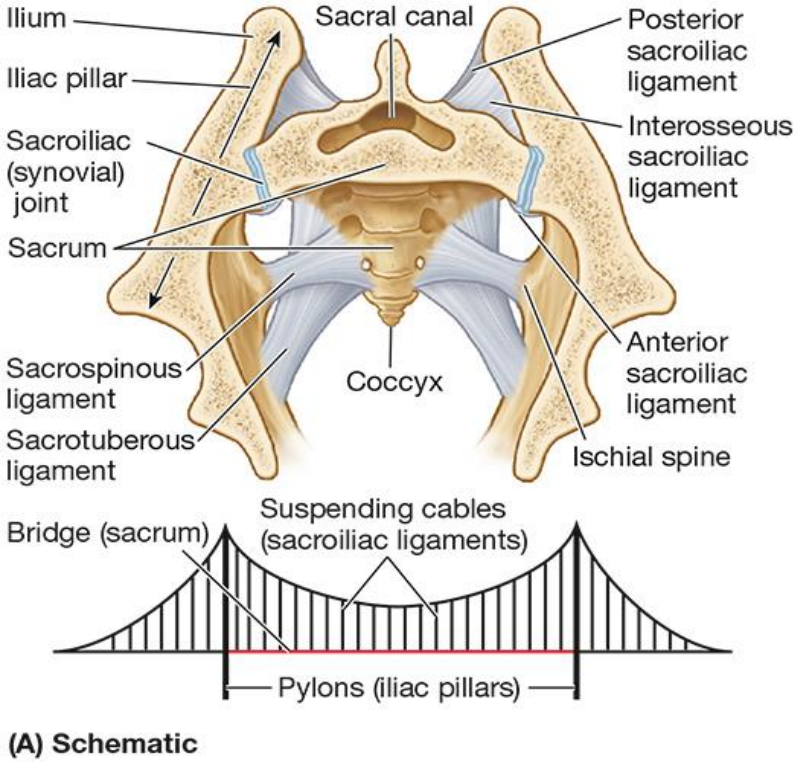
- Sacroiliac Joint – strong broad for weight bearing
- Compound Joint
  - Anterior regions are Synovial at auricular surfaces
  - Posterior region are Syndesmosis at tuberosities
- Synovial portions allows only limited movement
  - Gliding and sliding

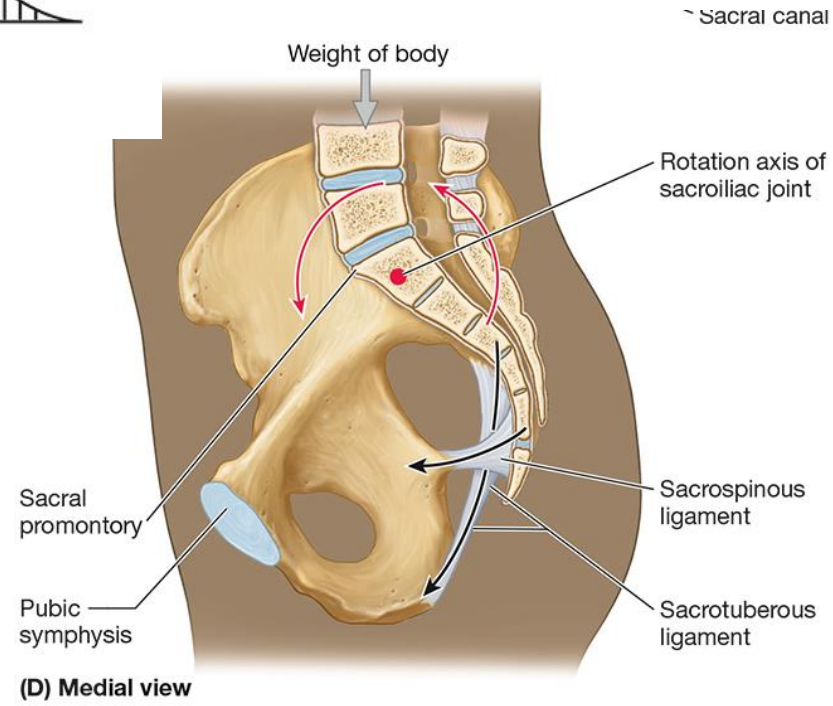


(B) Open book view

Sacroiliac ligaments

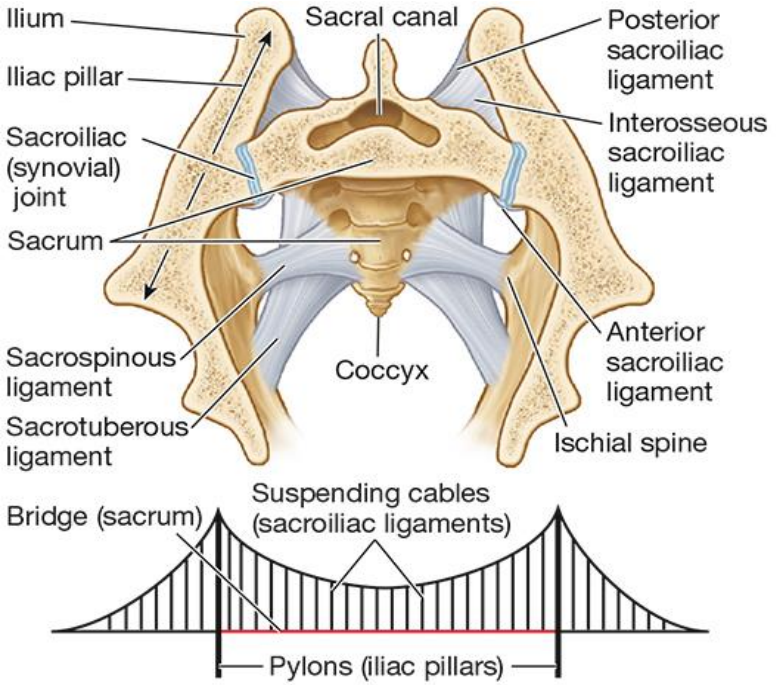
- Sacrum is suspended between hip bones by ligaments
  - Posterior and anterior sacroiliac and Interosseus sacroiliac
  - Major role in weight transfer



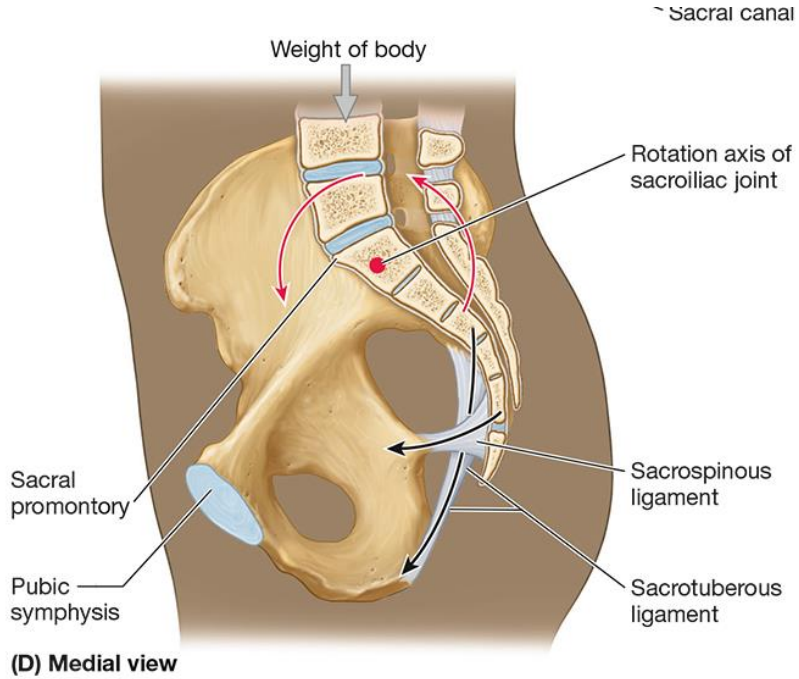


Ligaments Between Sacrum and Ischium

- Inferior end of sacrum supported by pairs of ligaments
- Sacrotuberous and Sacrospinous ligaments oppose movement (pivot) of pelvic girdle
  - Especially with sudden weight increases



(A) Schematic



(D) Medial view

Pelvic Foramen

Greater Sciatic Foramen

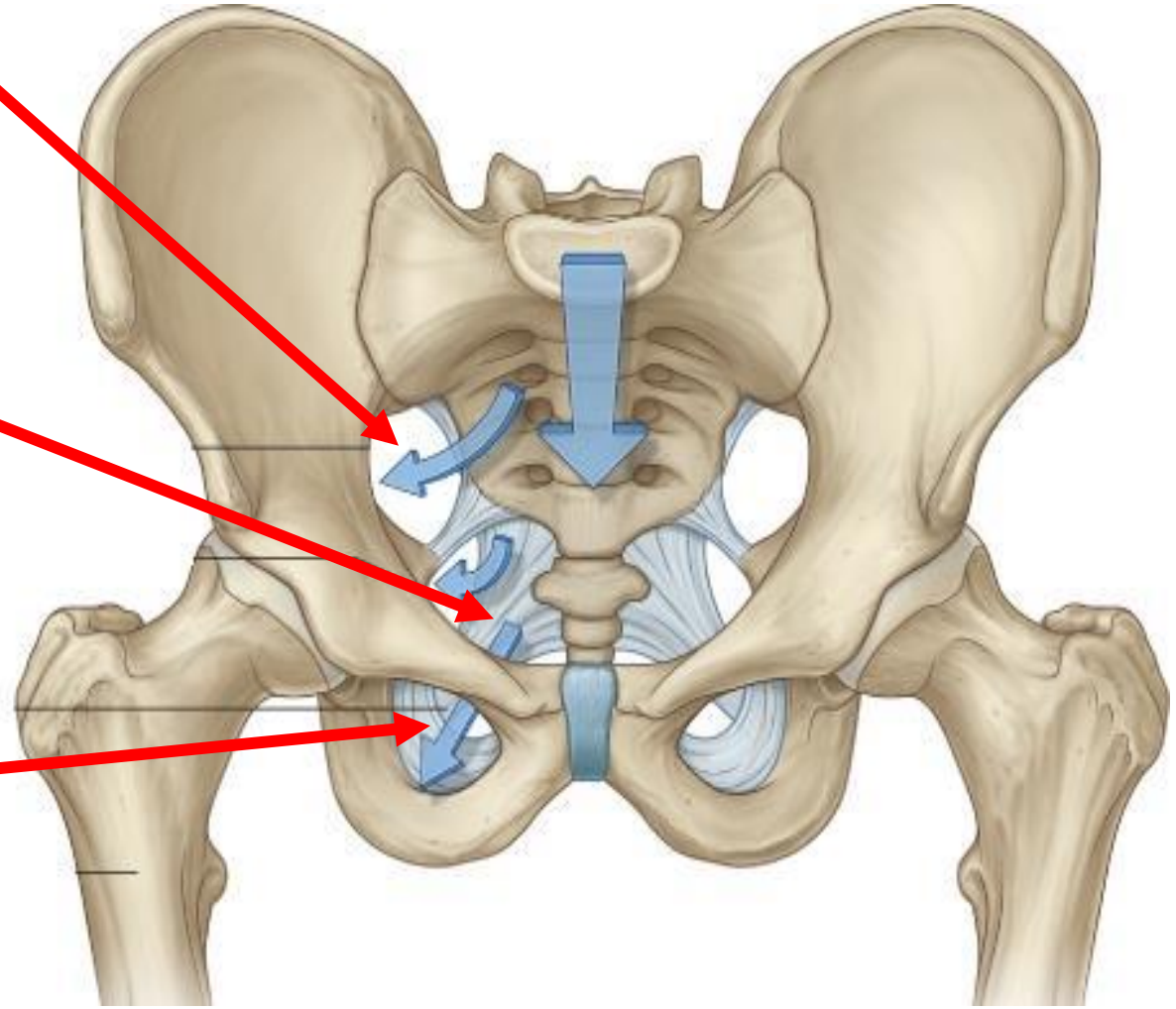
Allows structures to pass between the pelvic cavity and the gluteal region

Lesser Sciatic Foramen

Allows structures to pass between the pelvic cavity and the perineum

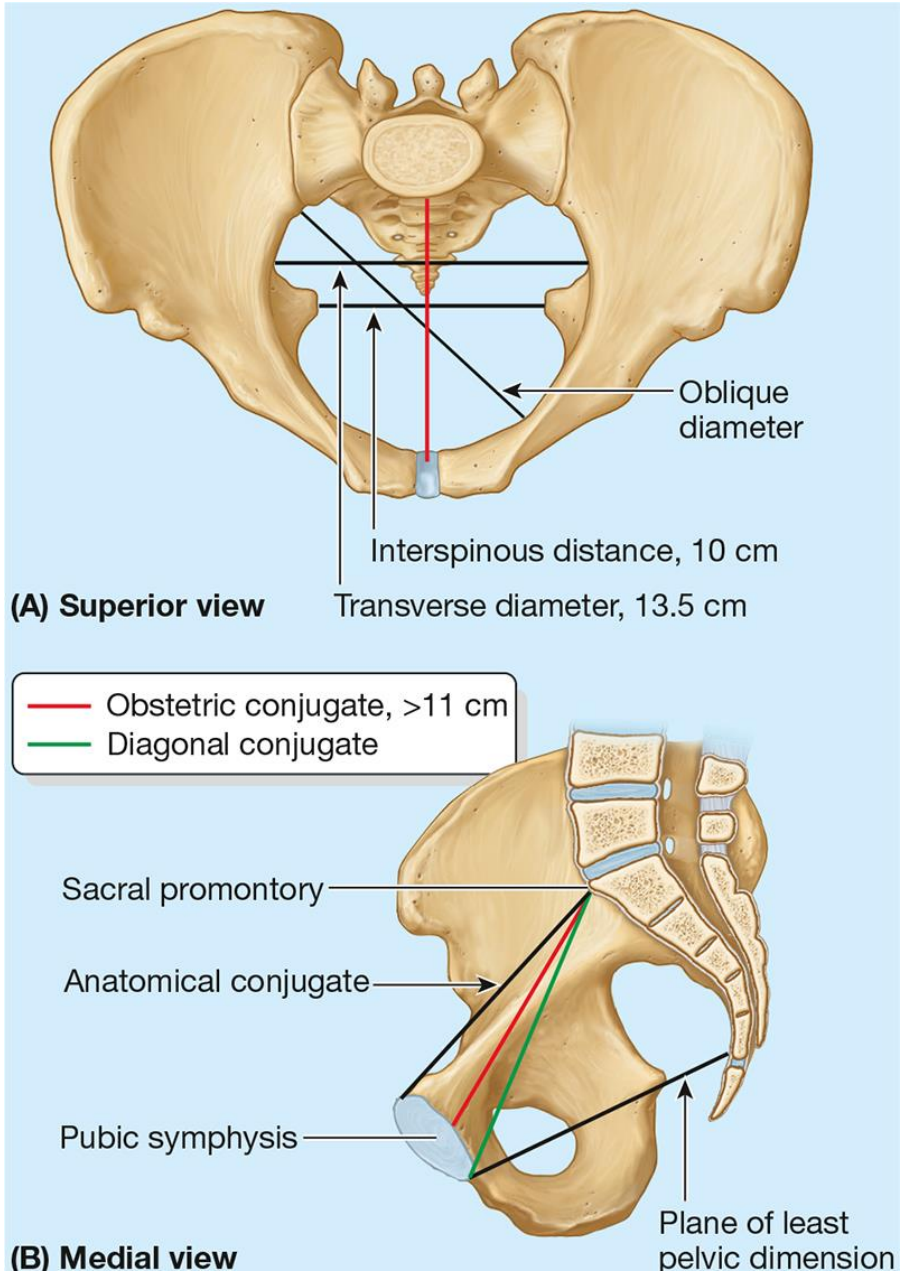
Obturator Foramen

Allows structures to pass between the pelvic cavity and thigh

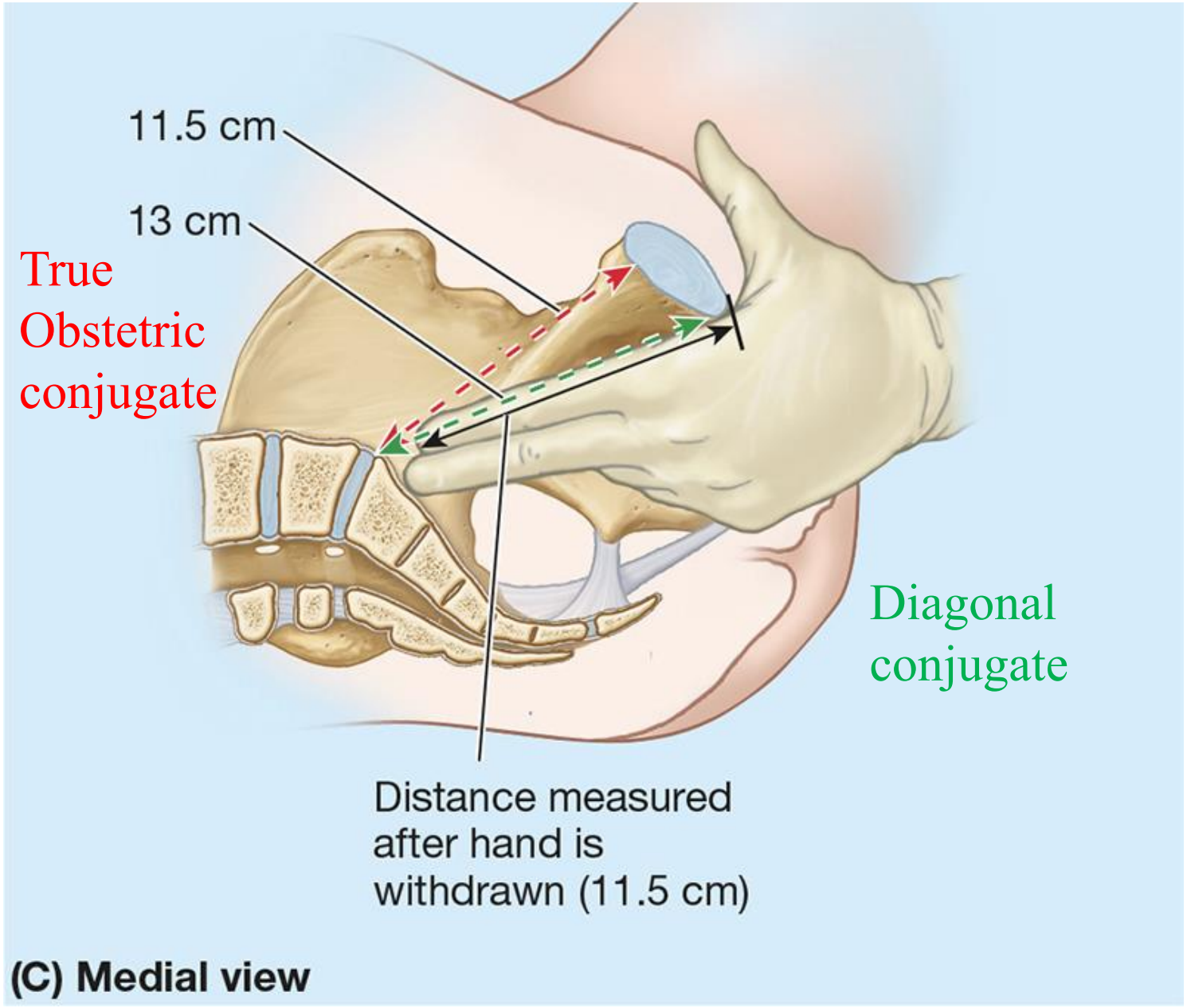


Pelvic Diameters

- **Diameters** are noted during pelvic examination
- **Size of lesser pelvis** is important for vaginal delivery
  - Considered adequate if three fingers can be inserted side by side in the vagina (between the ischial tuberosities)
- **True (obstetric) conjugate diameter**
  - The narrowest **fixed** distance the baby's head must pass through
  - Does not change with pregnancy hormones
  - Sacral promontory to posterosuperior margin of pubic symphysis
- **Diagonal Conjugate**
  - What is measured during a pelvic exam
  - Sacral promontory to inferior margin of pubic symphysis
    - ***Should be ≥ 11.0 cm***



Estimating Obstetric Conjugate



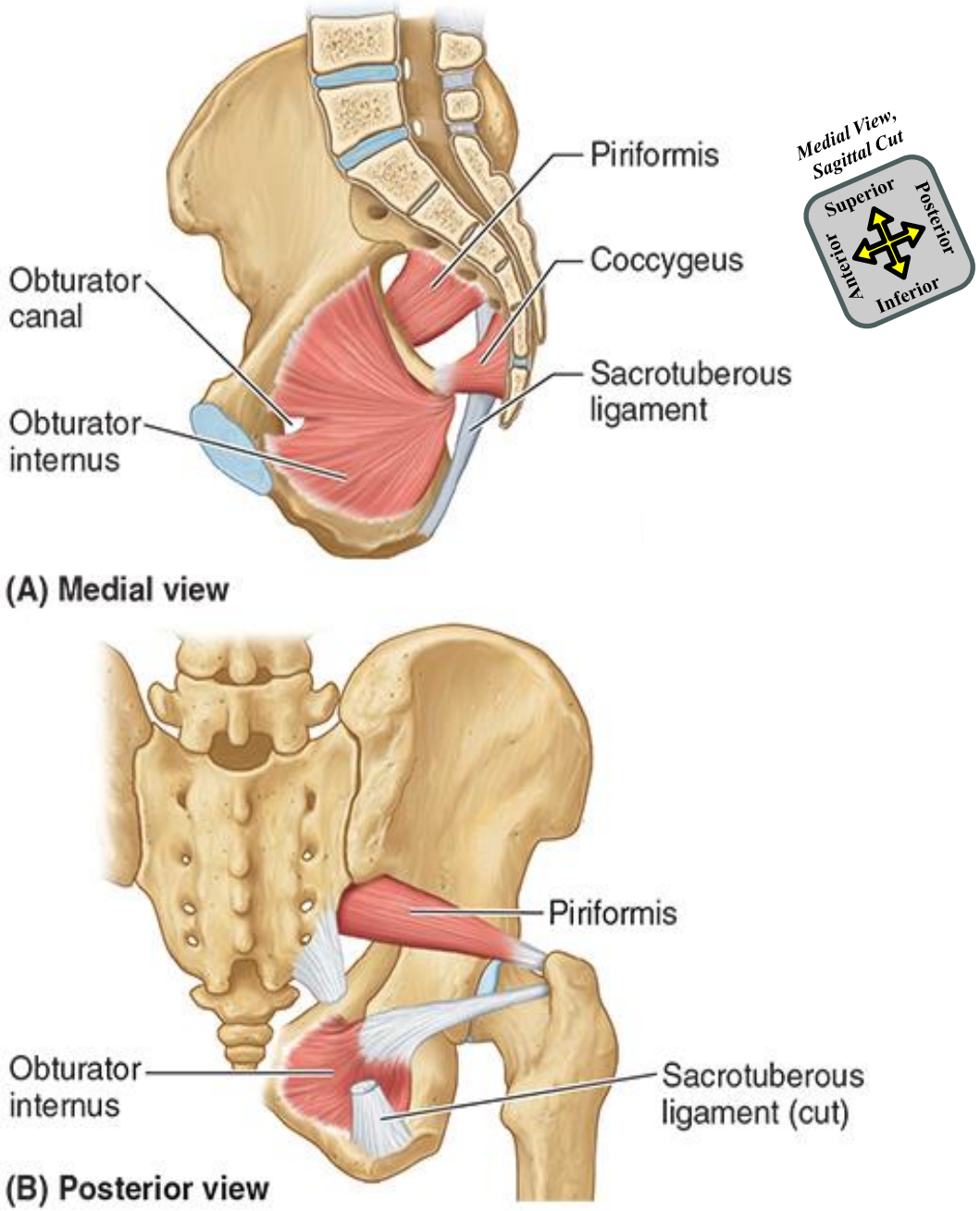
Learning Objectives

- Discuss the skeletal elements defining the pelvic cavity including osseous features, the differences driven by hormonally based development, and their clinical relevance
- Discuss the joints and ligaments of the pelvic girdle and their clinical significance
- Discuss structural and functional anatomy of muscles of the pelvic floor and the issues stemming from dysfunction
- Discuss the fascia defining and organs contained in the perineum

Obturator Internus Muscle

Obturator Internus

- Attachments within the pelvic cavity to the ilium, ischium, and obturator membrane
- Exits the pelvis through the **lesser sciatic foramen** to attach to the greater trochanter of the femur (posteriorly)
- L5-S2
- Laterally rotates thigh

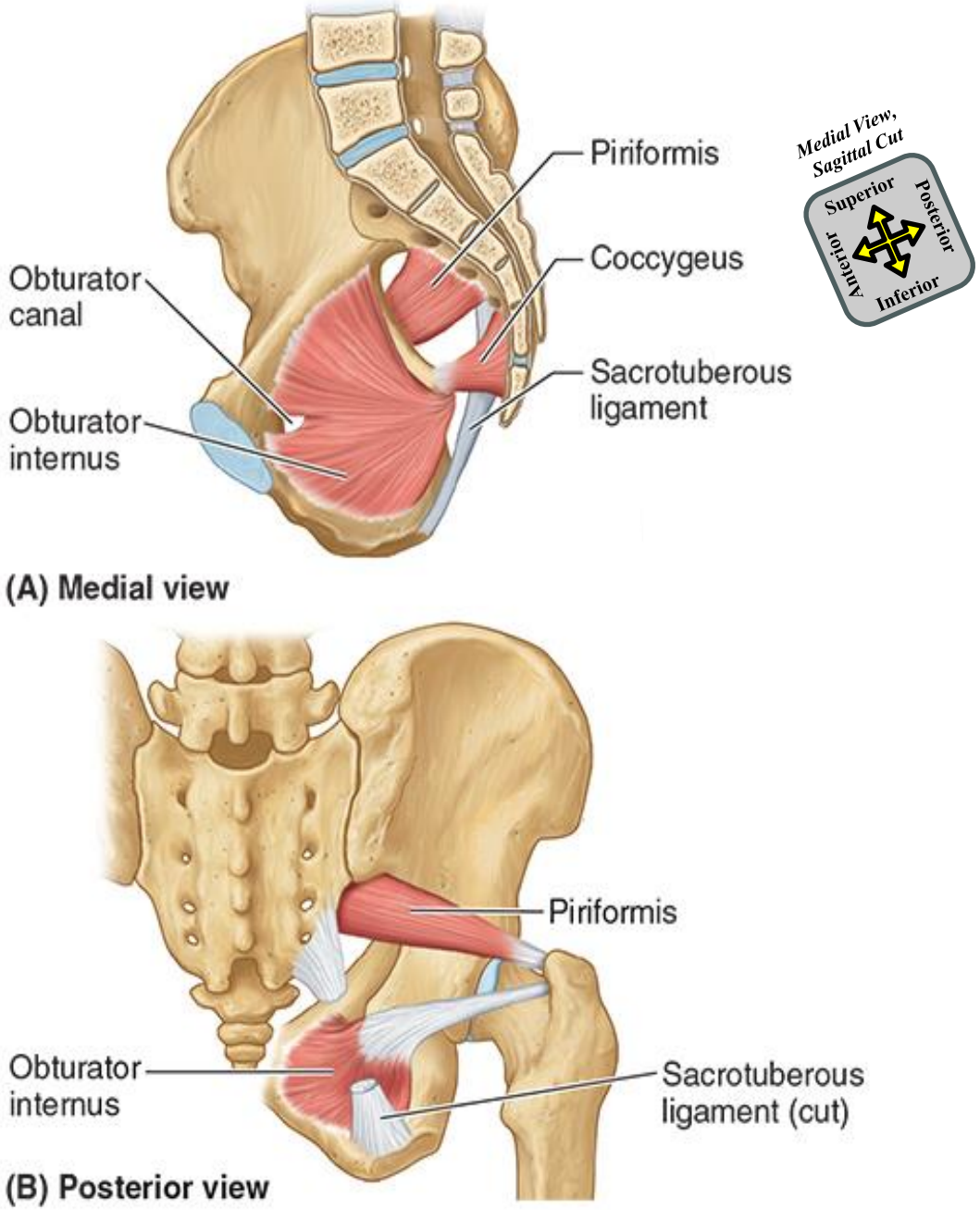


(Moore's, 9e., fig. 6.10)

Piriformis Muscle

Piriformis

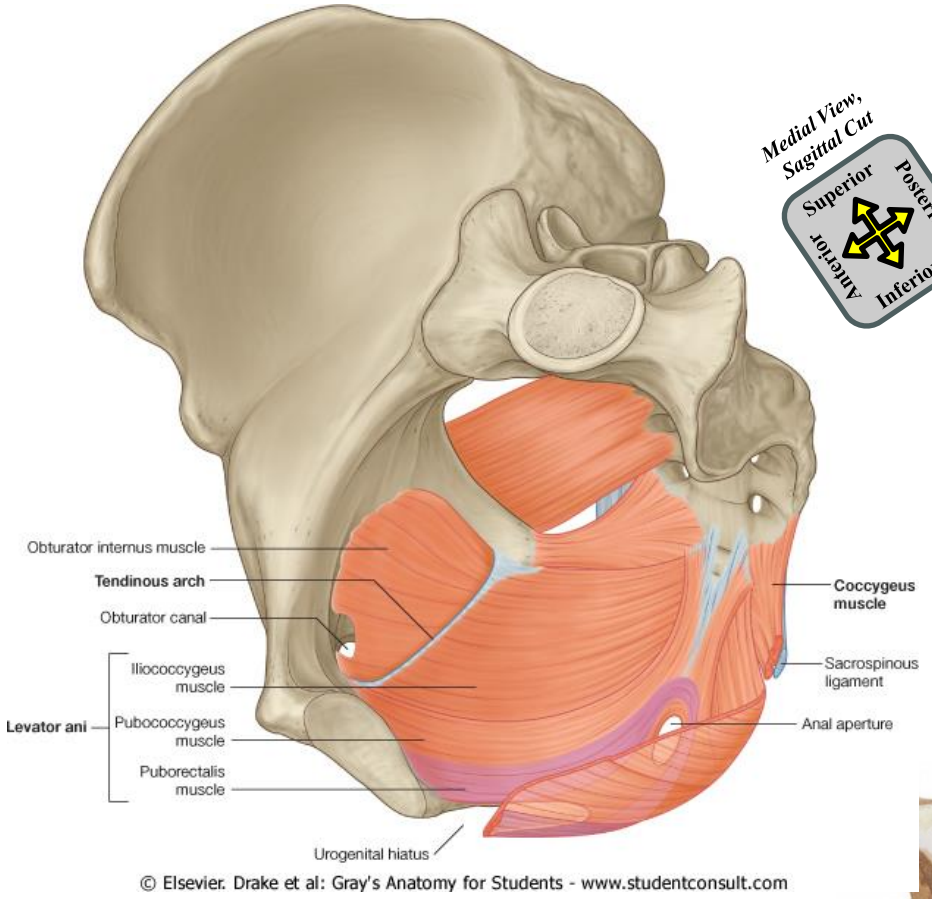
- Attachments within the pelvic cavity to the anterior sacrum
- Exits the pelvis through the **greater sciatic foramen** to attach to the greater trochanter of the femur (posteriorly)
- S1 and S2
- Lateral rotation, abduction, stabilization of femur in acetabulum

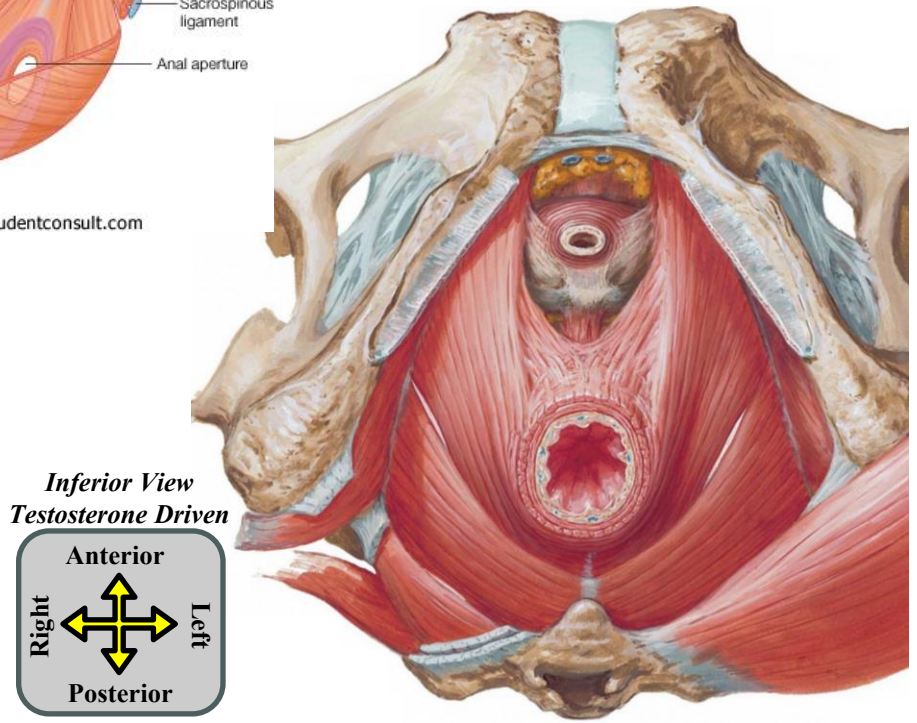


(Moore's, 9e., fig. 6.10)

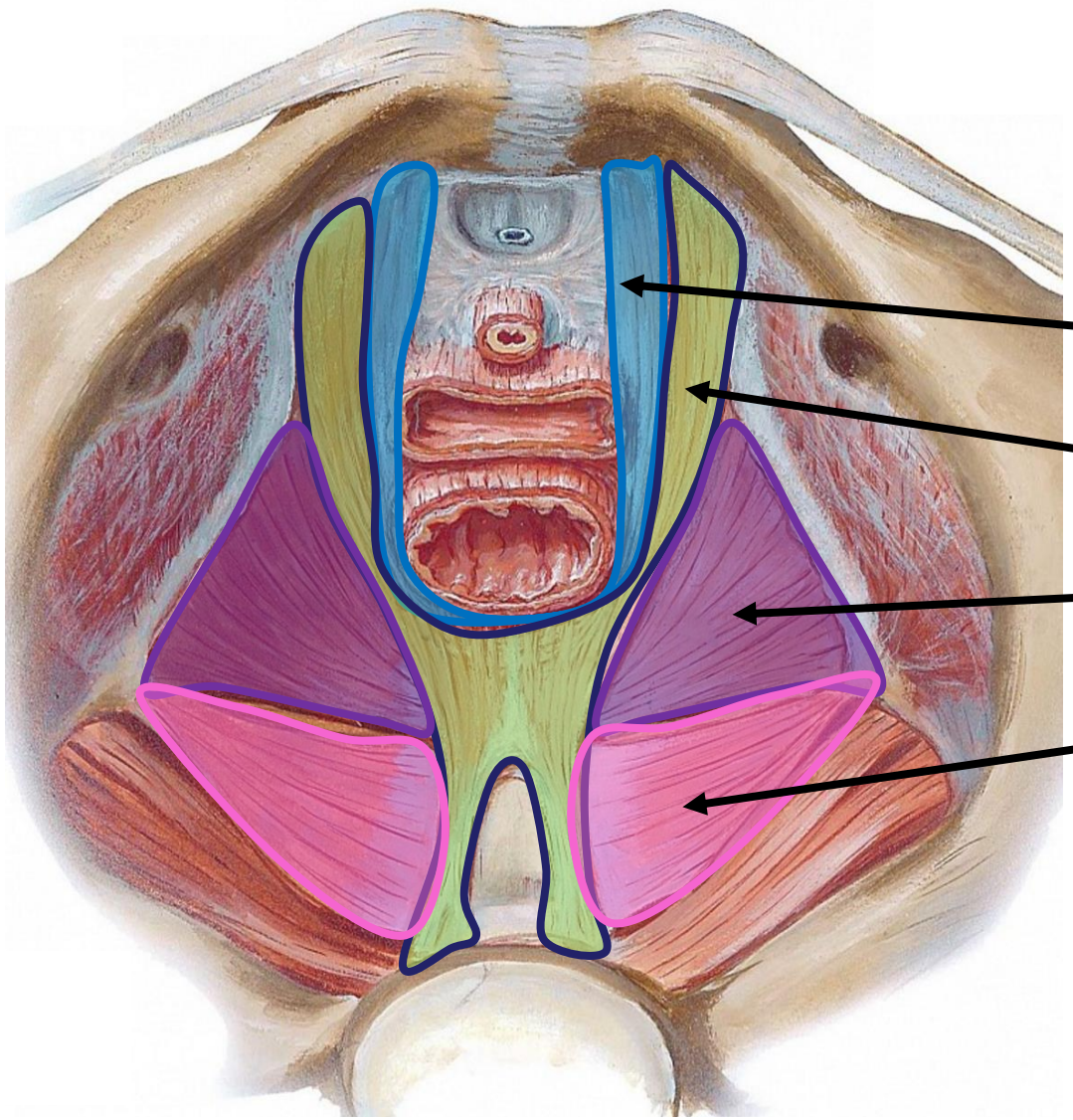
Pelvic Diaphragm

- Coccygeus muscle
  - Ischiococcygeus
  - Ischial spine to coccyx
- Levator ani muscle group
  - Iliococcygeus
  - Puborectalis
  - Pubococcygeus
- Origin of levator ani is the Tendonous Arch and Pubic Body
- Insertion in Perineal body, Coccyx, and Anococcygeal ligament





Pelvic Diaphragm



Pelvic Diaphragm

Levator Ani :

- **Puborectalis ***
- **Pubococcygeus ***
- **Iliococcygeus ***

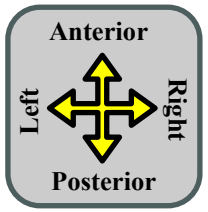
Coccygeus

Function

- Separates pelvic cavity from perineum
- Supports abdominal and pelvic viscera
- Actively contracts during coughing, sneezing, lifting
- Maintains urinary and fecal continence

* Boundaries not easily distinguished

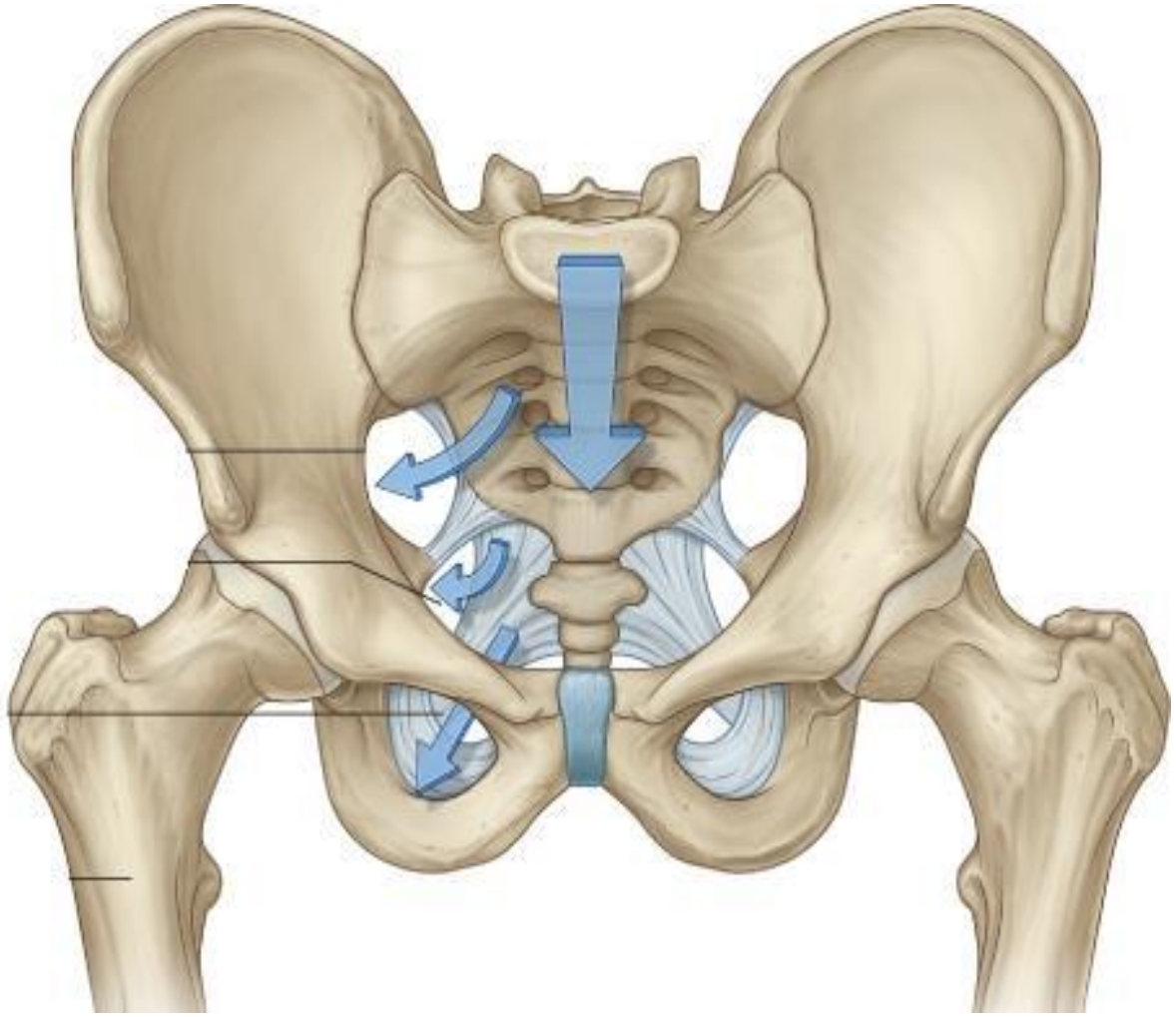
Superior View
Estrogen Driven



(Netter, 2014)

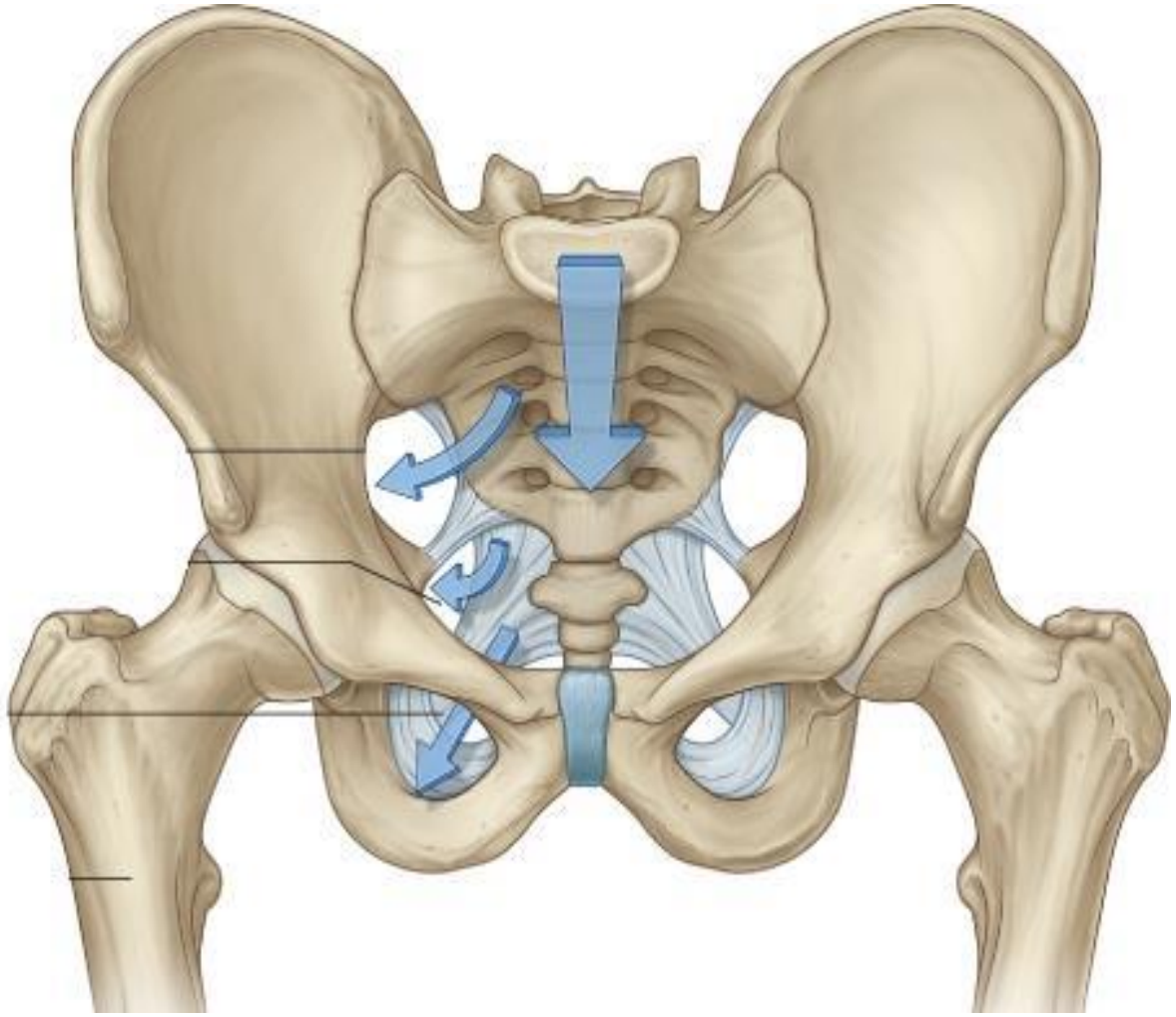
Apertures in the Pelvic Wall

- Greater Sciatic foramen
  - Formed by the Sacrospinous ligament
  - Piriformis runs through
  - Superior gluteal NV bundle
  - Inferior gluteal NV bundle
  - Sciatic nerve, nerves and vessels feeding the femoral muscles



Apertures in the Pelvic Wall

- Lesser Sciatic foramen
  - Formed by the Sacrotuberous ligament
  - Obturator internus muscle runs out
  - Pudendal nerves and vessels come in
- Obturator canal
  - Pierces through the obturator muscle and membrane
  - Obturator nerve and vessels going to medial thigh muscles



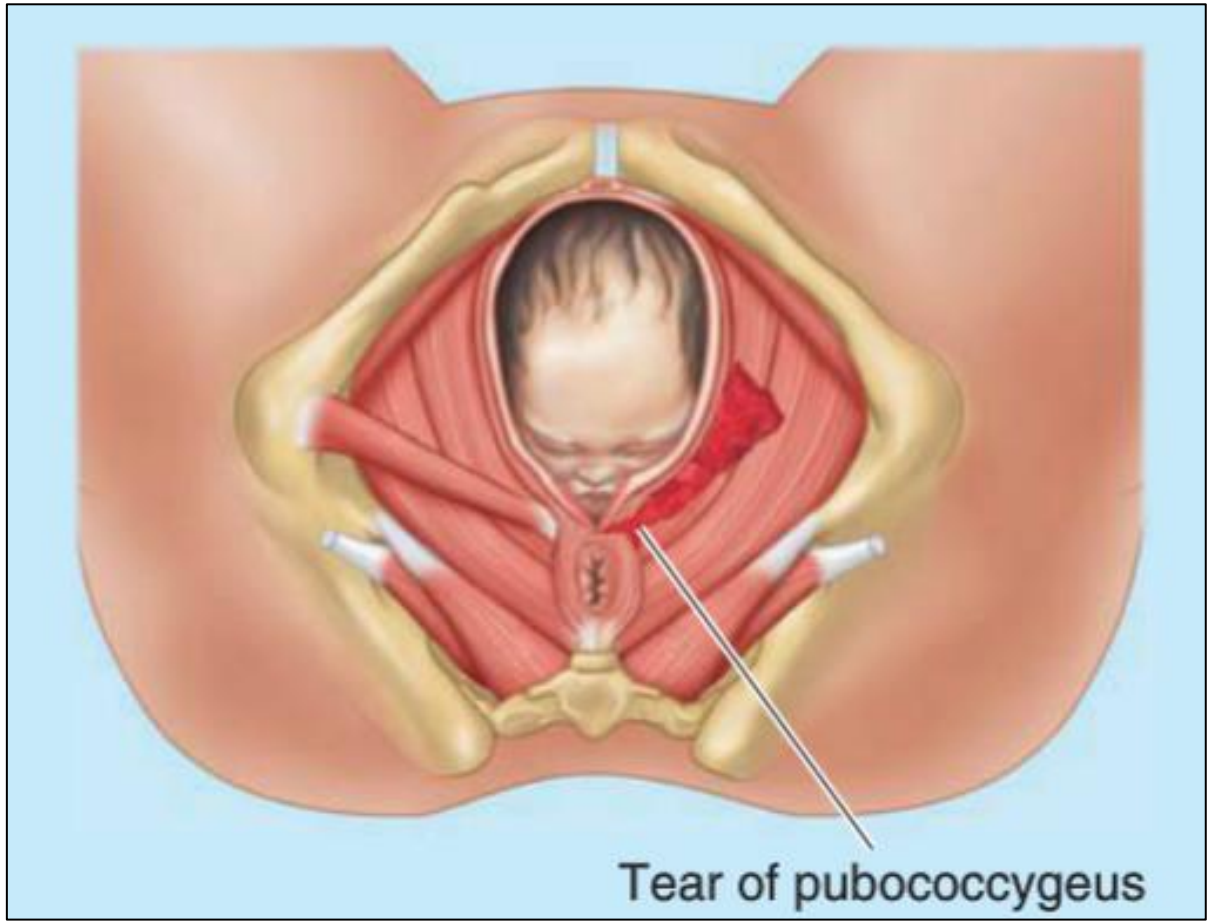
Vaginal delivery can result in tearing of levator ani muscles.

Most commonly injured during childbirth: pubococcygeus and puborectalis muscles

Weakening of pelvic floor (less support for pelvic viscera)

- Fecal and urinary incontinence
- Organ prolapse

Tear at the Pubococcygeus Muscle



(Moore et al., 2014)

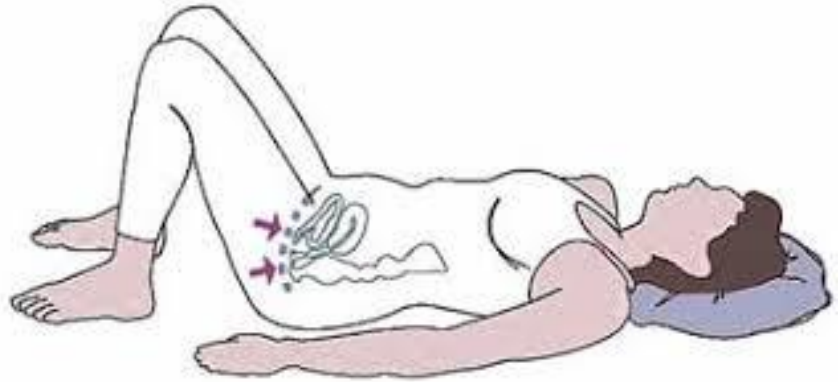
Pelvic Floor Dysfunctions

- Urinary incontinence
- Fecal incontinence
- Chronic pelvic pain syndromes
- Sexual dysfunction
- Visceral Prolapse (e.g. uterus, rectum)



Visceral Prolapse (uterus)

Pelvic Floor Strengthening Exercises



- Kegel exercises
- Developed by Arnold Kegel, MD in 1948
- These exercises involve contracting the Levator Ani muscles
  - The anus is elevated
    - Analogy: an elevator

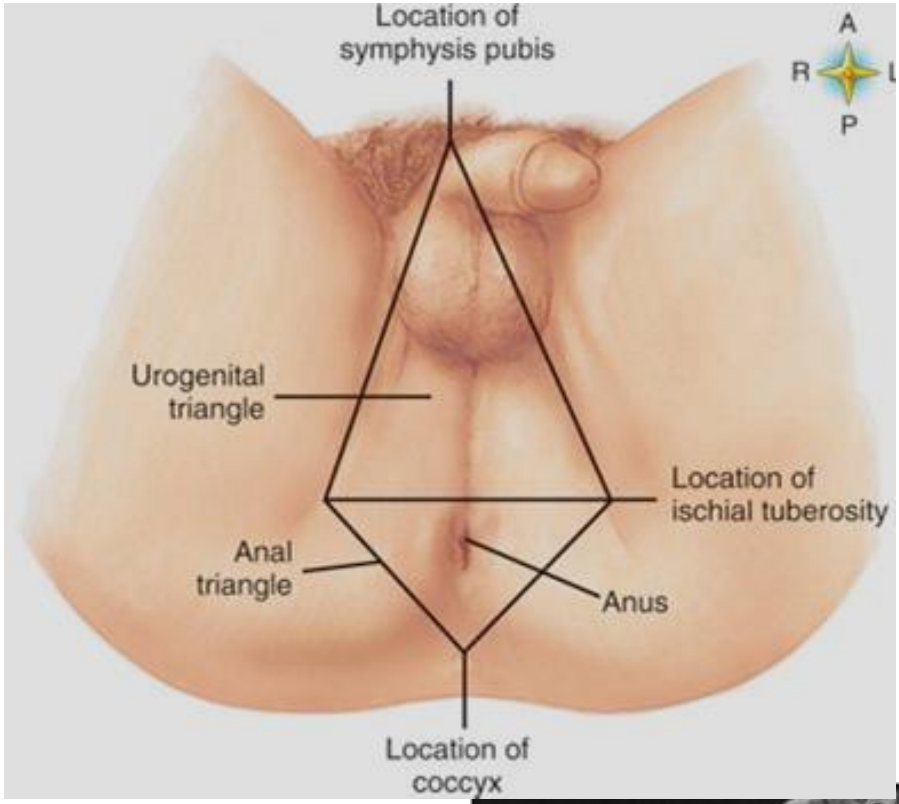


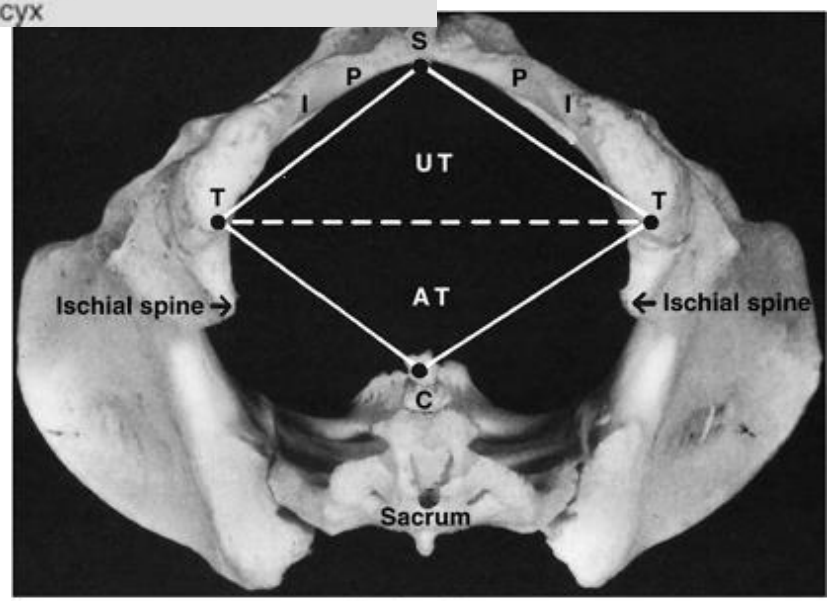
Learning Objectives

- Discuss the skeletal elements defining the pelvic cavity including osseous features, the differences driven by hormonally based development, and their clinical relevance
- Discuss the joints and ligaments of the pelvic girdle and their clinical significance
- Discuss structural and functional anatomy of muscles of the pelvic floor and the issues stemming from dysfunction
- Discuss the fascia defining and organs contained in the perineum

Perineum

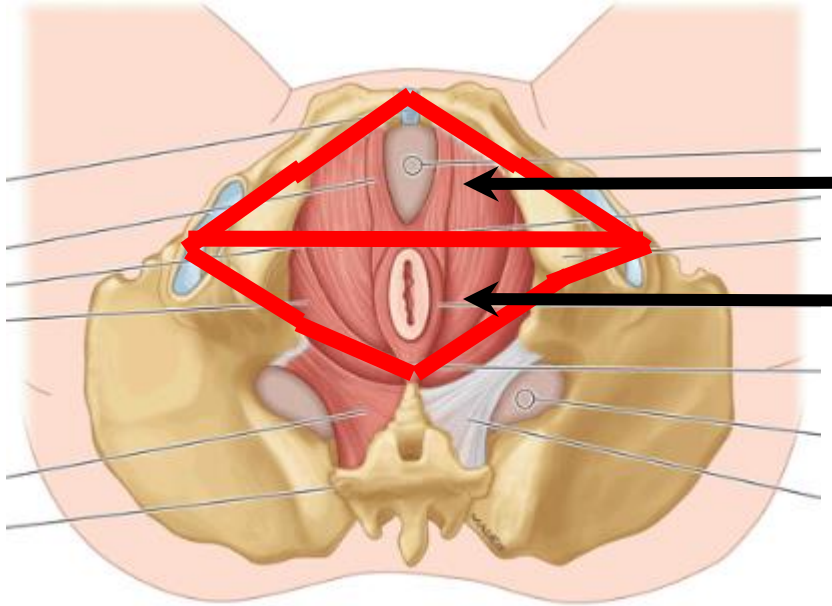
- Diamond shaped area
  - Points formed by the Ischial tuberosities laterally
  - Symphysis pubis anteriorly
  - Coccyx posteriorly
- Two triangles
  - Urogenital – External genitalia and urethral orifice
  - Anal – anal canal and the anus

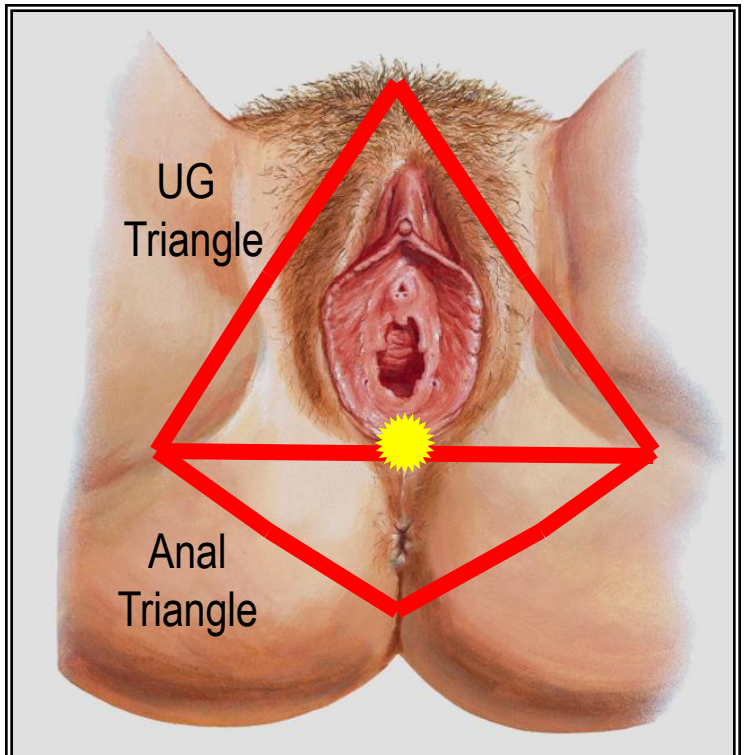




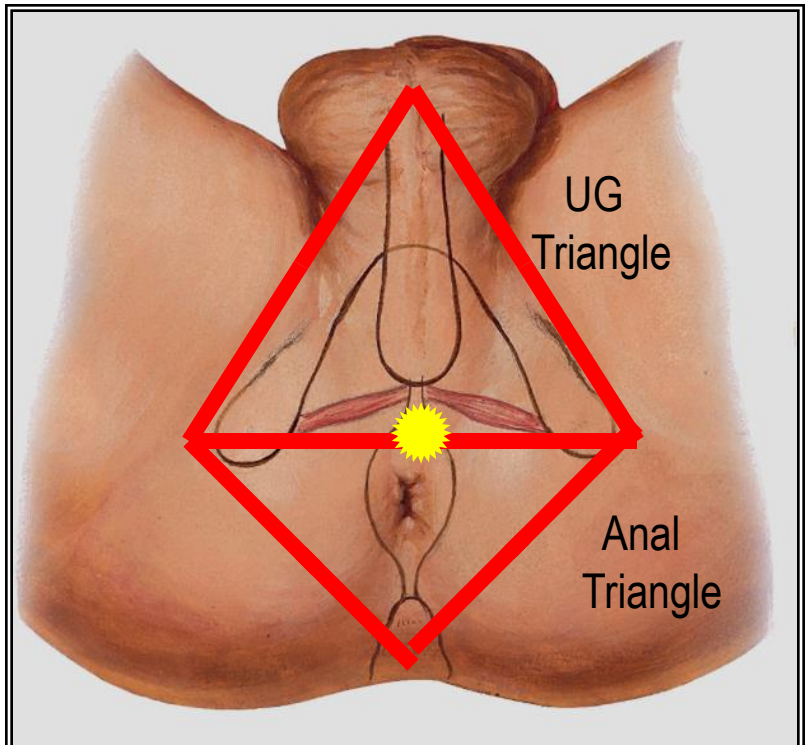
Inferior view

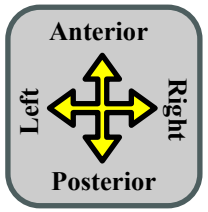
The Two Triangles of the Perineum





Perineal Body





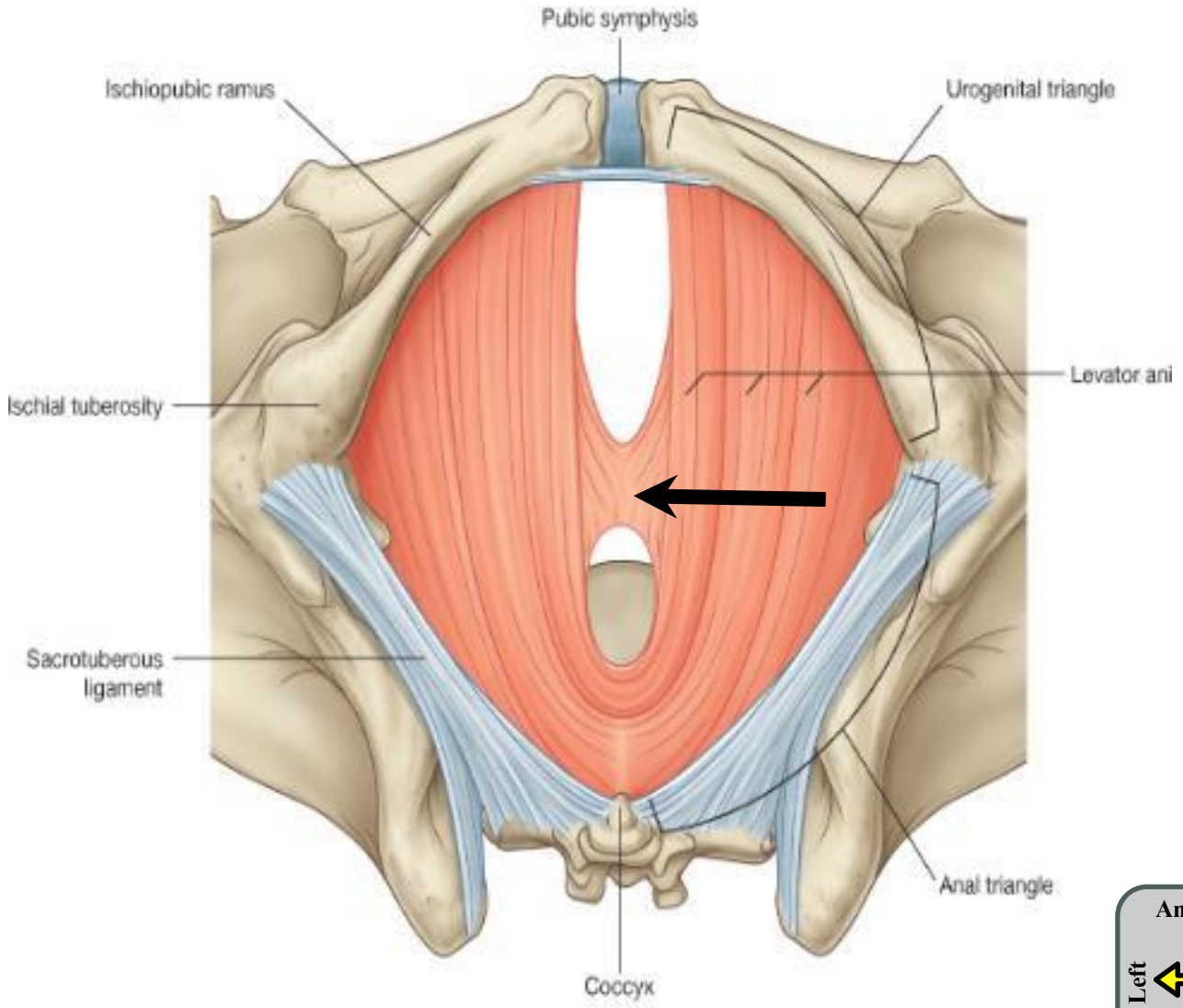
The Perineal Body

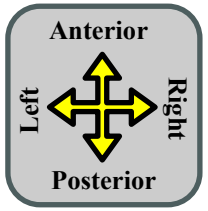
<u>A fibromuscular confluence</u>

- That lies at the center of the perineum midway between the ischial tuberosities

- All the membranes converge

- Many muscle sphincters converge here
  - e.g. urethra, vagina, and anus





Anatomy in Perineal Triangles

- Urogenital Triangle

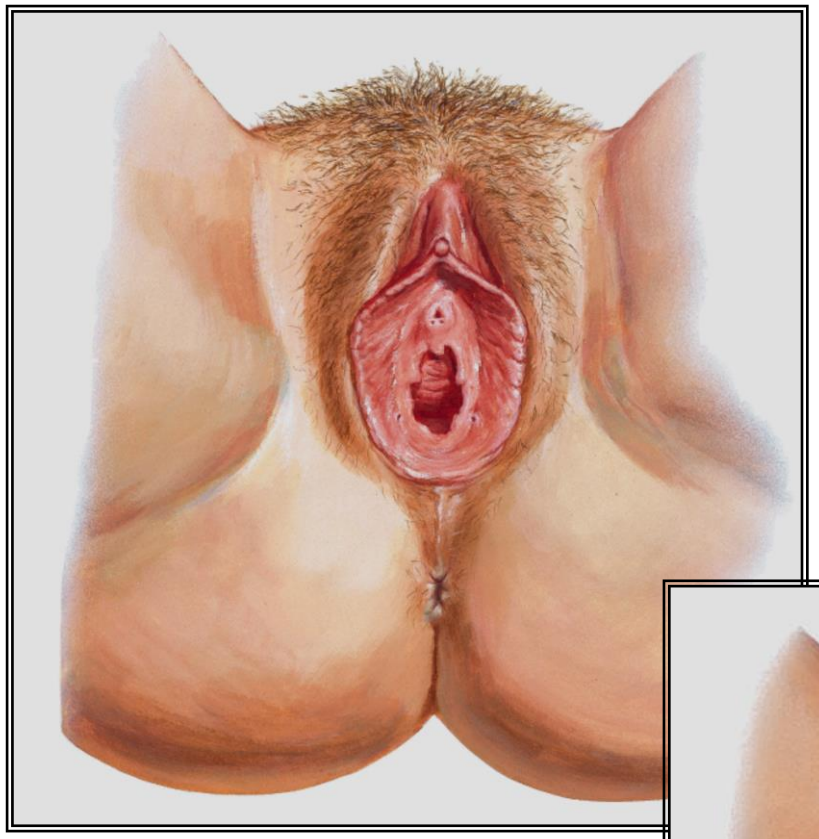
  - Vagina and vulva as external genitalia

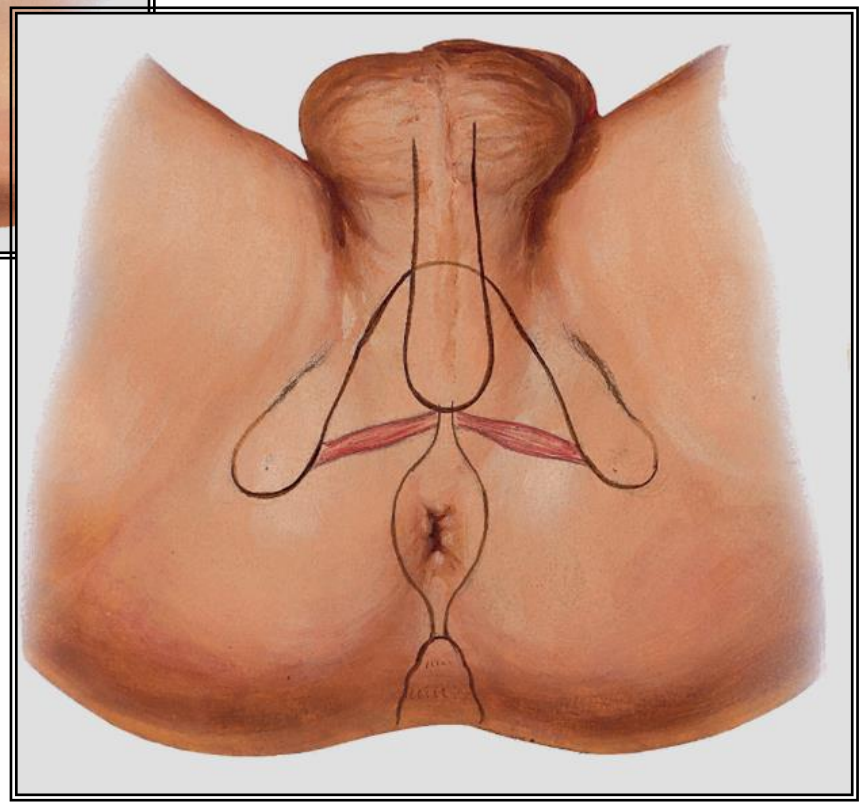
  - Scrotum and penis as external genitalia

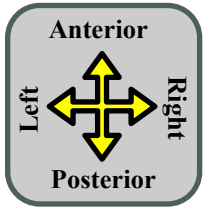
- Anal Triangle

  - Anal canal and anus

  - Ischioanal fat





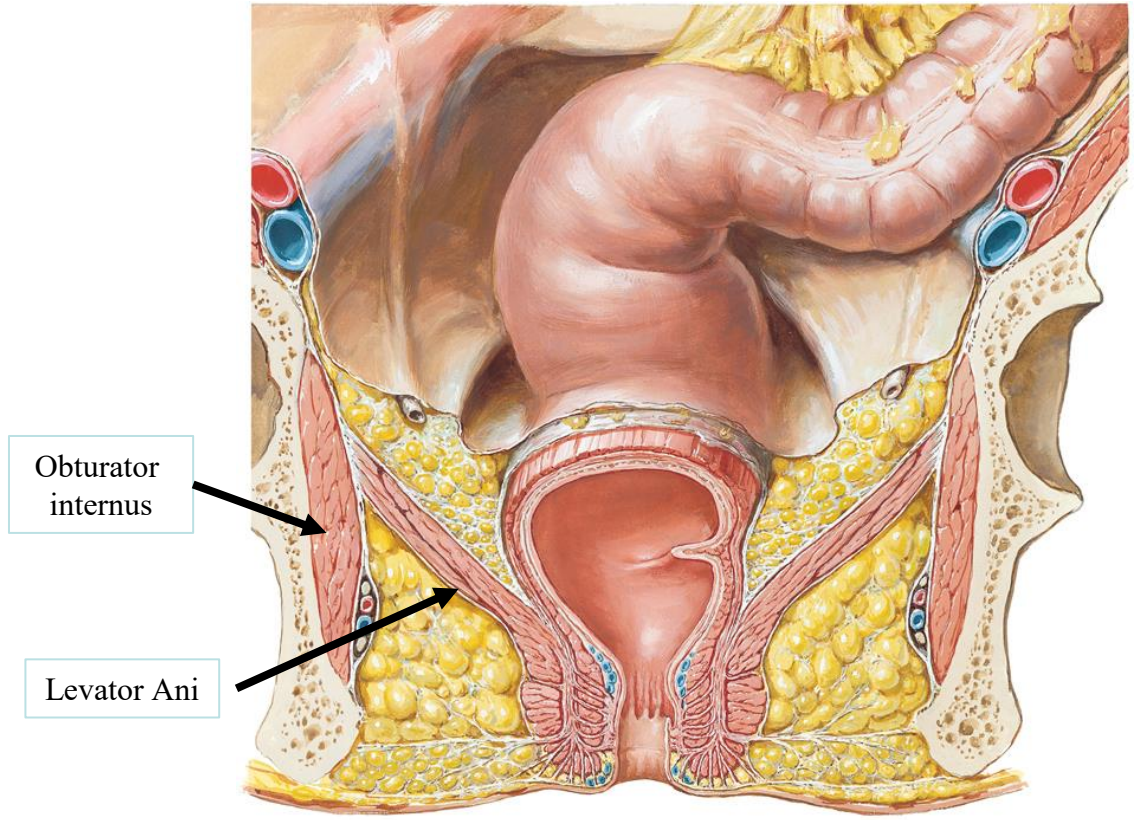


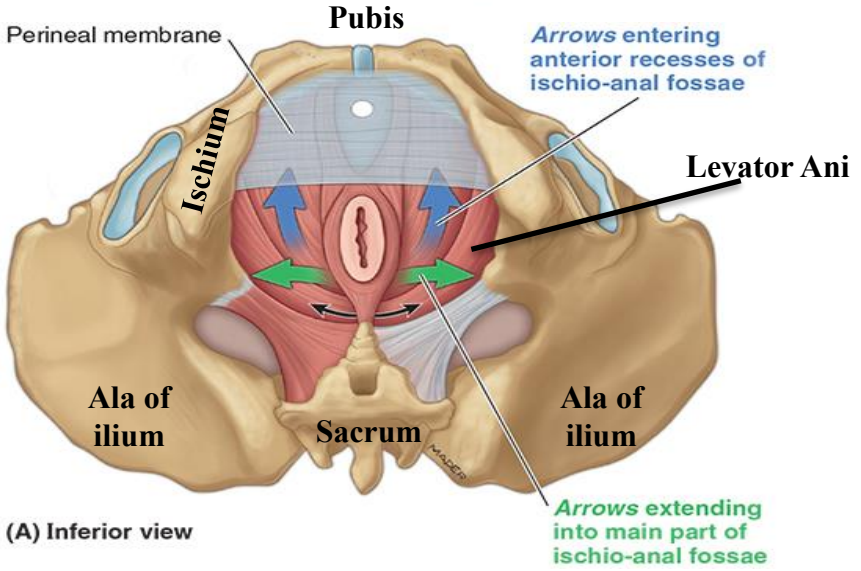
Anal Triangle

- Ischioanal fossa
  - Wedge shaped regions around wall of anal canal and deep to the pelvic diaphragm
- Filled with fat and fibrous tissue

Boundaries

- Laterally – Obturator internus muscle and fascia
- Medially – External anal sphincter,
- Posteriorly – Sacrotuberous ligament and gluteus maximus muscle
- Anteriorly – fat extends anterolaterally to pubis





Clinical Considerations
Anal Triangle

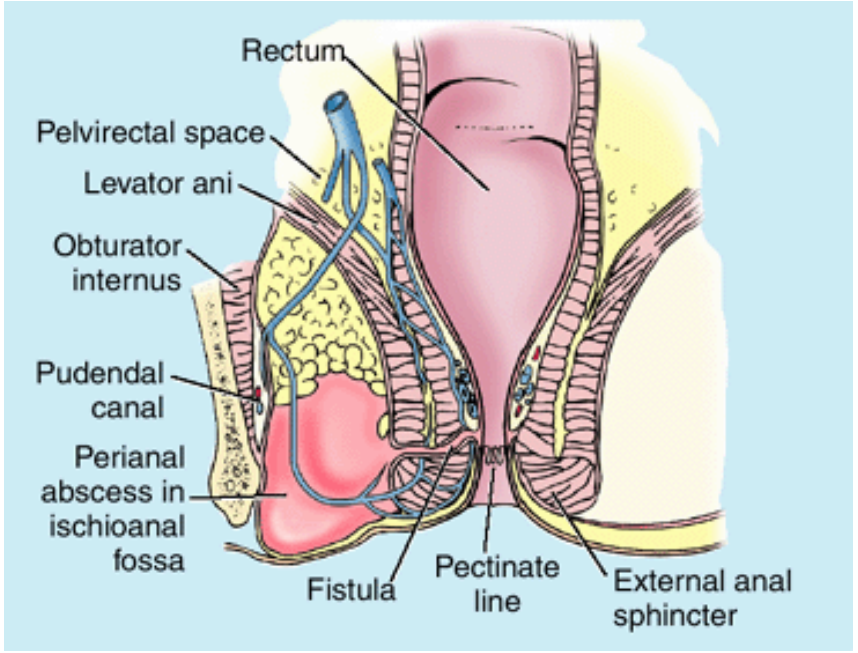
Abscess in ischioanal fossa

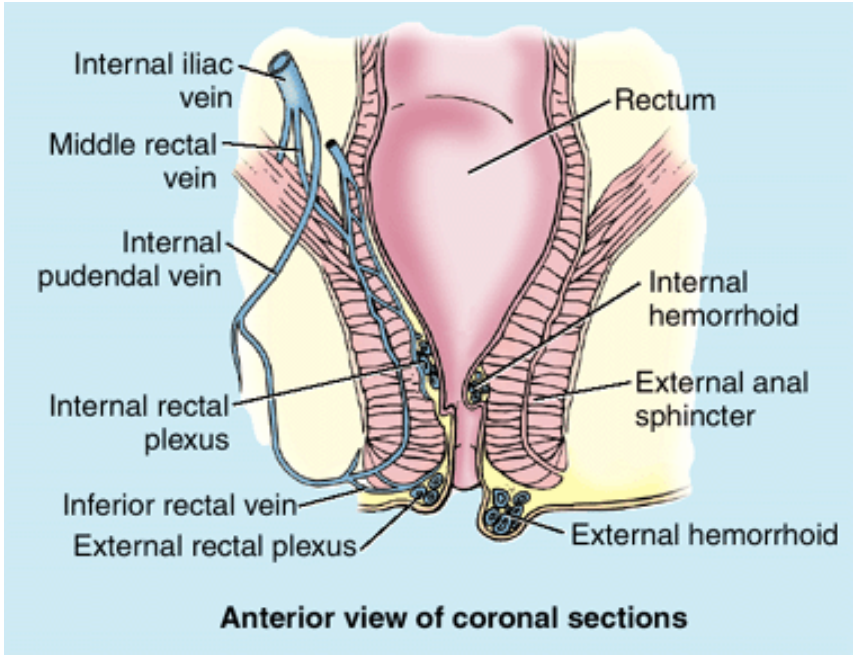
- Infection in fossa

- Rupture if pressure builds
  - Anal sphincter muscles leading to formation of fistula
  - Or rupture through skin

Hemorrhoids

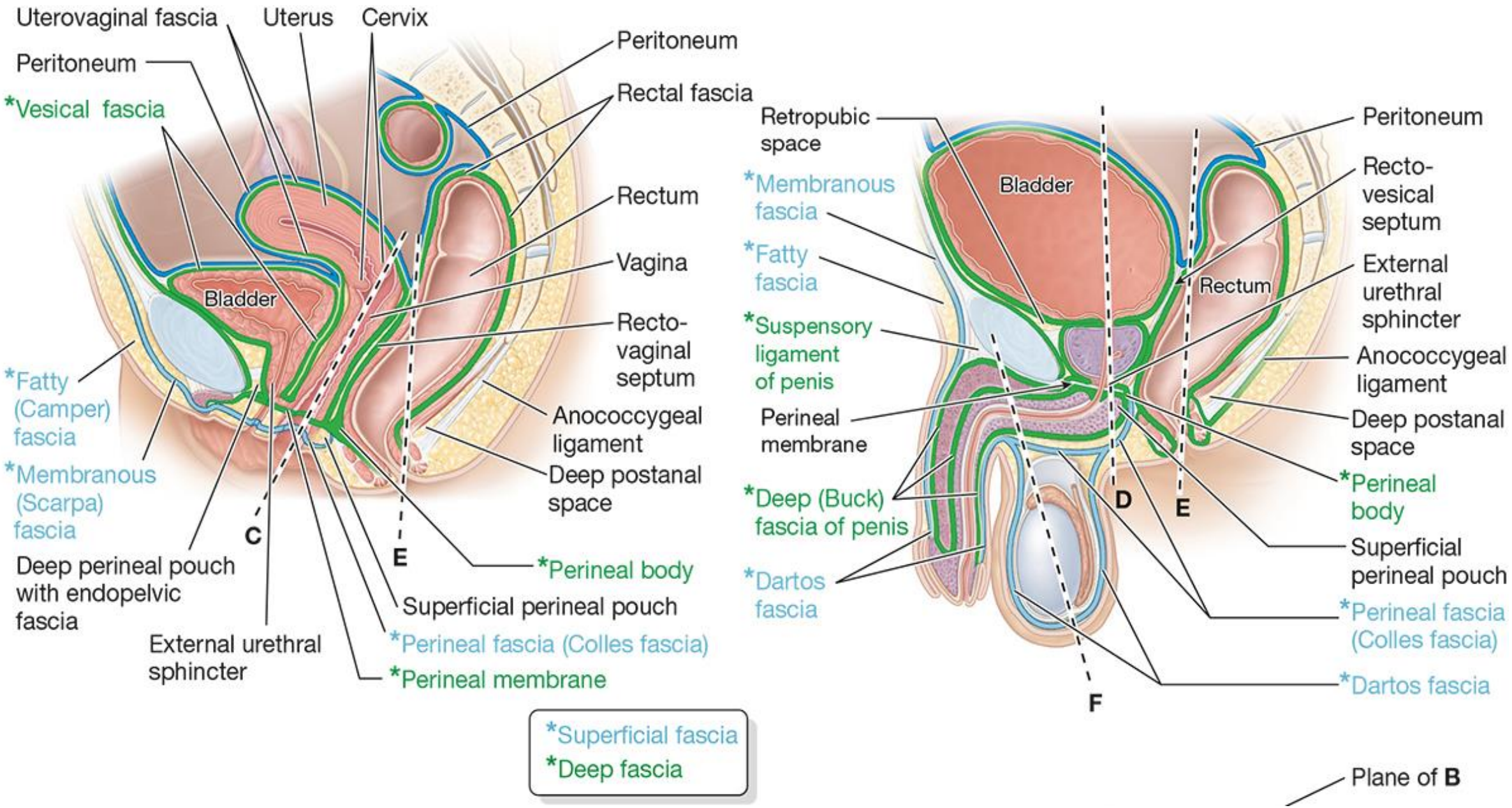
- Internal
  - Bright red blood in stool
  - No pain/above pectinate line
- External
  - Clot in the external rectal plexus
  - Caused by impediments to normal venous return
    - Constipation, Intra-abdominal pressure
  - Very painful





Fascia of the Urogenital Triangle

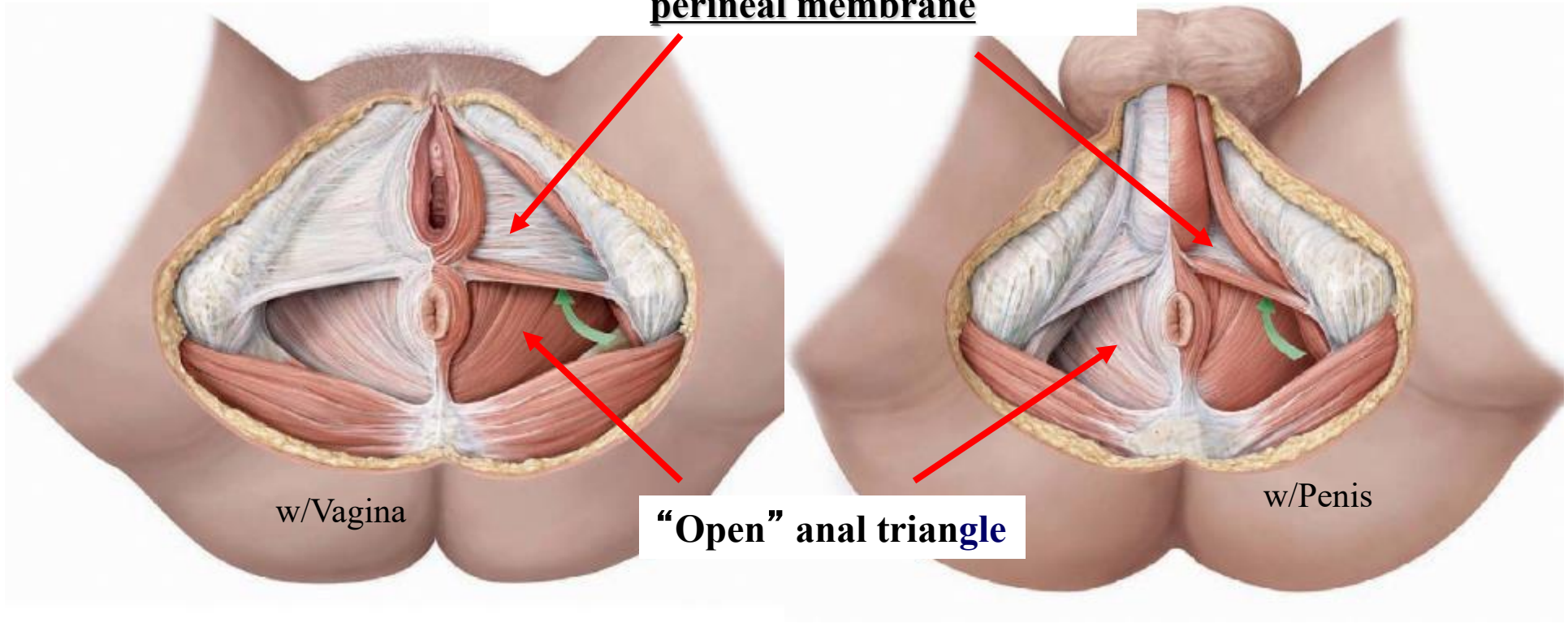
- Superficial layer
  - Continuous with abdominal wall skin
  - Camper fascia – subcutaneous fat under skin of perineum
  - Scarpa fascia – Colles fascia of the perineum in both sexes. Continuous with investing fascia of thigh
  - Becomes Dartos fascia around the scrotum
- Deep Layer
  - Continuous with deep fascia of the abdominal wall muscles
  - Perineal membrane in both sexes
  - Continuous with Buck's fascia around the penis
- Perineal Body – in both sexes fusion of all membranes/muscles/fascia


Uterovaginal fascia
Uterus
Cervix
Peritoneum
Peritoneum
*Vesical fascia
Rectal fascia
Rectum
Vagina
Bladder
Recto-vaginal septum
*Fatty (Camper) fascia
Anococcygeal ligament
*Membranous (Scarpa) fascia
Deep postanal space
C
E
Deep perineal pouch with endopelvic fascia
*Perineal body
Superficial perineal pouch
External urethral sphincter
*Perineal fascia (Colles fascia)
*Perineal membrane
*Superficial fascia
*Deep fascia
Retropubic space
Peritoneum
*Membranous fascia
Bladder
Recto-vesical septum
*Fatty fascia
Rectum
External urethral sphincter
*Suspensory ligament of penis
Anococcygeal ligament
Perineal membrane
Deep postanal space
*Perineal body
D
E
*Deep (Buck) fascia of penis
Superficial perineal pouch
*Dartos fascia
*Perineal fascia (Colles fascia)
F
*Dartos fascia
Plane of B

Perineal Membrane

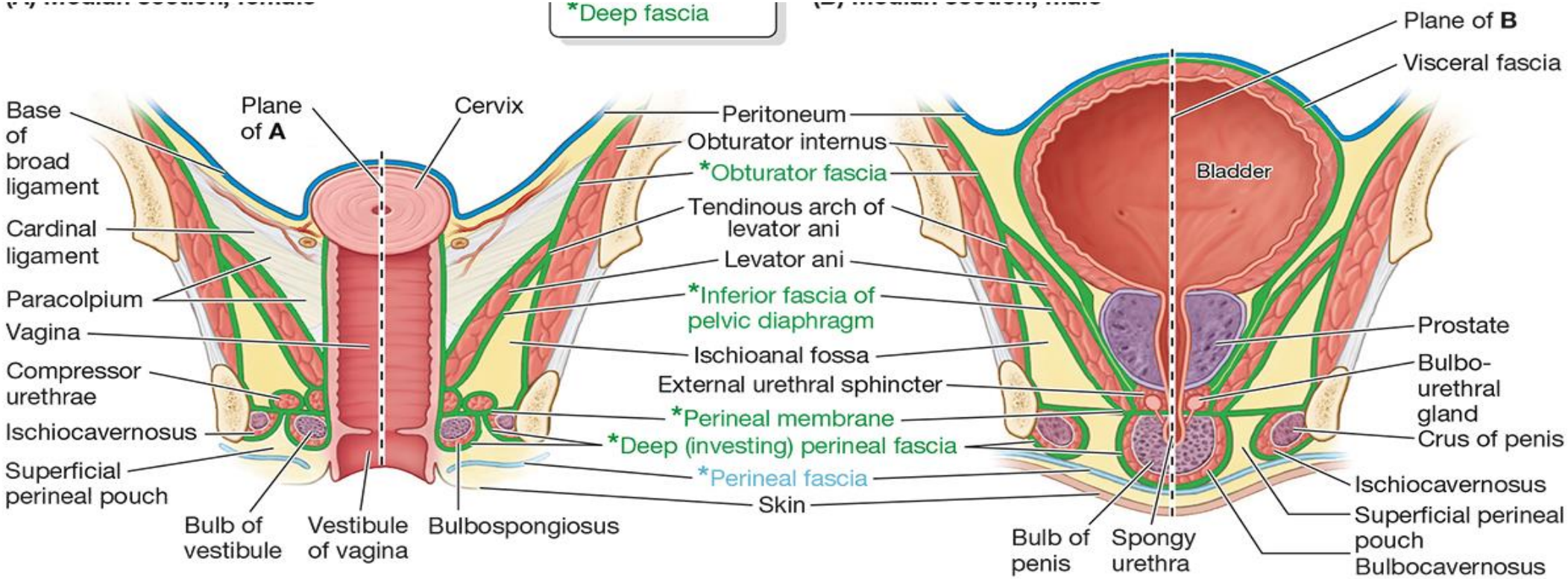
- A tough fascial sheet
- Attaches to ischiopubic rami
- Supports the urogenital triangle
- Perforated by the urethra in both sexes and by the vagina when present

"Closed" Urogenital triangle by the perineal membrane

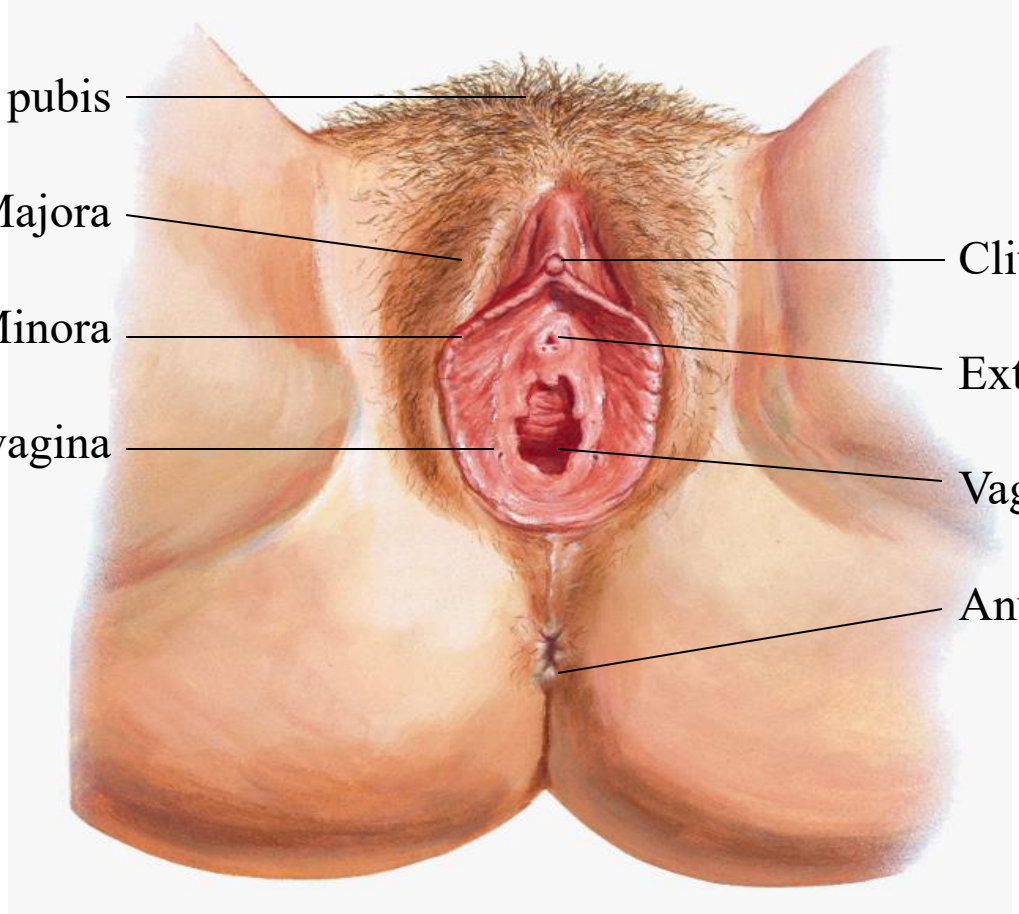


Perineal Pouches

- Perineal membrane divides the urogenital triangle into two pouches
  - Superficial between the skin and perineal membrane
  - Deep between the perineal membrane and the pelvic diaphragm

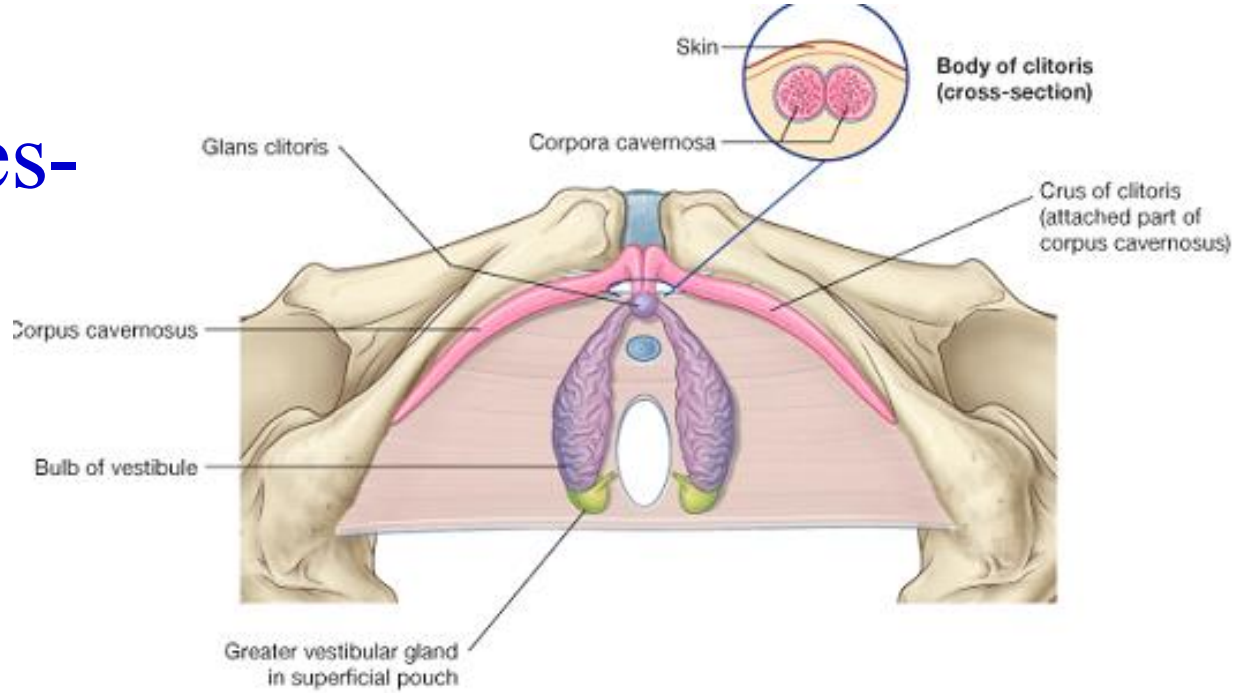


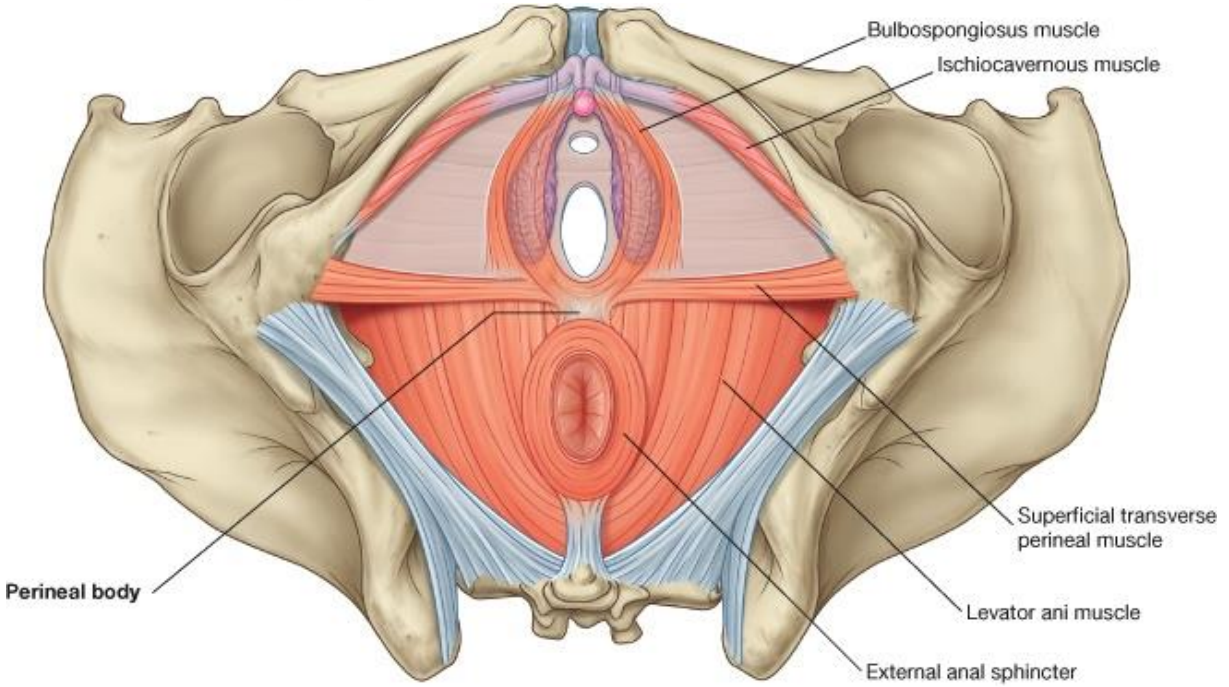
Surface Anatomy
Estrogen-Dependent External Genitalia



Estrogen Dependent Structures- Superficial Pouch

- Crura of clitoris and covering muscle - ischiocavernosus
- Bulbs of vestibule and covering muscle - bulbospongiosus
- Superficial transverse perineal muscles
- Greater vestibular glands
- Branches of internal pudendal vessels
- Branches of pudendal nerves

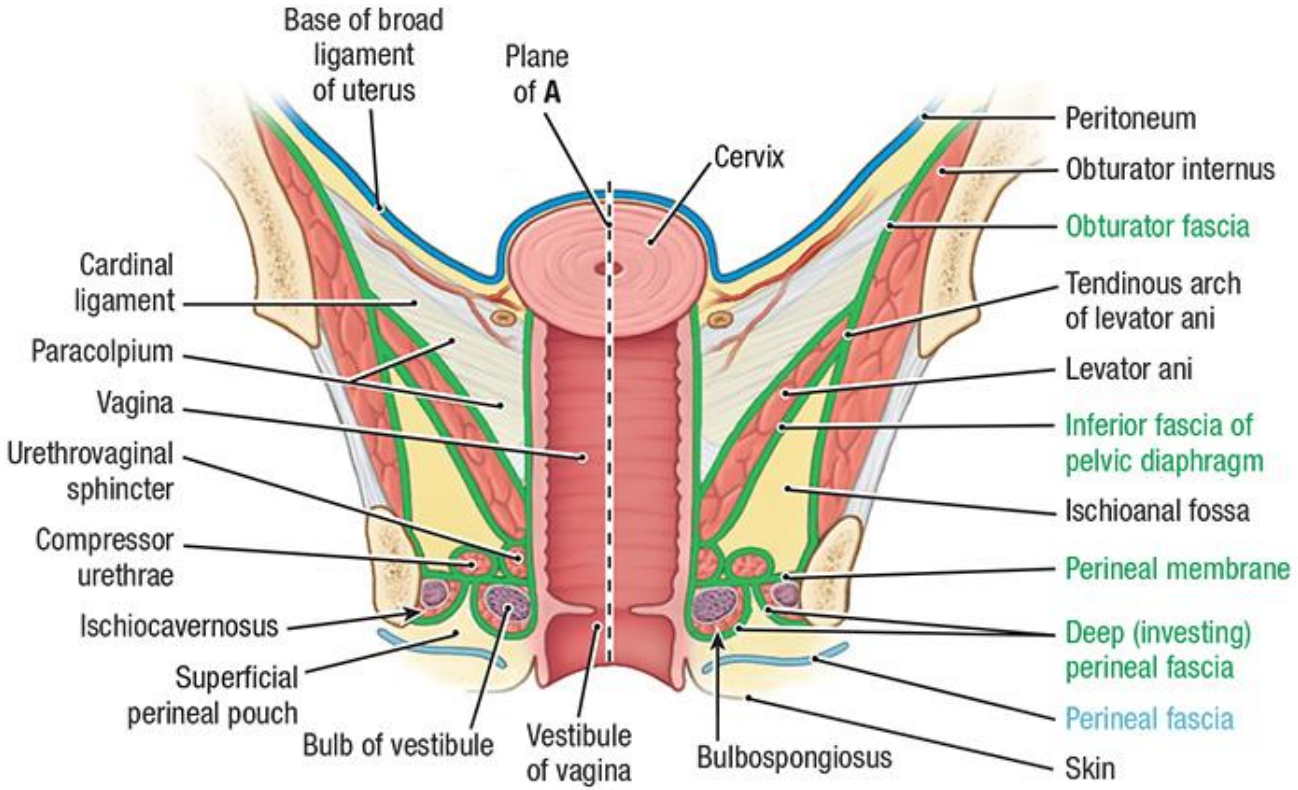




Estrogen-Dependent Deep Perineal Space

Contents

- Portion of the urethra
- External urethral sphincter muscle
- Portion of the vagina
- Deep transverse perineal muscles
- Branches of pudendal nerve, internal pudendal arteries and veins
- Anterior recesses of the fat filled ischioanal fossae

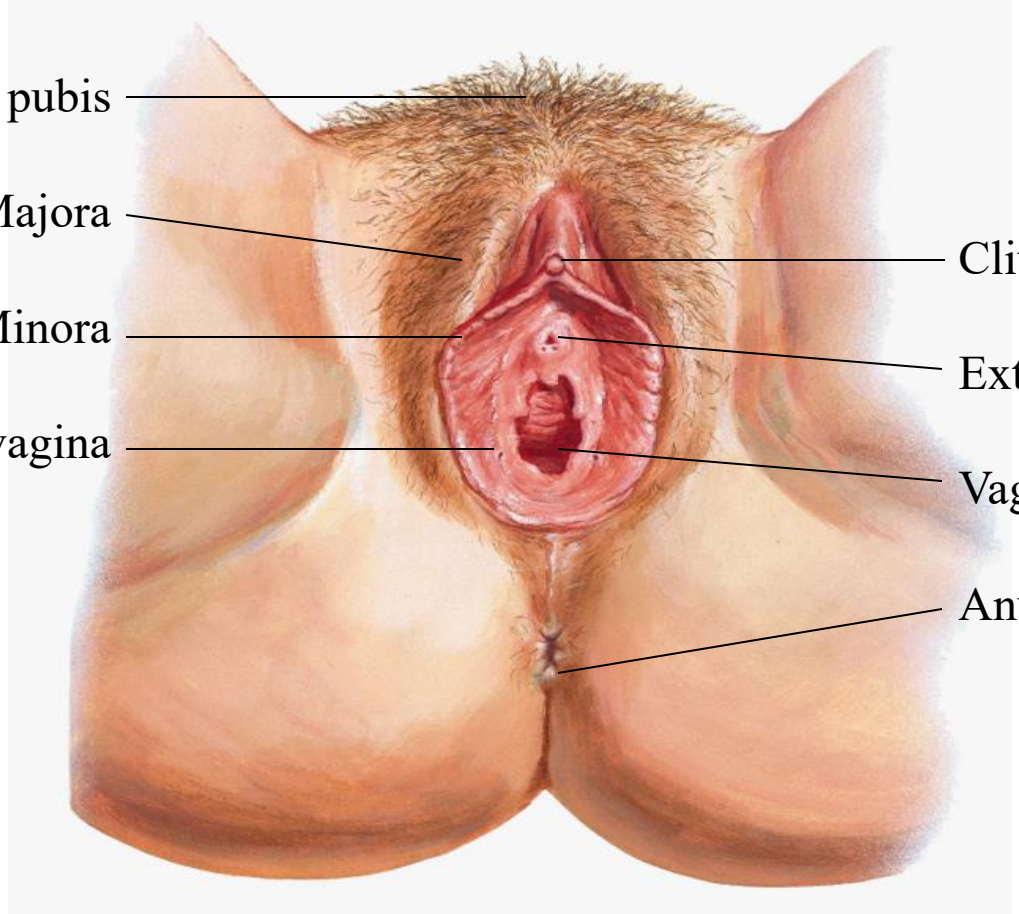


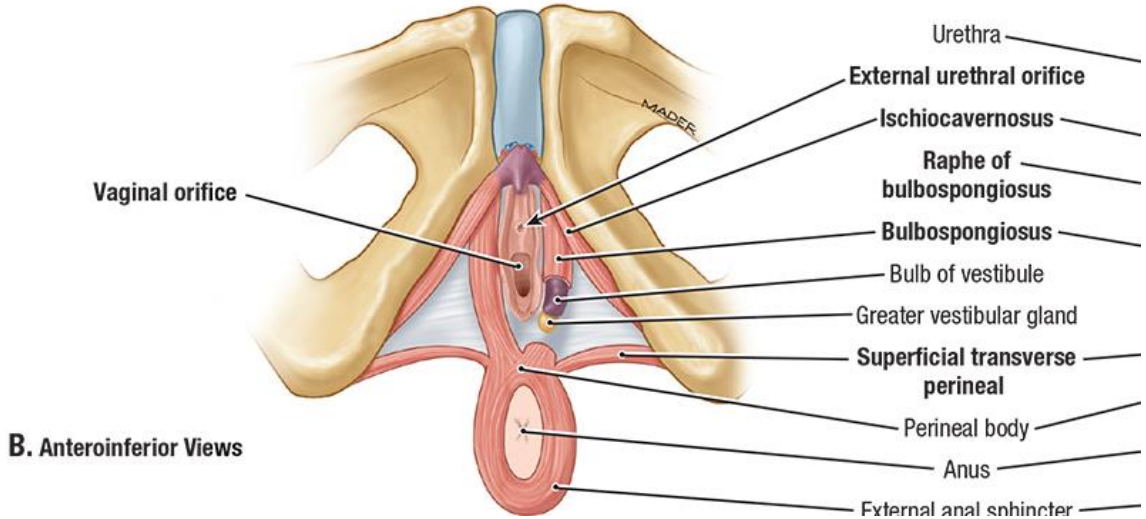
C. Coronal Section in Plane Indicated in *Part A*

Urogenital Diaphragm consists of:

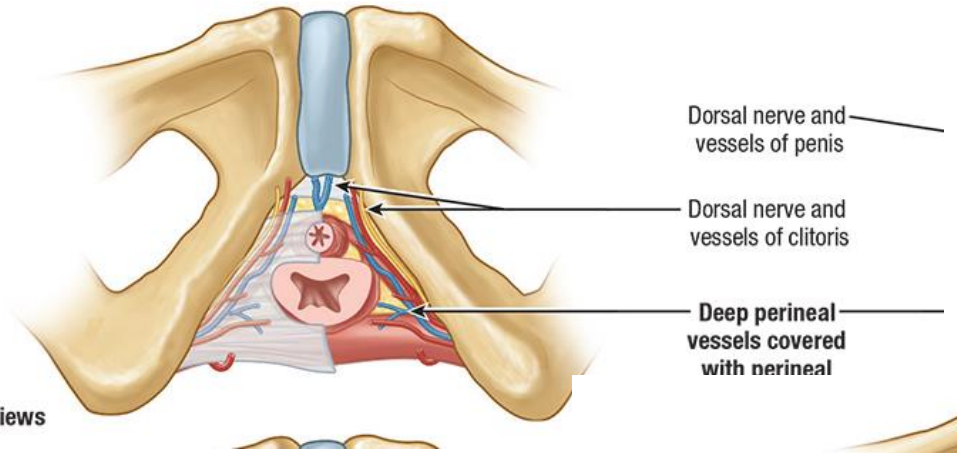
- The floor of the deep pouch (Perineal membrane)
- Muscles directly superior to the perineal membrane in the deep perineal pouch

Surface Anatomy
Estrogen-Dependent External Genitalia

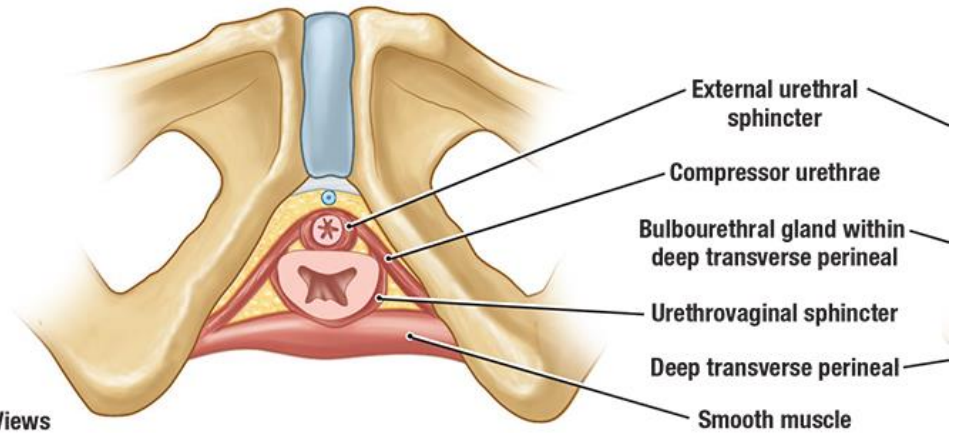




B. Anteroinferior Views



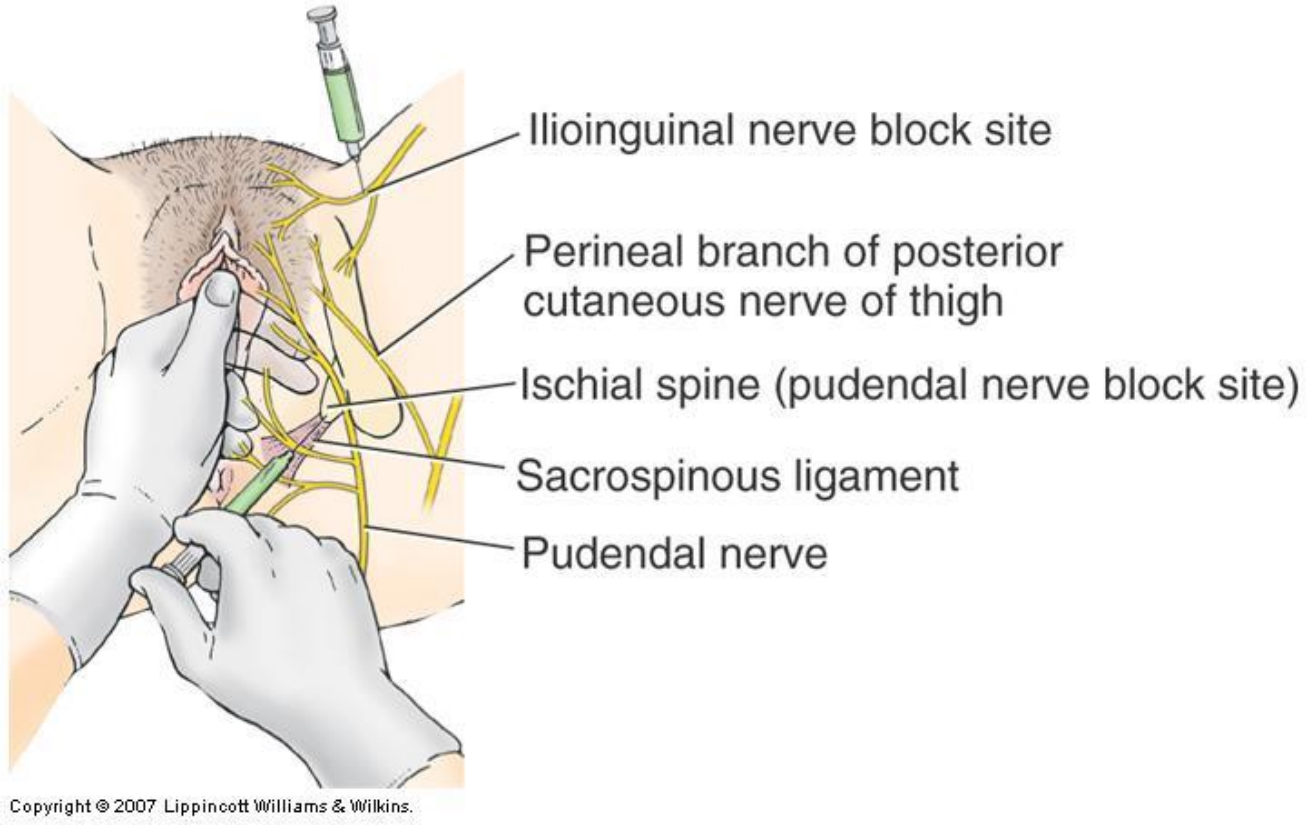
C. Anteroinferior Views



D. Anteroinferior Views

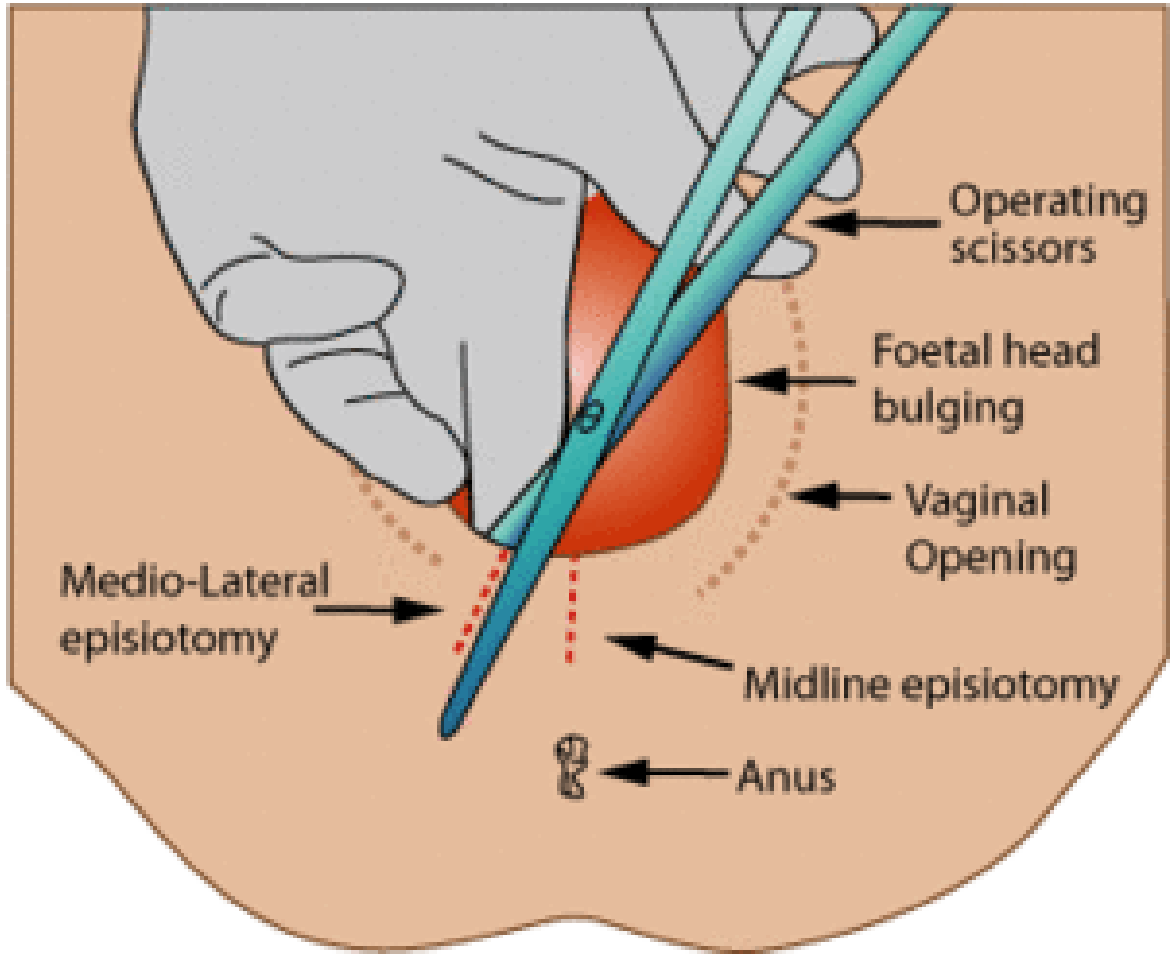
Clinical Considerations - Vulva

- Infected Greater Vestibular glands
  - **Bartholinitis**
  - From numerous pathogenic organisms
  - Swollen up to 4-5 cm in diameter
  - Very painful; impacts wall of rectum
- Nerve blocks in females
  - Pain of childbirth
  - Ilioinguinal
  - Pudenal blocks

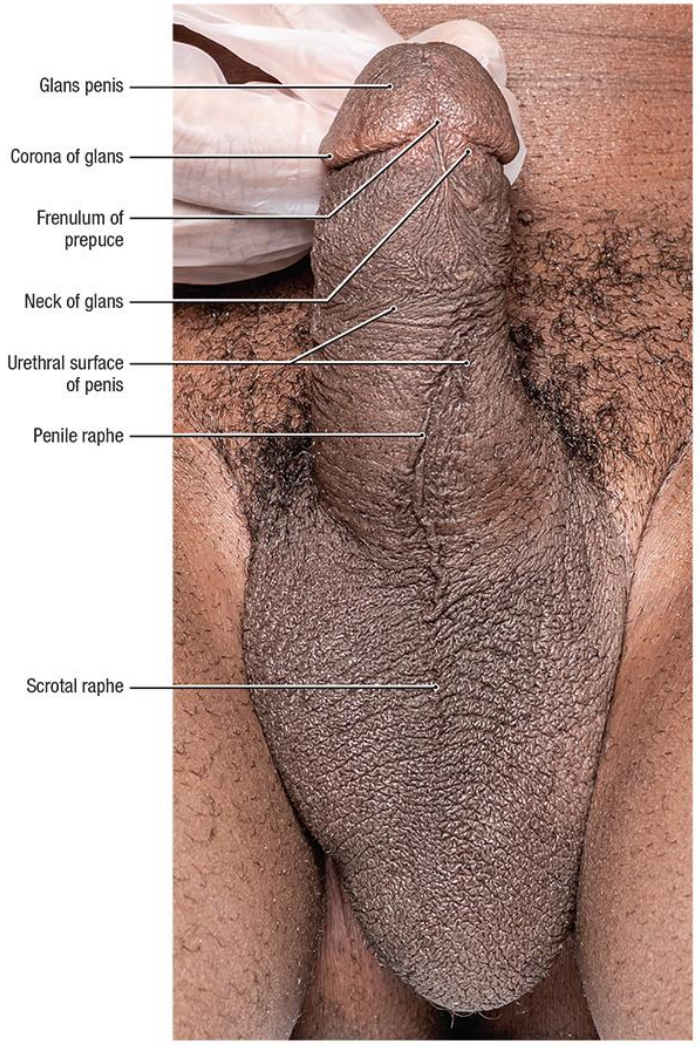


Clinical Considerations - Vulva

- Episiotomy during childbirth
  - Surgical incision of the perineum and vaginal wall
  - Prevention of a jagged tear
  - Midline most common in US
    - Prevalence decreasing
- Disruption of the perineal body
  - Problem in females following vaginal childbirth
  - Stretching weakens the support in this region
  - Prolapse of the pelvis viscera



Surface Anatomy
Testosterone-Dependent External Genitalia

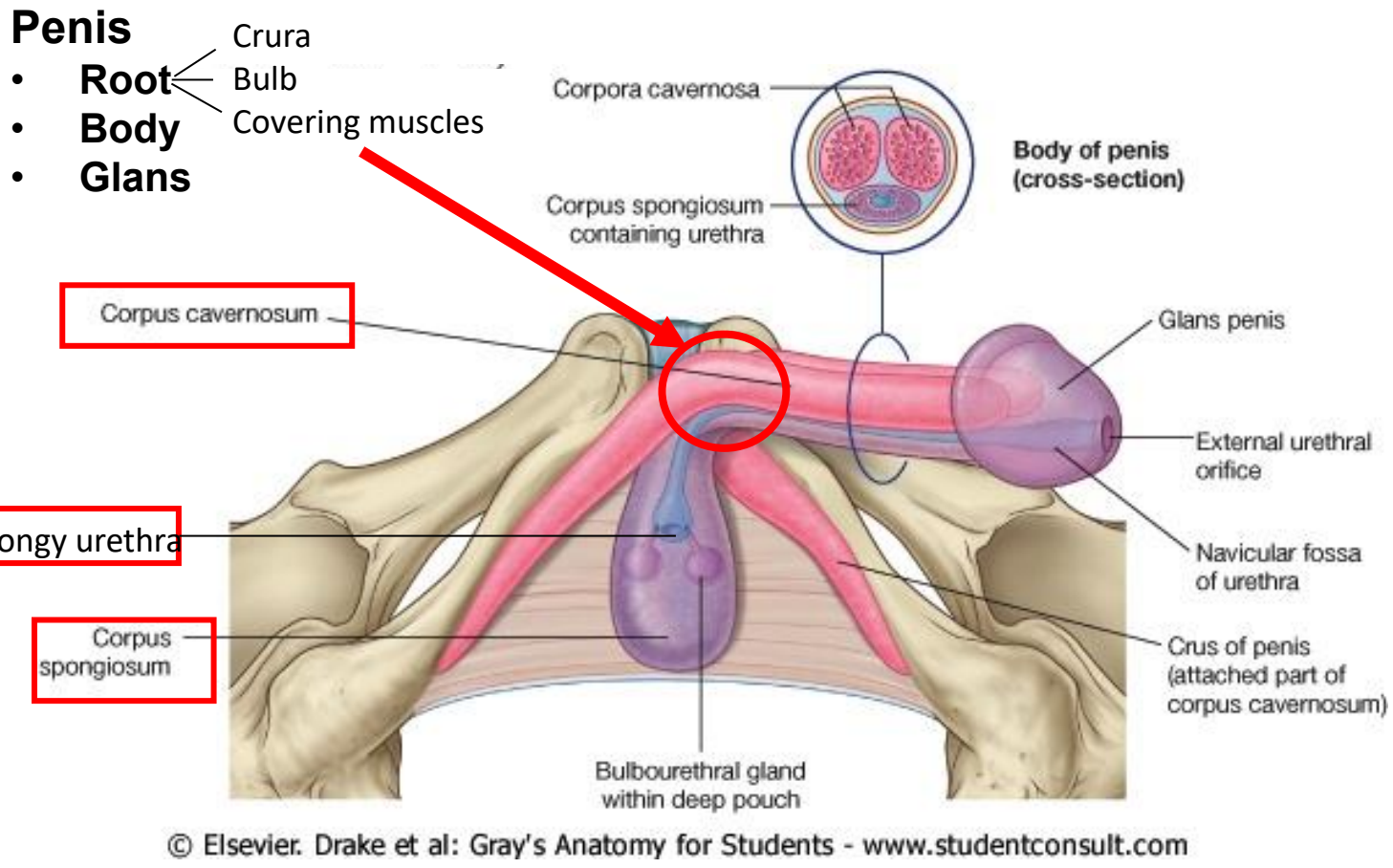


A. Inferior View, Penis Retracted Anteriorly



Testosterone Dependent Structures- Superficial Pouch

- Bulb and crura of the penis and muscles
  - Bulbospongiosus
  - Ischiocavernosus
- Superficial transverse perineal muscles
- Branches of internal pudendal vessels
- Branches of pudendal nerves



Testosterone-Dependent Contents of Superficial Pouch

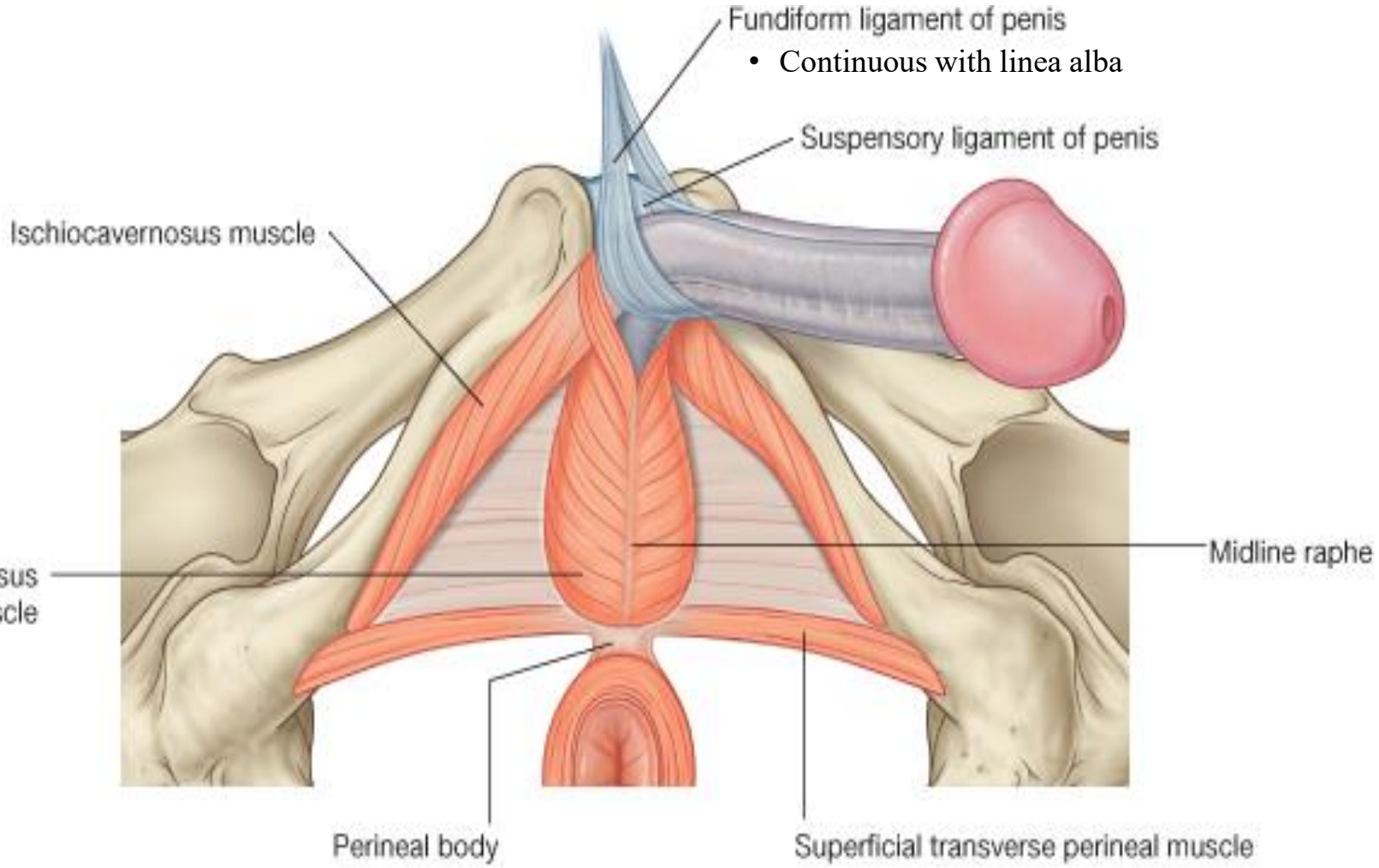
Ischiocavernosus Muscle

- Innervation: perineal n.,
  - a branch of pudendal nerve (S2-S4)
- Compresses deep dorsal vein
- Forces blood into distal penis
- Helps maintain erection

- More developed related to penis vs. clitoris

Bulbospongiosus

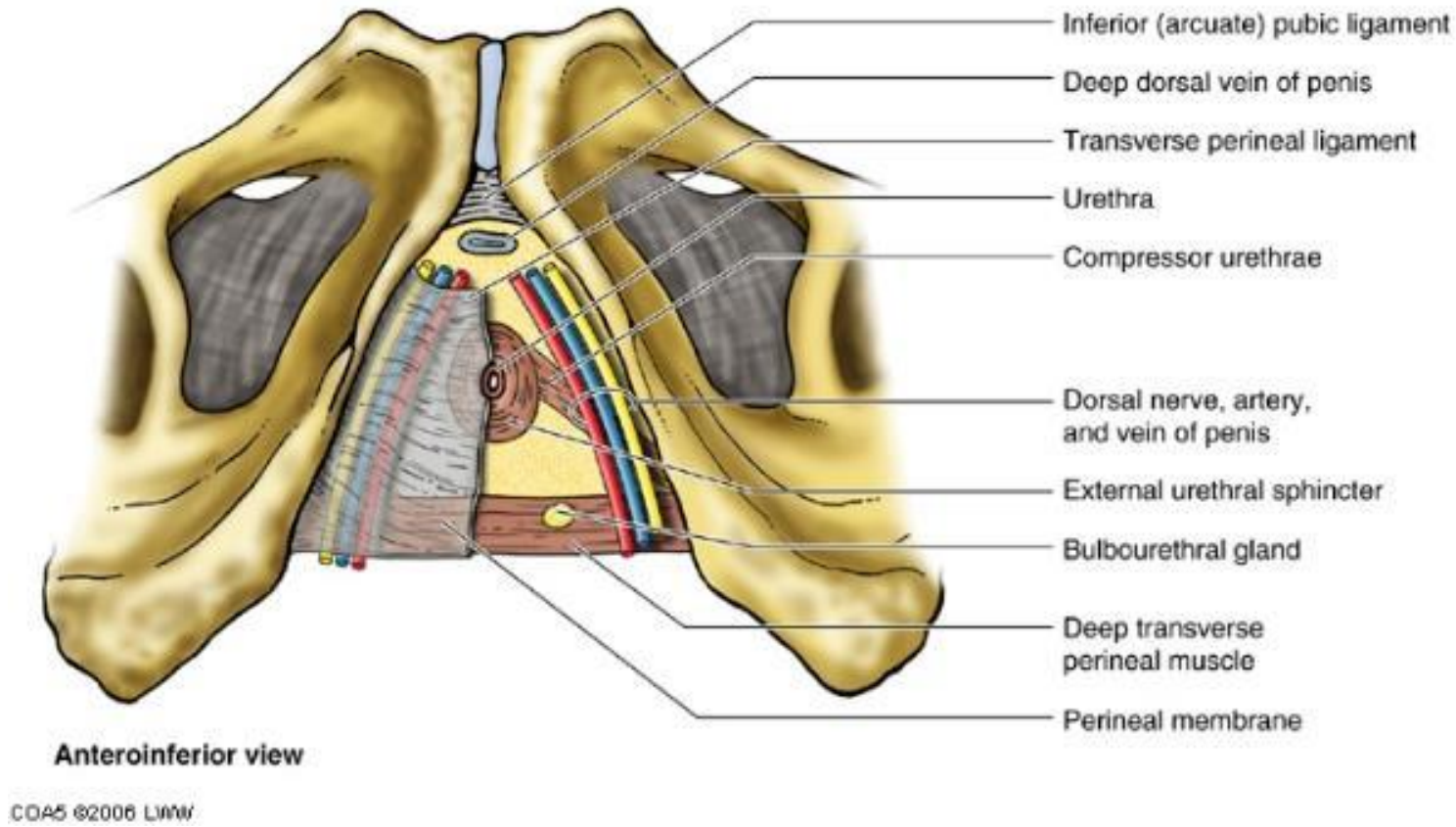
- Innervation: perineal n.,
- branch of pudendal nerve (S2-S4)
- Expels last drops of urine/semen



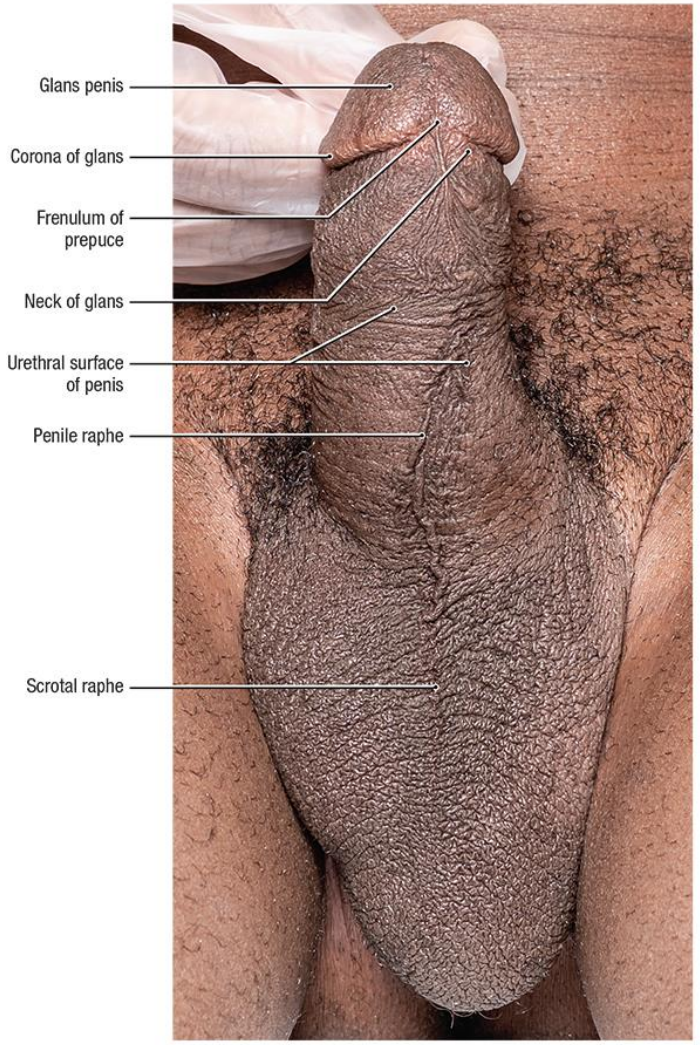
Testosterone-Dependent Contents of Deep Perineal Spaces

Contents

- Membranous portion of the urethra
- External urethral sphincter muscle
- Bulbourethral glands
- Deep transverse perineal muscles
- Branches of pudendal nerve, internal pudendal artery and vein
- Anterior recesses of the fat filled ischioanal fossae

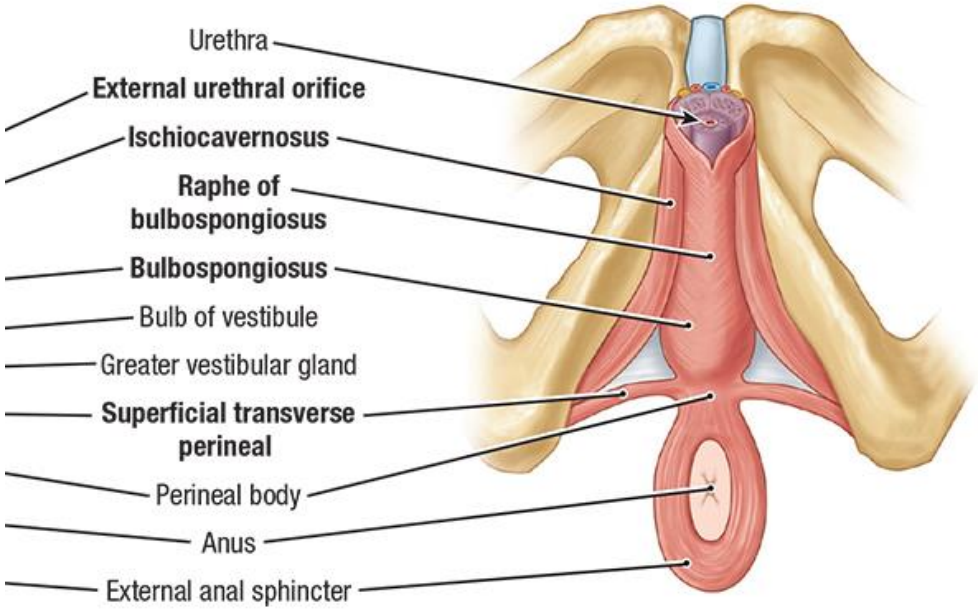


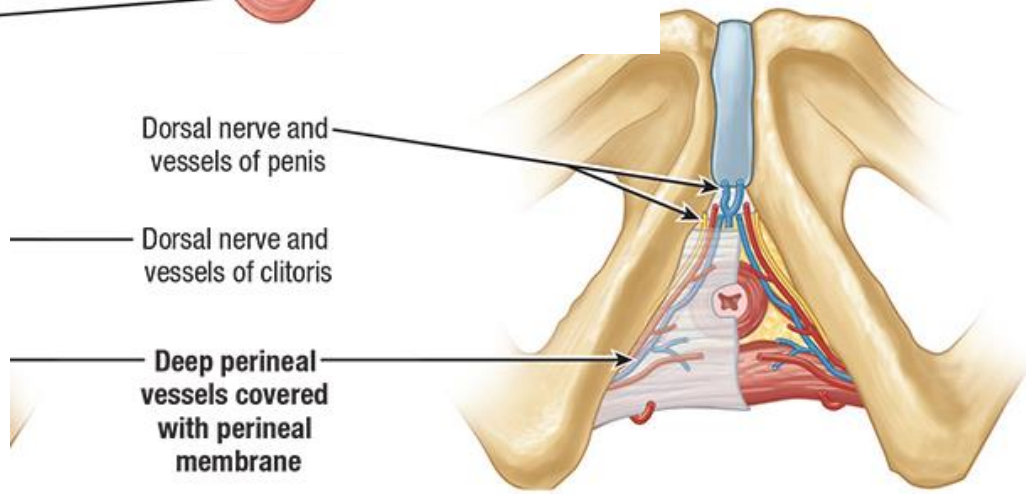
Surface Anatomy
Testosterone-Dependent External Genitalia

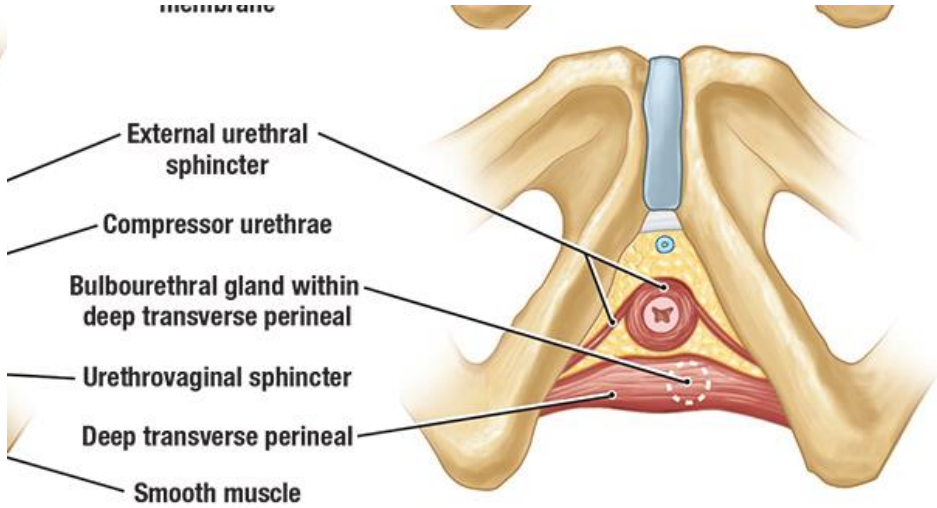


A. Inferior View, Penis Retracted Anteriorly

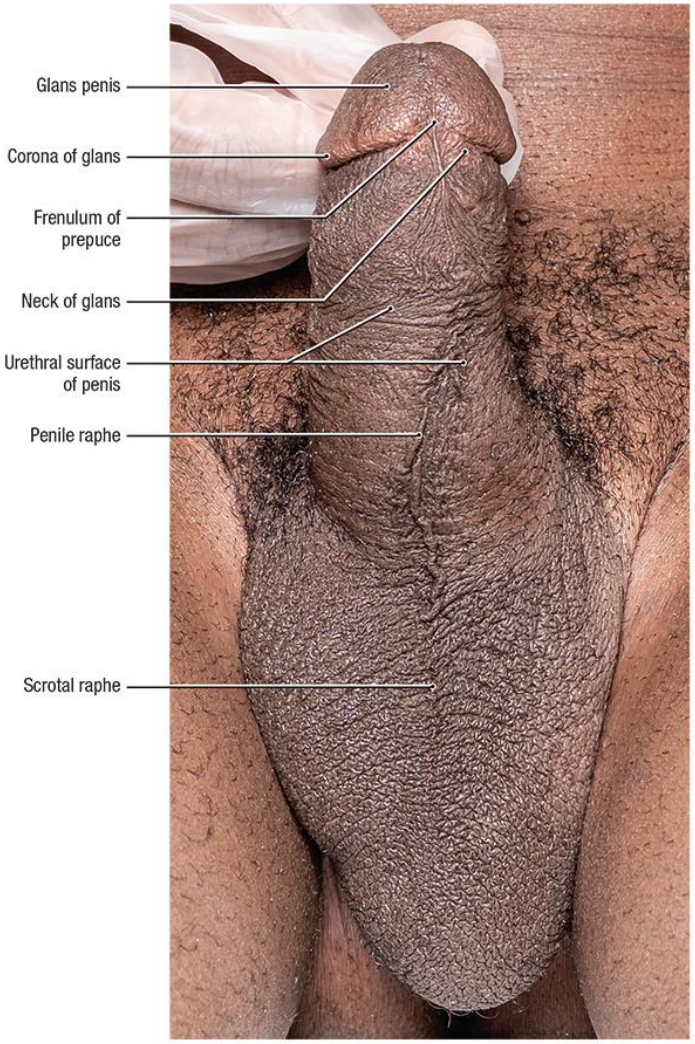



Urethra
External urethral orifice
Ischiocavernosus
Raphe of bulbospongiosus
Bulbospongiosus
Bulb of vestibule
Greater vestibular gland
Superficial transverse perineal
Perineal body
Anus
External anal sphincter


Dorsal nerve and vessels of penis
Dorsal nerve and vessels of clitoris
Deep perineal vessels covered with perineal membrane


External urethral sphincter
Compressor urethrae
Bulbourethral gland within deep transverse perineal
Urethrovaginal sphincter
Deep transverse perineal
Smooth muscle

Surface Anatomy
Testosterone-Dependent External Genitalia

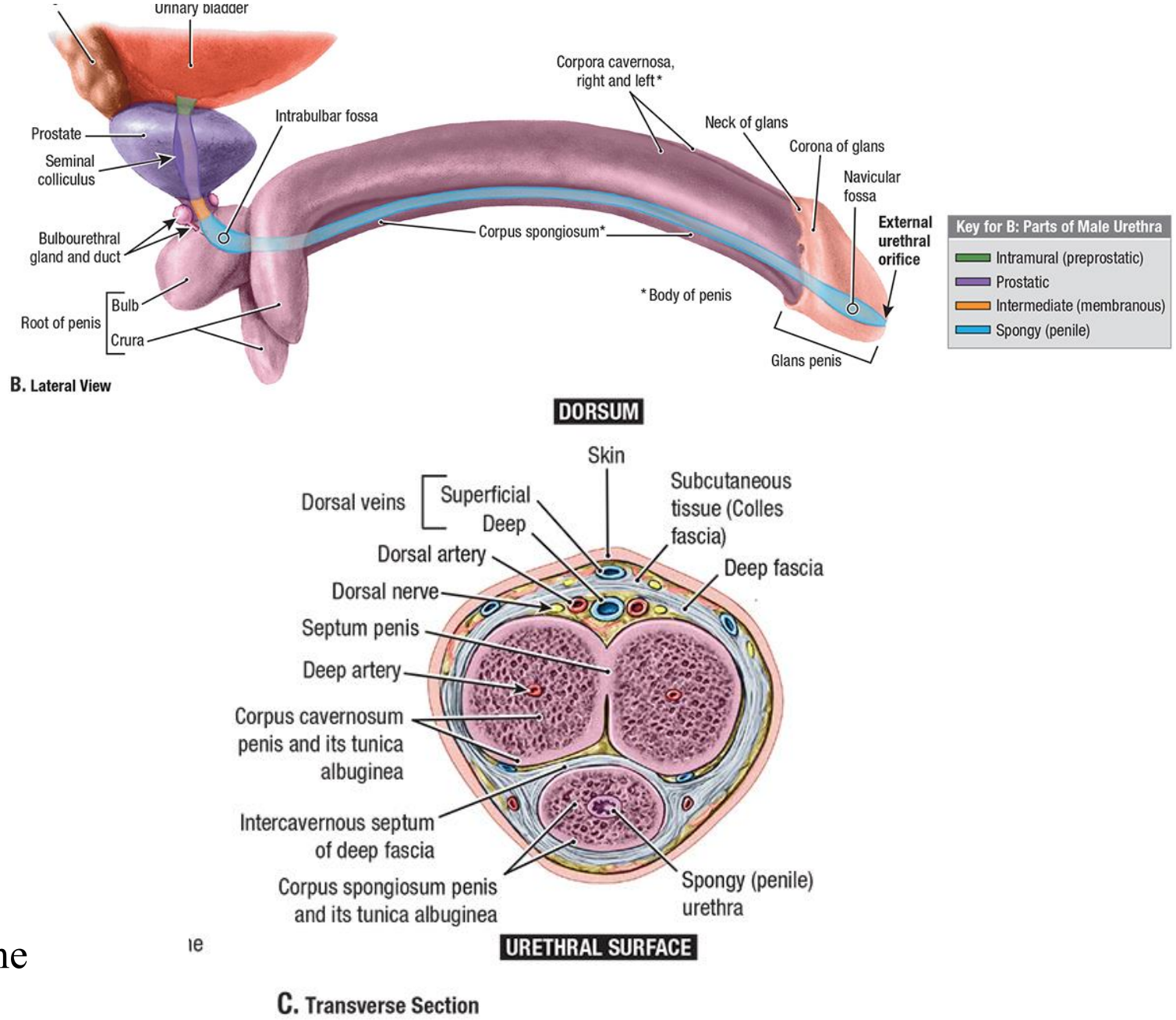


A. Inferior View, Penis Retracted Anteriorly



Penis

- Root
  - Base aligned with Symphysis pubis
  - Several perineal muscles
  - Bulbospongiosus
  - Ischiocavernosus
- Body
  - Three cylindrical structures
    - Each enclosed in a connective tissue tunica albuginea
  - Paired corpus cavernosum
  - Ventral corpus spongiosum
    - Houses the urethra
- Glans Penis
  - Distal expansion of the cavernosa
  - Corona extends beyond the ends of the cavernosum

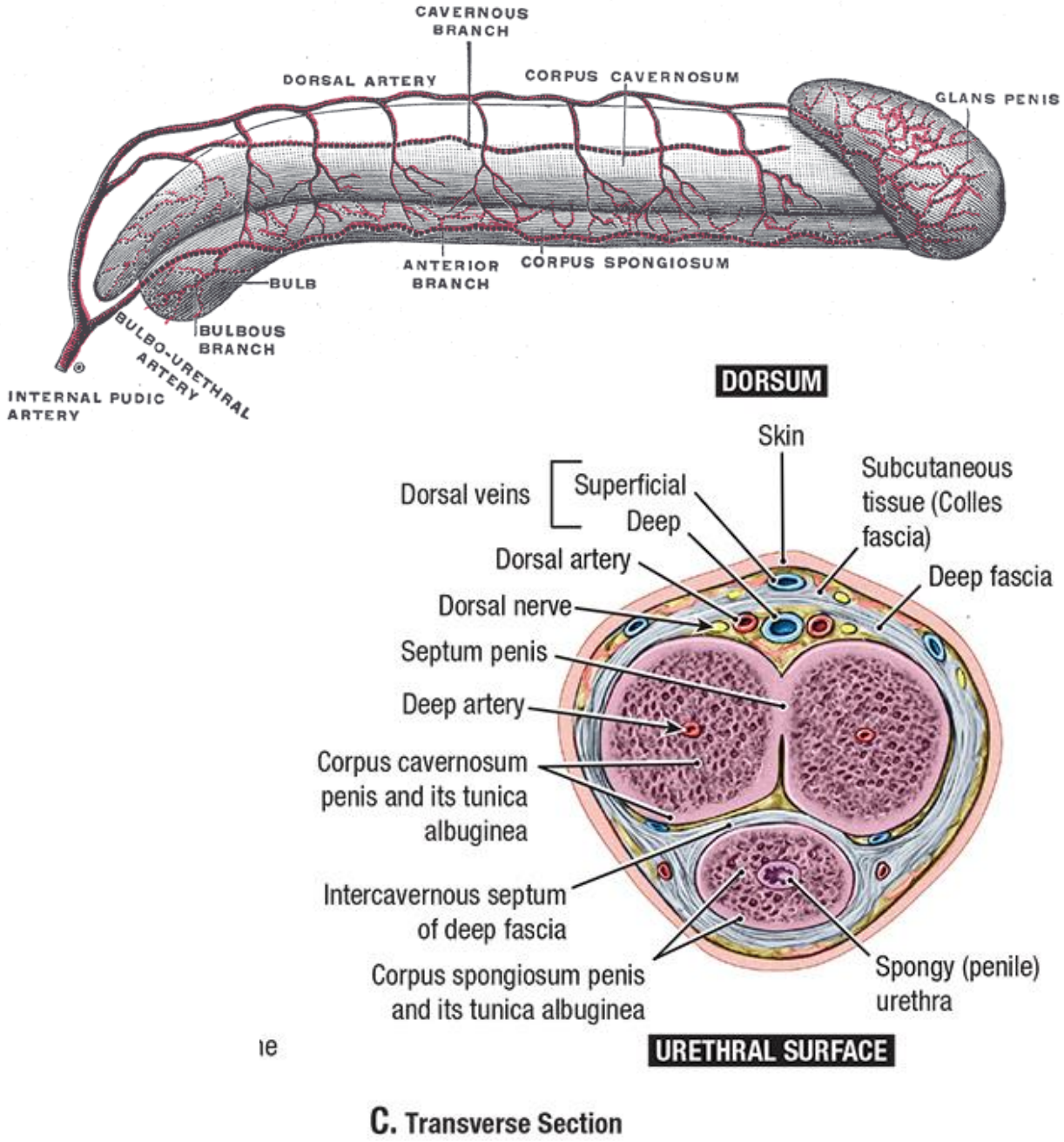


Neurovasculature of Penis

- All internal Penile structures are fed from branches of the Internal Pudendal arteries
- Dorsal artery of the penis
  - Beneath deep fascia
- Deep arteries of the penis
  - Runs in the cavernosum
  - Helicine artery branches feed the cavernous spaces
  - When Flaccid they are coiled to block blood flow
- Bulbourethral artery serves spongiosum

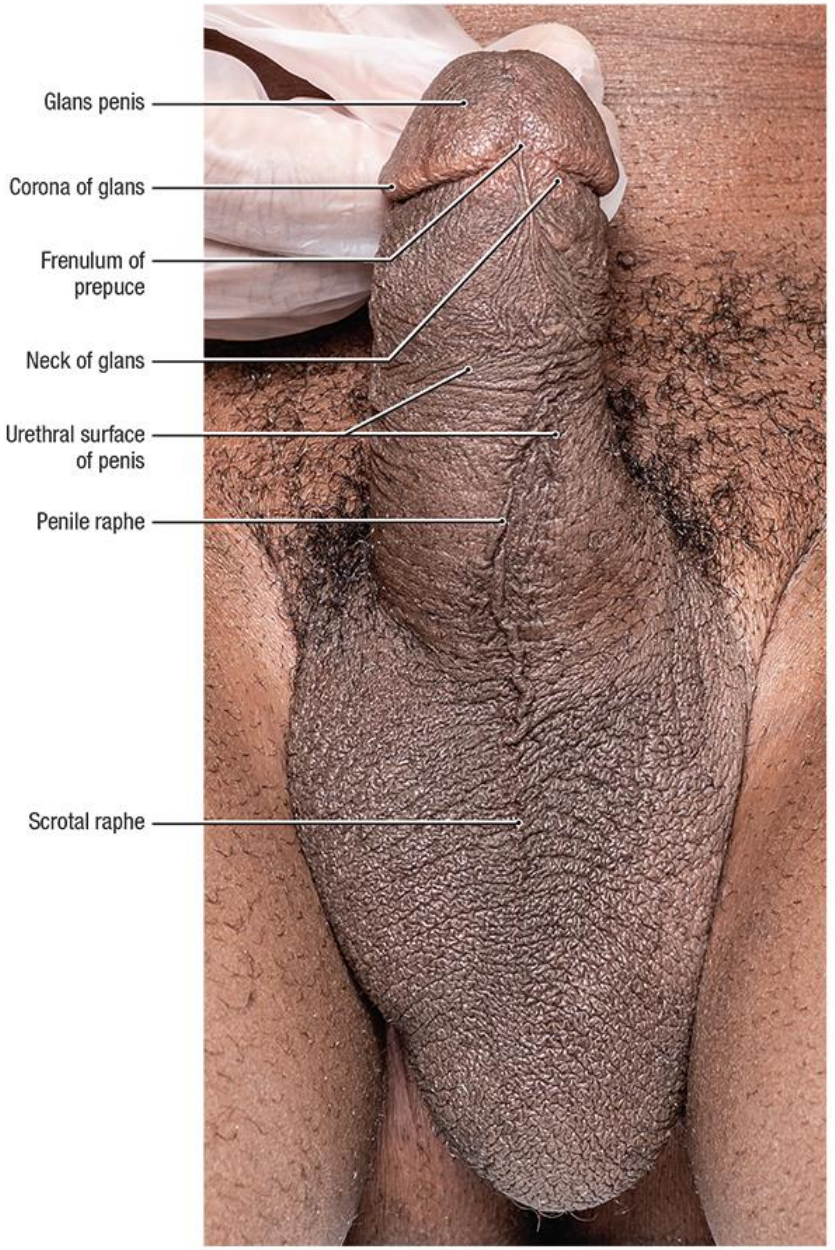
Dorsal Veins drains the penis

- Deep dorsal drains the erectile tissue
- Superficial dorsal drains skin

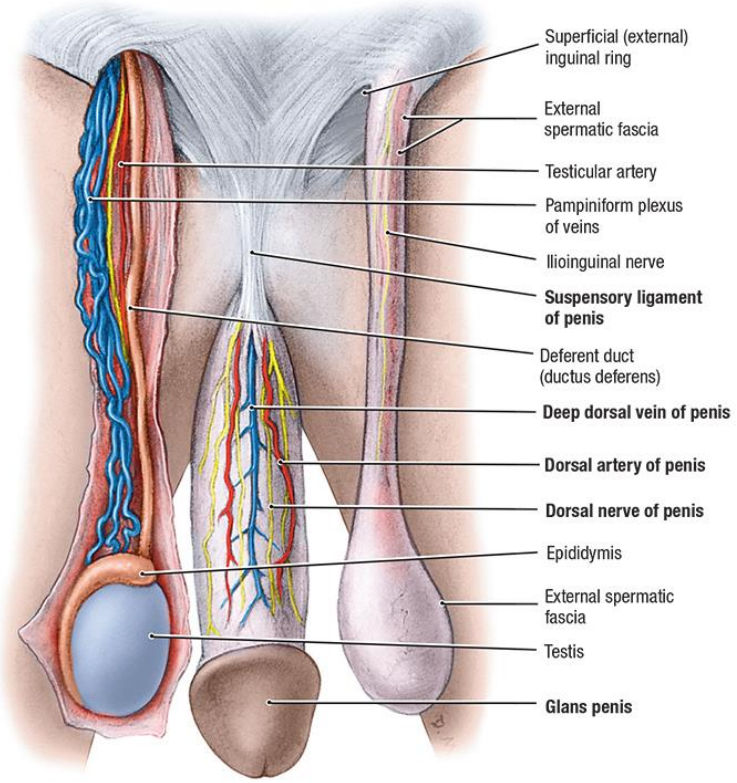


Scrotum

- Scrotum is a Fibromuscular sac that houses the testes
  - Inferior to the pubic symphysis
- Bilateral symmetry
  - Ventral Midline bisected by the scrotal Raphe
  - Continuous w/penile and perineal raphes



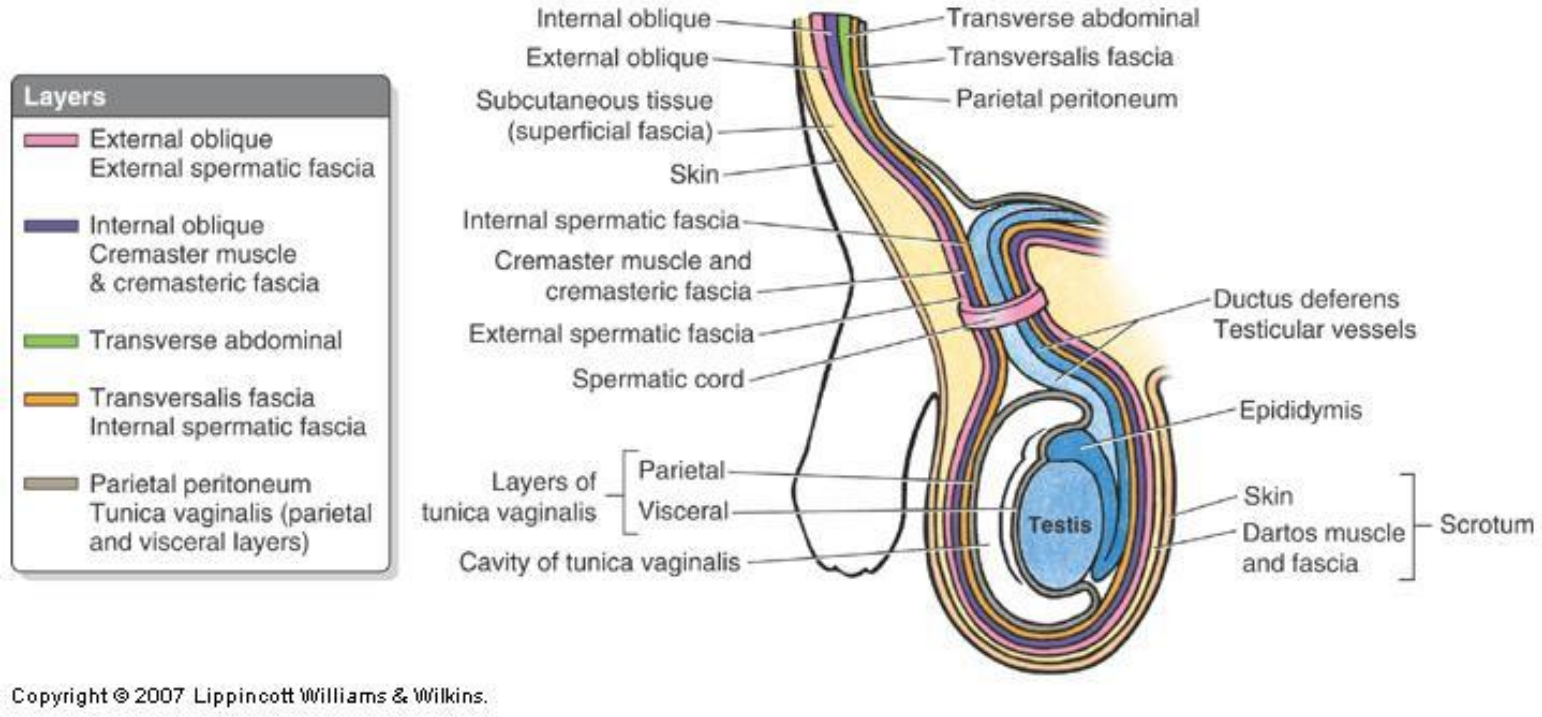
A. Inferior View, Penis Retracted Anteriorly



C. Anterior View

Cord Coverings and Fascia

- Internal Spermatic Fascia from the Transversalis Fascia
- Cremasteric Fascia from the internal oblique fascia
- Cremasteric Muscle from the internal oblique muscles
  - Retraction of the testicles
- External Spermatic Fascia from the external oblique aponeurosis and fascia
- Tunica Vaginalis from the parietal peritoneum



Copyright © 2007 Lippincott Williams & Wilkins.

Penile/Scrotal Clinical Considerations

- Rupture of the urethra in males
  - Extravasation of urine into the deep perineal pouch when intermediate region involved
  - Into superficial space if spongy urethra involved
    - Can collect in the connective tissue around the scrotal sac

